Economic and Psychological Returns to Social Relationships: Alleviating Constraints to Network Formation in Malawi*

Gabriella Fleischman†

September 16, 2026

Click here for the latest version

Abstract

What are the causal effects of new social relationships, and do returns to link formation vary within versus across social class? I conduct a field experiment among women in rural Malawi who relocate to their husband's home village, thereby disrupting their preexisting social ties. I randomly assign low socioeconomic status (SES) women with the opportunity to send meal invitations to neighboring women willing to receive them. I experimentally vary whether the names appearing on the list are all low-SES, all high-SES, or both. I cross-randomize the invitation-sending arms with an in-kind meal subsidy, which weakens financial constraints to meal-sharing. Eighty-one percent of women send invitations across all treatment arms, while the meal subsidy shifts the composition of invitations towards high-SES women. There are substantial but differential returns to cross- and same-SES linkages: one year later, participants with the opportunity to invite higher-SES guests are 20% more likely to earn income from self-employment and report a 0.23 SD improvement in food security, while women with the opportunity to invite guests within-SES experience a 0.22 SD (29%) reduction in depressive symptoms. These results highlight a trade-off in network formation, where social ties that offer economic advancement may come at the expense of relationships that improve psychological well-being.

*I am deeply grateful to Marcella Alsan, Emily Breza, Reshmaan Hussam, and Eliana La Ferrara for their guidance and support. I am indebted to James Mkandawire, Davie Chitenje, Tombozgani Mhango, the Invest in Knowledge Initiative (IKI) data collection team, and Stella Idadi, without whom this work would not have been possible. I am thankful for useful comments and suggestions from Alice Danon, Laura Derksen, Akib Khan, Lauren Rice, Daniel Suryadarma, Christina Qiu, Sarah Vincent, and participants at numerous seminars and conferences. Reagan Curtis and Samira Mathur provided excellent research assistance. I gratefully acknowledge financial support from the Weiss Fund, the Adrienne Hall Fund at the Harvard Kennedy School Women and Public Policy Program, the Institute for Quantitative Social Sciences at Harvard University, the Center for International Development at the Harvard Kennedy School, the Graduate School of Arts and Sciences at Harvard University, the Center for African Studies at Harvard University, and SurveyCTO. This study received IRB approval from Harvard University protocol number IRB22-0484 and the Malawi National Committee on Research in Social Sciences and Humanities protocol P.05/23/760. The experiment was pre-registered under RCT ID AEARCTR-0013644.

†Fleischman: University of Pittsburgh, gfleischman@pitt.edu

1 Introduction

Economic theory posits that social connections are both fundamental for market functions and intrinsically utility-generating (Becker, 1974; Bentham, 1789). Empirically, the economics literature has linked social interactions with employment, economic mobility, financial inclusion and increased savings (Banerjee et al., 2013; Beaman, 2012; Chetty et al., 2022a,b).[1] Social connections might be even more critical in low-income contexts, where informal risk-sharing provides services to fill missing markets (Coate and Ravallion, 1993; Fafchamps and Lund, 2003; Karlan et al., 2017; La Ferrara, 2003; Ligon and Schechter, 2003, 2012; Munshi, 2014; Townsend, 1994; Udry, 1994). At the same time, a wealth of descriptive evidence, largely in fields outside economics, finds that social relationships are a powerful predictor of well-being (Banerjee et al., 2023; Algren et al., 2020; Morrish et al., 2022; Rico-Uribe et al., 2018).

What are the causal effects of forming new relationships, and are there different returns to interacting with someone in the same social class or in a higher social class? Despite the view that social interactions matter for markets and well-being, endogenous link formation makes the causal effects of social connection difficult to study. There is even less economics research identifying what encourages or constrains social connections from arising, or characterizing trade-offs between social ties.[2] This study asks these questions among a sample of women who moved across rural villages in Malawi, usually for marriage. Having been uprooted from their families and existing social networks, these women are in a circumstance that creates the conditions for social isolation, loneliness, and economic vulnerability. Given that 73% of societies worldwide are traditionally patrilocal (women move to their husband's villages), and there is evidence that patrilocality is becoming increasingly common, this reflects a common experience for rural women (Murdock, 1967; Shenk et al., 2019).

I conduct a randomized controlled trial that exogenously provides these women with opportunities to link with each other. I randomly assign 1,200 low socioeconomic status (SES) female migrants to receive a list of the names of neighboring women willing to

[1] There is a rapidly growing base of empirical evidence on the role of social networks for economic outcomes, especially within development economics. A large body of work takes the social network as a fixed object through which information, technologies, or norms diffuse (Banerjee et al., 2013, 2019, 2024b; Beaman, 2012; Beaman and Magruder, 2012; Beaman et al., 2021; Breza and Chandrasekhar, 2019; Breza et al., 2024; Chandrasekhar et al., 2018, 2022; Conley and Udry, 2010; Foster and Rosenzweig, 1995; Jensen, 2007; Iacobelli and Singh, 2020; Kondylis et al., 2023). A smaller literature evaluates shocks to network structure, such as introducing people through group-based or mentoring programs (Asiedu et al., 2025; Duflo et al., 2023; Feigenberg et al., 2013; Schechter and Vasudevan, 2024; Vasilaky and Leonard, 2018; Vasilaky, 2013), or asks how interventions lead to social network changes (Banerjee et al., 2024a; Derksen and Souza, 2024).

[2] Classic works in sociology and political science addressing theoretical trade-offs across network links and structures include Blau (1977); Bourdieu and Passeron (1977); Coleman (1988); Granovetter (1973); Lin (2001); Portes (1998); Putnam (2000); Tilly (1998).

receive meal invitations and the opportunity to send them invitations. I experimentally vary whether the names appearing on the list are all other low-SES women, all high-SES women, or a random mix of both low-SES and high-SES women. I cross-randomize the invitation-sending arms with an in-kind meal subsidy, which weakens financial constraints to meal-sharing. To establish which women are willing to receive invitations, I first ask participants if they would like their name to be on a list from which others can send them an invitation for a shared meal, which 99% of women sign up for. I then show treated women (*Inviters*) six names of their neighbors from that list (*Guests*) and offer to send from zero to five invitations delivered through an enumerator. Although women in the experiment are acquainted with one another, the intervention serves as a genuine shock to status quo social interactions: 81% of Inviters send an invitation through the intervention, 99% of which are to women outside their self-reported baseline networks.

One year later, the pooled Inviter treatment leads to a 0.14 standard deviation (SD) increase in an index of food security and a 0.16 SD (23%) reduction in experiencing mild-to-severe depressive symptoms. The treatment effects on food security are significantly larger among Inviters who I randomize to only interact with high-SES women (*Inviters with the High-SES Guest List*) relative to Inviters who only interact with low-SES women (*Inviters with the Low-SES Guest List*) (a 0.23 SD increase versus a 0.07 SD increase, $p = 0.025$). Conversely, the treatment effects on depressive symptoms are significantly larger among Inviters with the Low-SES Guest List relative to Inviters with the High-SES Guest List (a 10.4 percentage point reduction versus a 4.5 percentage point reduction, $p = 0.060$). The economic effects are driven by Inviters with the High-SES Guest List earning income from new self-employment activities within just one month of the intervention, while there are no impacts on borrowing or food transfers. These results are especially impactful given the context: Malawi was in a severe food shortage during the study—the fifth that Malawi has experienced in the twenty-first century—where an estimated 28% of the population suffered acute food insecurity (UNICEF, 2025).

Given that extraordinary benefits appear to arise from little more than a nudge, what prevents these interactions from happening organically? I first ask if there are barriers that preclude individuals from interacting with specific types of people. The probability of sending an invitation or sharing a meal is not different across the groups of women who are constrained by the experiment to send an invitation only to low-SES or high-SES women, implying that women are *not* restricted to interact only with people within their own social class, for example, through social norms. At the same time, I do find that interaction with low-SES and high-SES women yield meaningfully different outcomes.

Is social interaction restricted by financial costs? To answer this question, I cross-randomize a voucher that is redeemable at a local butcher. Meat is a well-liked yet expensive and rarely consumed good in this setting, so the voucher effectively reduces

the price of a nice meal to the same price as an everyday meal. The voucher does not change invitation-sending or meal-sharing in either of the groups with the Low-SES-only or High-SES-only Guest Lists. Even when a high-quality meal is unaffordable, sending invitations to either type is better than nothing, suggesting that the binding constraint to initiating relationships is what the Inviter treatment alone resolves for everyone: women do not know who to invite, and it requires some prohibitive margin of effort.

While the financial cost of sharing a high-quality meal does not limit women from sending invitations, it does affect the composition of who they invite. Among Inviters who can invite any type of woman (*Inviters with the Random Guest List*), receipt of the voucher significantly increases the likelihood of inviting high-SES women, driven by Inviters who also invite a low-SES woman. When women are budget-constrained and the option to invite low-SES guests is available, the scenario that best approximates choices outside of the experiment, being unable to afford serving a nice meal *does* constrain interactions across social class. However, reduced prices never generate cross-SES connections at the expense of within-SES connections, underscoring the value of the psychological benefits of within-SES links.

I interpret these findings through the lens of a conceptual model of social network formation under budget constraints, where relationships have both intrinsic and instrumental value. I make two critical assumptions: first, that low- and high-SES relationships are each better at producing one type of benefit, and second, that the benefits of high-SES relationships are more sensitive to financial investment in social interaction than the benefits of low-SES relationships. The model generates two key insights: first, when people value both economic and psychological well-being, cross-SES linking is optimally achieved through *economically diverse* networks (rather than supplanting within-SES networks with cross-SES networks), since low-SES relationships yield important intrinsic utility benefits that are not substitutable for high-SES links; and second, the high prices of the most valuable social activities makes economically diverse linking unaffordable and unattractive relative to homophily, implying that income-based homophily—which is defined by network sorting by the ability to afford high-priced goods—is self-reinforcing.

The primary contribution of this paper speaks to the economics literature on the value of social ties. I present novel evidence that the economic and psychological returns from social relationships do not come from the same people, implying that, when social linking is constrained, individuals face highly consequential trade-offs in determining with whom to interact.

The finding that relationships generate economic and psychological value are each independent and meaningful contributions. First, I identify, to my knowledge, the first causal evidence that social relationships generate long-term improvements in mental health.

This finding extends a large body of evidence documenting a powerful relationship between social connection and well-being. The evidence to date is descriptive (Algren et al., 2020; Morrish et al., 2022; Rico-Uribe et al., 2018), or experimentally evaluates the impacts of one-time social interactions with strangers (Epley and Schroeder, 2014; Epley et al., 2022). My results suggest that the social isolation accompanying low-income may be an important mechanism underlying the relationship between low-income and poor mental health and loneliness (Banerjee et al., 2023; Ridley et al., 2020; Surkalim et al., 2022), a pathway that has been understudied empirically relative to scarcity and stress mechanisms (Annan and Archibong, 2023; Haushofer and Salicath, 2023). Furthermore, I provide a practicable intervention that reduces long-term depressive symptoms among low-income and marginalized women by 30%.[3]

Second, I find that exogenously formed high-SES network links generate large impacts on food security that are consistent with seasonal consumption smoothing, implying that risk-smoothing is limited by frictions in creating new social relationships. A large and classic literature in development economics evaluates the presence and efficiency of risk-smoothing—across people, time, plots, and places—within villages in low-income countries (Attanasio et al., 2012; Beaman et al., 2014; Ligon, 1998; Ligon et al., 2002; Kinnan and Townsend, 2012; Kinnan et al., 2024; Mobarak and Rosenzweig, 2013; Rozensweig and Stark, 1989; Santos and Barrett, 2011; Townsend, 1994). This literature is predicated on the idea that social ties are an important mechanism through which people share risk, but it is rare to find and evaluate exogenous network shocks, especially without technological innovation (Jack and Suri, 2014; Riley, 2018). The fact that the economic benefits I identify come from high-SES ties speaks to a growing body of work describing the positive relationship between peers' economic outcomes and one's own economic outcomes (Bergman et al., 2024; Cannon et al., 2024; Chetty et al., 2016, 2022b; Laschever, 2013; Linde and Egede, 2023; Ludwig et al., 2013; Mallah, 2026; Patel and Wolfe, 2024; Sacerdote, 2011). This paper significantly expands the external validity of the causal evidence base in this literature, as a paper that is experimental, in low-income country, and outside of an educational context.[4]

Lastly, I contribute to the social networks literature. While a nascent body of evidence

[3]There remains a great need to find novel cost-effective mitigation strategies. Many of the interventions that positively impact mental health are either not cost-effective for that specific outcome (such as cash transfers), or are cost-effective but explicitly target mental health disorders (Angeluci and Bennett, 2024; Bhat et al., 2022; Blattman et al., 2023; McKelway et al., 2023; Patel et al., 2007), which comes at the risk of low take-up or stigma (Pescosolido, 2013; Ridley, 2025; Smith, 2024) (see Haushofer and Salicath (2023) and Ridley et al. (2020) for a review).

[4]Much of the evidence documenting the prevalence and correlates of economic connectedness (how connected low-income individuals are with high-income individuals) is from the United States. The causal evidence of economic connectedness to date either bundles economic connectedness with other associated goods (such as neighborhoods or classrooms), or analyzes the impacts of economic connectedness that happens through the randomization of peer groups, often in classrooms or dorm rooms.

in economics shows that network externalities can causally limit social network formation (Anders and Pallais, 2025; Banerjee et al., 2024a), my experiment demonstrates how individual-level constraints to meaningful social interactions can inhibit social linking even when meeting opportunities are available and people want to link, with large economic consequences. Furthermore, my experiment presents causal evidence identifying a mechanism underlying homophily in choices over with whom to interact. Despite the widespread prevalence of homophily and its economic relevance, we have minimal causal evidence identifying *why* social networks become and remain homophilic (Jackson, 2025; McPherson et al., 2001; Verbrugge, 1977).[5] I show that the financial cost of valuable social interaction inhibits lower-income women from initiating economically valuable relationships with higher-income women, and that, when women face the least constraints, many of them seek to form new relationships both within- and across-SES.[6]

The paper proceeds as follows: Section 2 discusses the cultural context, Section 3 describes the experimental design, Section 4 presents first-stage results of the experiment (invitation-sending decisions), Section 5 presents treatment effects on second stage outcomes, Section 6 interprets the results through a conceptual model of social network formation, and Section 7 concludes.

2 Context

2.1 Rural Migration

This project is set in Mchinji, a rural district in Central Malawi. Eighty-five percent of my sample moved across rural villages for marriage, which separates women from their families and preexisting networks. Marriage migration is an exceedingly common phenomenon, especially for women. Women traditionally move to the area where her husband and his kin live (patrilocality) in 71% of societies (Murdock, 1967), and more societies have shifted towards patrilocality than away (Shenk et al., 2019).

In the 2015 Malawi Demographic and Health Survey, 57% of women in Mchinji lived outside their home village. This is a fairly representative district in Malawi—43% of

[5]Empirically, economic research shows that homophily can hurt economies by stymieing the flow of information or goods (Burchardi and Hassan, 2013; Chaney, 2014), and can limit social cohesion (Rao, 2019). On the other hand, homophily can improve loan targeting and health adherence through reduced information asymmetries and improved trust (Alsan et al., 2019; Fisman et al., 2017).

[6]Furthermore, identifying a causal link between meat consumption and strategic social network investments adds context to the literature in development economics on nutritional poverty traps (Dasgupta and Ray, 1986). A wide body of evidence finds that individuals consistently choose to spend excess income on foods that are expensive per calorie, rather than cheap calorie-dense foods, which has brought the nutritional poverty trap model into question (Jensen and Miller, 2008; Subramanian and Deaton, 1996). I provide evidence that consuming expensive calories can be a rational *social investment decision* with substantial economic returns—in addition to being a health input or luxury consumption good.

the women in all other districts in Malawi lived outside their home village, though with regional variation (Appendix Figure A.1). Land ownership is an important predictor of moving villages, even conditional on ethnic and regional kinship norms (Appendix Table A.1).[7] Overall, this sample is triply disadvantaged with regards to social inclusion on the basis of their gender, migrant status, and income status.

2.2 Correlates of Social Inclusion and Socio-Economic Status

I construct an index of socio-economic status, using proxies of household assets (a detailed description of how I construct the index is in Section 3.3.1). Importantly, the women who I code as 'high-SES' are poor on a global scale. On average, only 20% of them have a single household member who has completed secondary school, and they eat fewer than three meals per day even right after the harvest (Appendix Table A.2). However, they are still significantly more advantaged than the women I code as 'low-SES', with almost double the landholdings, and almost double the likelihood of operating any business.

I find descriptive evidence that income is a social cleavage in my setting. Socio-economic status and years in the village are both positively and significantly correlated with measures of social inclusion, such as having more friends, being more likely to belong to a women's church group, and reporting less loneliness (Figures A.2 and A.3). These measures improve with time in the village very slowly—women who have been in the village for twenty years have, on average, only one more friend than women who arrived this year—suggesting that frictions to social network linking may not resolve with time. Inclusion in economic relationships is also related to socio-economic status. Women in the top decile of socio-economic status are almost three times more likely to belong to a Village Savings and Loan Association (VSLA), an important source of credit in this setting, compared to women in the bottom decile (Figure A.3).

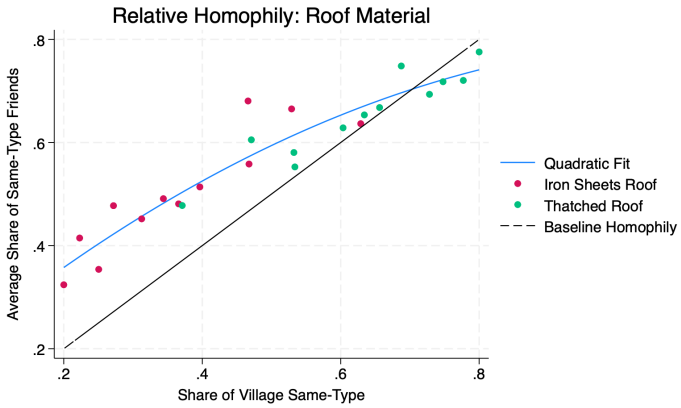
2.2.1 Baseline Networks

To elicit networks, enumerators ask participants to list each person with whom she engages with in a series of activities. At baseline, participants have six network connections (mean = 6.2, median = 6, IQR=2). Notably, however, almost half of the network is comprised of extended family from the husband's side, while women only have 2.2 network links who are their own friends. The most common activity to engage with network links in is meal-sharing, but usually with the husband's family: at baseline, only 56% of women list any non-relative who they report sharing a meal with in the past year.

Baseline networks are homophilic with respect to income. In every village, participants

[7] While the Chewa tribe, a predominant tribe in my sample, is traditionally matrilocal, norms have been largely disrupted and patrilocality is common even within traditionally matrilocal tribes.

Figure 1: Income-Based Homophily



Notes: Each red point represents the average of participants within a village who have an iron-sheets roof (proxy for high-SES). Each green point represents the average of participants within a village who have a thatched roof (proxy for low-SES). The x-axis is the percent of participants in the village with that same roof material, while the y-axis is the average percent of people in the network with the same roof material. The 45-degree black line represents "baseline homophily" – the percent of network members with the same roof material that we would expect if people linked randomly within the village.

with iron sheets roofs are more likely to link with other people with iron sheets roofs (a proxy for higher-income) than if they linked randomly within the village; and in all but four villages, participants with thatched roofs are more likely to link with other people with thatched roofs than if they linked randomly within the village (Figure 1).[8]

2.2.2 Depression and Loneliness

I measure depressive symptoms using the CESDR-10 (Center for Epidemiological Studies Depression Scale revised 10-item questionnaire). Consistent with convention, I denote anyone who scores 10 or greater as at risk of mild-to-severe depression. I measure loneliness using the UCLA-3 questionnaire, a three-item loneliness scale. Consistent with convention, I denote anyone who scores 6 or greater as at risk of loneliness. Among my sample of low-income migrant women, 27% are likely depressed at baseline and 14% are likely lonely. Although depression and loneliness are correlated, the correlation is quite weak ($\rho = 0.113$) and only 6% of the sample is both likely depressed and lonely.

Consistent with recent work documenting high levels of depression among the elderly in low- and middle-income countries (Banerjee et al., 2023), depression risk is starkly increasing in age (Figure A.4, left-side panel). Conversely, loneliness is U-shaped: the prevalence of loneliness is highest among the oldest and youngest deciles, and lowest in middle age.[9] Surprisingly, while there is a significant correlation between the SES index

[8]Many of my respondents' network connections are outside my sample, and I do not have detailed information about their household income. As an easily observable and salient income proxy, participants are able to report the roof material of each network connection.

[9]Twenty-three percent of the youngest decile (ages 18-20), versus 36% of the oldest decile (ages 42-69), experience depressive symptoms. Sixteen percent of women ages 18-20 and ages 42-69 experience loneliness, while the least lonely cohort is women ages 37-41, where 9% experience loneliness.

and loneliness, there is *no correlation* between the SES index and depression.

There are two variables that emerge as predictors of both loneliness and depression: low food consumption and the difficulty of returning to a woman's home village. Consuming on average three meals a day (38% of the low-SES sample) and having consumed any meat in the past month (77% of the low-SES sample) are each significantly associated with reductions in both depression and loneliness.[10] I construct a PCA index of how challenging it is for a participant to return to her home village using the price of transport, time that a trip home takes, and visits home per year. This index is significantly correlated with both depression and loneliness, which reiterates the importance of migrant status to these women's disadvantage (Figure A.4, right-side panel).

2.3 Meal-Sharing

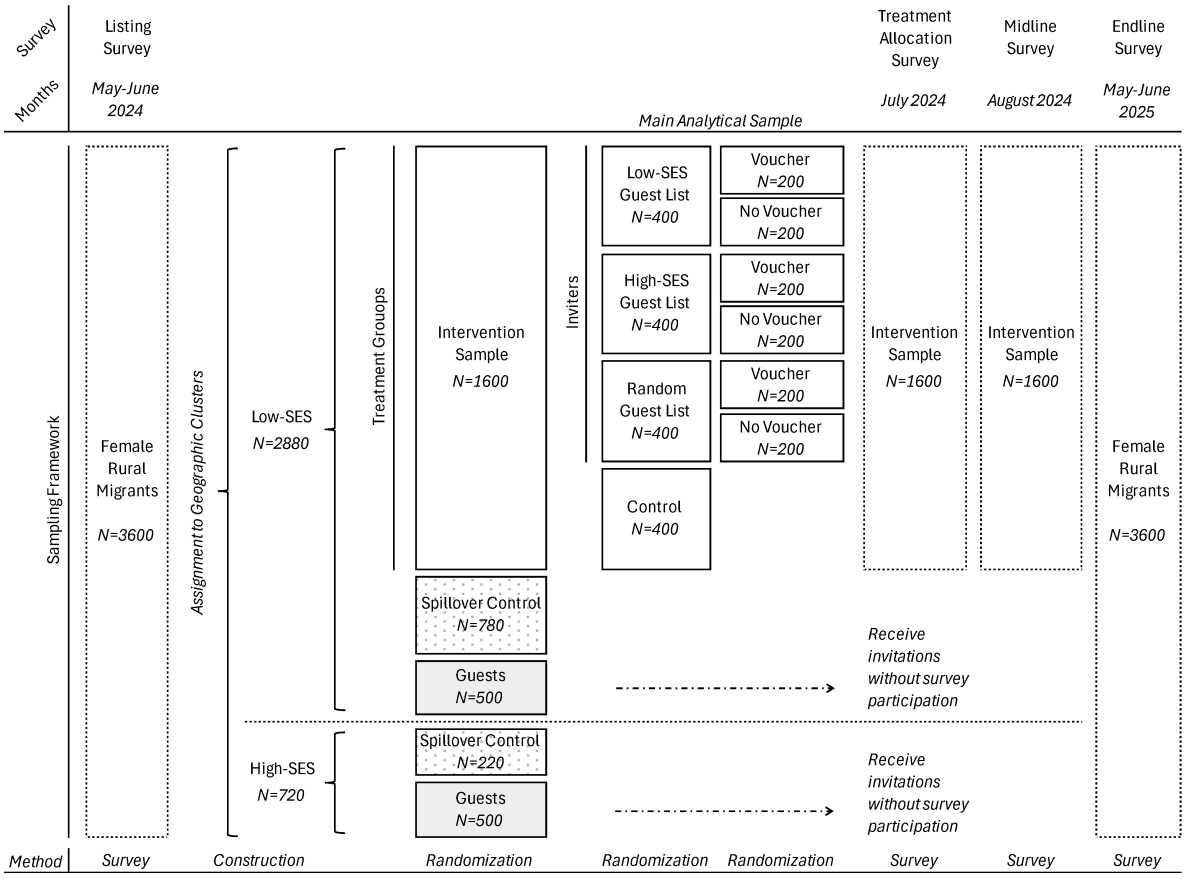
Meal-sharing is a common social activity. At baseline, 90% of respondents report that they shared a meal with somebody outside their household in the previous week, with 68% of respondents both visiting and hosting somebody at least once. Participants report hosting others more often than visiting others, though both are common. For shared meals that participants host and shared meals where they visit others, it is equally common to go after being invited and to initiate a shared meal without an invitation.[11]

Why is meal-sharing such a common activity? While *food*-sharing is a common method of risk-sharing, *meal*-sharing also encompasses sitting down together and eating with one another. In a sample of sixty-eight survey participants in rural Lilongwe (the district adjacent to Mchinji), I ask participants how personal a conversation could become across four settings: a shared meal, while working, while doing chores, and while at church. Both male and female participants overwhelming and consistently rate a shared meal as the setting where a conversation could become the most personal, regardless of if they imagine the shared meal to be with somebody who they typically share meals or with, or somebody they know from working together (Appendix Table A.3). Despite the high prevalence of meal-sharing, 72% of respondents report that they would prefer to share more meals per week than they shared last week (Appendix Figure A.5).

[10] The SES index is constructed from variables of productive assets and roof quality, rather than consumption, so these correlations are not mechanical.

[11] Participants report that they hosted 2.3 meals and visited others for 1.6 meals in the past week on average. Anecdotally, shared meals that happen without an invitation are a result of visiting a friend in her home while she is cooking, and receiving a spontaneous invitation to join.

Figure 2: Study Design



Notes: Women are eligible for the Listing Survey if they are at least 18 years old, moved to the village after the age of 14, and have been there for fewer than 20 years. Geographic clusters are constructed using participant house coordinates from the Listing Survey and k-means clustering. Geographic clusters are based only on spatial proximity, and do not represent political or administrative boundaries. Women are coded as "Low-SES" if they fall below the 80th percentile on the SES-index within their geographic cluster, and are coded as "High-SES" if they are in the top quintile of the SES-index within their geographic cluster. The SES index is constructed using Anderson (2008) inverse covariance weighting of: household roof material, an indicator for if the woman lives in her own house, acres of land owned, number of garden plots owned, and a business capital index. Low-SES women are randomly assigned to be in the Intervention Sample (N=1600); to be a Spillover Control group member (N=780); or to be a Guest (N=500). High-SES women are randomly assigned to be in the Spillover Control group (N=220), or to be a Guest (N=500). The Intervention Sample participate in every survey and constitute the main analytical sample in the paper. The Spillover Control Group and Guests are only surveyed in the Listing Survey and Endline, and Guests receive invitations during the Treatment Allocation Survey. The Intervention Sample is randomly split into Inviters (N=1200) and Control (N=400). Inviters are randomly assigned to have a Low-SES Guest List (they only view the names of Low-SES Guests on their Sharing List); a High-SES Guest List; or a Random Guest List (they view a mix of high- and low-SES women's names). Conditional on being within 800 meters of the Inviter, conforming to the Guest List sub-randomization, and appearing on no more than ten lists total, six Guests are randomly assigned to each Inviter's Sharing List. I cross-randomize a Voucher for a butcher across all Inviter treatments.

3 Experimental Design and Empirical Specifications

3.1 Sample

The intervention design is summarized in Figure 2. I recruit a sample of 3,600 women who moved to the village after 14 years old, no more than 20 years ago. During the recruitment Listing Survey, every eligible participant can add her name to a list of women who are interested in sharing meals with other women in the village more often (Sharing List), knowing that at a later stage some women in the village will see the list and have the

opportunity to invite people from the list. Almost every woman signs up for the list (99.6%), and almost all agree to have their photo taken and displayed on the list so that women who do not know them by name can still send them an invitation (96.4%).

I use my listing sample data to construct geographic clusters and a continuous SES index, draw quintiles of the index within geographic clusters, and then assign women to high- or low-SES using these geographic-cluster-specific SES quintiles (SES index construction is described in Section 3.3.1). I randomly select 1,600 women from the bottom 80% of the SES index for the intervention (henceforth, the Intervention Sample), since high prices are more likely to be prohibitive for lower-income women. Of the women in the top 20% of the SES index, I randomly select 500 to be a Guest if they agree. Of the women who are in the bottom 80% of the SES index, but are *not* selected for the Intervention Sample, I also randomly select 500 the opportunity to be a Guest. The women who sign up for the Sharing List *and* are randomly selected to be a Guest have the opportunity to actually appear on the Sharing List. The remaining participants serve as a Control group.

3.2 Treatment Arms

The RCT includes two intervention arms: an inviting service (*Inviter* treatment) and a meat price subsidy (*Voucher* treatment). The Inviter treatment includes information about who in the village would like to share meals, which reduces the believed probability of rejection. Enumerators then send invitations for a shared meal on the participants' behalf, which reduces the true effort costs of sending invitations, such as the emotional costs of face-to-face rejection and the hassle of finding women in the village to invite.

The Intervention Sample, which is entirely low-SES, is randomized into three groups:[12]

1. Inviter Treatment (N=600): Participants view six names from the Sharing List and can invite zero to five of them to share a meal (invitation delivered by enumerators).
2. Vouchers + Inviter Treatment (N=600): Participants receive 7,000 MWK vouchers (about 4 USD), redeemable at a local butcher, and the Inviter Treatment.
3. Control (N=400): This group does not receive any intervention.

Within treatments 1 and 2, participants view one of three types of Sharing Lists: Low-SES Guest List (all women on the list are also in the bottom 80% of the SES index), High-SES Guest List (all women on the list are in the top 20% of the SES index), or the Random Guest List (names on the list are randomly selected across both the high-SES and low-

[12]I do not have a Voucher-only treatment arm in my study because I seek to test if, *conditional* on the access to initiate relationships with different sets of individuals, the price of meat serves as a barrier to social interaction. The Voucher treatment does not affect the probability of sending invitations, suggesting that fixed effort costs of social interaction, rather than financial costs, are the first-order constraint that inhibits women from initiating social relationships.

SES subsamples). Respondents and enumerators are blinded to this sub-randomization, and are only informed that Guests are randomized to an Inviter's Sharing List based on proximity. There are 200 participants within each of the smallest subgroups (Figure 2).

The SES index is constructed from variables I collect in the Listing Survey, and has two advantages: because it is continuous, I can draw precise quantiles of income within villages and control the geographic saturation of High- and Low-SES women;[13] and the Anderson (2008) weighting allows me to reward uncorrelated ways of holding wealth within a single index. Randomization is at the individual level, stratified by geographic-area clusters and quintile of the SES index. This stratification ensures that the density of the treatment arms does not differ spatially and create differential cultural shifts; and that average income differences between the Inviters and Guests are constant across treatment arms.

One at a time in a randomly assigned order of Inviters, I randomly select one Guest (among those eligible to appear on each specific Inviter's list) to appear on that Inviter's list. A Guest is eligible to appear for a specific Inviter if she conforms to the Inviter's treatment (for example, a high-SES woman cannot appear on the list of an Inviter randomized to see only low-SES Guest names), lives within 800 meters of the Inviter, and has not already appeared on ten lists. On average, Inviters report that they have spoken to 3.6 women on their list before the intervention, and are significantly more likely to have previously spoken to the women who live relatively closer to them.[14]

3.3 Data

I conduct four rounds of data collection: a Listing Survey with all 3600 participants (May 2024), a Treatment Allocation Survey with the Intervention Sample (July 2024), a Midline Survey with the Intervention Sample one month later (August 2024), and an Endline Survey with the full sample one year later after the following harvest (May 2025).

3.3.1 Listing Survey

The Listing Survey was conducted in May 2024. In this survey, all participants are informed that some people will be selected for a project where they might receive a voucher and the opportunity to invite other women to share a meal. Enumerators explain to participants that, while we cannot say if they will be selected for the program, they have the opportunity to put their name on a list that indicates their interest in receiving

[13]The design requires precise portions of participants to be classified as High- and Low-SES, so we do not conflate the individual effects of cross- versus within-SES relationships across sub-arms with non-random treatment saturation. Varying treatment saturation at the geographic cluster level does not confound this interpretation, because treatment saturation variation is random, and individual treatment assignment is random, so treatment saturation is orthogonal to the individual effect of cross-versus within-SES relationships.

[14]See Appendix Section A.3 for more details about the Sharing List construction.

an invitation for a shared meal from a neighbor. They are informed that the list will not be printed, that it will only be used if at least ten women from within their area sign up, and that signing up for the list does not guarantee that they will receive an invitation or appear on a list.[15] Enumerators then request to take a photo of the participant in front of her house, to help others identify her when they see her name on the list.

Defining Socio-Economic Status

I construct an index of socio-economic status from data collectd in the Listing Survey. The SES index is the inverse-covariance-weighted index (Anderson, 2008) of the following variables: an indicator for having a high-quality roof material, an indicator for whether the respondent lives in her own home, acres of farmland owned, number of garden plots owned, and an index of business capital.[16] There are two main classes of income-generating assets in this region: farmland and businesses. Land ownership serves as a proxy for farming assets, and I create a proxy for business capital based on the type of self-employment that each woman's household engages in.[17]

I code women in the top 20% of the SES index as high-SES (quintiles drawn within geographic clusters), and women in the bottom 80% of the SES index as low-SES. Note that these women are *relatively* lower- and higher-income than one another, but are still all very poor on a global scale, and even within the context of Malawi. That said, there are still stark differences between the high- and low-SES samples (Appendix Table A.2).

Since participants have network connections who are outside the sample, for whom I do not have data, I cannot use the SES index to measure the income of participant's network. Roof material is an observable household feature, and a salient income proxy. I ask participants to report the roof material for each network link, and use the composition of the roof material of network ties as a proxy for network economic connectedness. While roof material is a coarse proxy, and is not as good at identifying the highest-income individuals because 38% of the sample has a higher-quality roof, it approximates the high- and low-SES sub-samples within my sample fairly well (Appendix Table A.2).

[15]Recall that signing up for the Sharing List does not guarantee that they are randomized to be a Guest. This clarification also ensures that Guests who do not receive an invitation are unaware that they ever appeared on any Sharing List. Since almost half of the women who sign up for the Sharing List do not participate in meal-sharing through the experiment, it is unlikely that women attribute non-participation to being excluded by Inviters, rather than by the experimenter. Qualitatively, women who do not participate in meal-sharing through the intervention attribute this exclusion to random group assignment rather than to exclusion by the community.

[16]See Appendix Section A.1 for a detailed description of how I construct the business capital index. To account for cases where roof material is not one's own asset (for example, staying with relatives while the home is under construction), I include a variable indicating if the respondent lives in her own home.

[17]I use the Anderson (2008) index because it down-weights correlated variables, rewarding both the stock and the diversity of assets.

3.3.2 Treatment Allocation Survey

The Treatment Allocation survey was conducted in July 2024. Women are randomized into treatment groups using a computer program after the Listing Survey. Women in the Intervention Sample all participate in the Treatment Allocation survey. First, women are informed if they are recipients of the Voucher treatment. After receiving their Voucher treatment status, women in the Inviter treatment see a list of up to six names and photos and receive the opportunity to send invitations to up to five women on the list.

I use the following three comparisons to test for baseline balance: all Inviters versus Control participants in the Intervention sample (Appendix Table B.1), Inviters with the High-SES Guest List versus Inviters with the Low-SES Guest List (Appendix Table B.2), and Inviters with the Random Guest List and the Voucher versus Inviters with the Random Guest List without the Voucher (Appendix Tables B.4).[18] Across all three of these tests, two unbalanced variable emerges: the Inviters with the Low-SES Guest List are rent more garden plots than Inviters with the High-SES Guest List (0.170 SD difference), and Inviters with the High-SES Guest List are more likely than Inviters with the Low-SES Guest List to name their husband in their baseline network (0.068 SD difference) (Appendix Table B.2). Thus, in all specifications, I control for these two variables.[19]

3.3.3 Midline and Endline Surveys

I collect outcome data one month after the Treatment Allocation Survey in a Midline Survey (August 2024, with the Intervention Sample only), and one year after the Treatment Allocation Survey in an Endline Survey (May 2025, with the full sample).

Main Outcomes

I analyze three primary downstream outcomes: a food security index, total farm yields in USD value, and an indicator for experiencing mild-to-severe depressive symptoms.[20]

Food Security Index: The food security index is the first PCA component of the typical number of meals participants eat daily and the frequency of animal protein consumption. Food and protein consumption are both particularly consequential in this sample, 50% of whom report pregnancy, child-birth, or breast-feeding during the intervention. Variables

[18] I present comparisons of all other sub-groups in Tables B.2 and B.3 for completeness. To avoid overfitting, I limit my the balance tests that I use to select control variables to the three main comparisons that I use in analytical inference.

[19] Reassuringly, this imbalance goes in the opposite direction of treatment effects. Inviters with the High-SES Guest List, the group with fewer baseline garden plots, improve more in their food security. Inviters with the Low-SES Guest List, the group less likely to socialize with their husbands at baseline, are suggestively the most likely to improve in their marital relationships. Controlling for these unbalanced variables, women who actually share meals are also balanced on all other variables across all groups (Appendix Table B.7), reaffirming that it is very unlikely that selection on these variables drives results.

[20] See Appendix Section E.1 for explanations of deviations in main outcomes from the Pre-Analysis Plan, and a populated Main Outcomes table as described in the Pre-Analysis Plan.

in the midline food security index are estimated over the past month, and I exclude meat consumption because it is mechanically affected by the Voucher treatment. One year later, a positive shock to consumption during the "lean season"—the annual period of lowest consumption, which falls between the midline and endline surveys—is potentially more welfare-enhancing than over the past month because it has implications for consumption-smoothing. Bearing in mind the value of food security in the population broadly, and the augmented importance of food security in the lean season, the one-year food security index incorporates measures of the frequency of daily meals and the frequency of consuming animal proteins (including meat), both in the past month and during the lean season (more details about variable definitions are in Appendix Section A.4).

Farm Yields in USD Value: Respondents self-report their yields for each crop, estimating their expected yields if they have not finished harvesting all of their crops (the endline survey overlaps with the end of the harvest). Participants report yields in the unit of measurement that they harvest each crop, and I convert their total yields into kilograms. Then, for the most frequently cultivated crops—maize, beans, groundnuts, soybeans, sunflower seeds, tobacco, sugarcane, pumpkin, and sweet potatoes—I multiply their total harvest by the market rate per kilogram in USD, as reported by local news articles. I only collect this outcome in the Endline Survey, since there is an annual harvest and the Midline Survey was before the agricultural season.

Mild-to-Severe Depressive Symptoms: I measure depressive symptoms using the CESDR-10 tool (Center for Epidemiological Studies Depression Scale revised 10-item questionnaire). Consistent with convention, I denote anyone who scores 10 or greater as at risk of depression. Consistent with convention, "Mild depressive symptoms" is a score from 10-15; "Moderate" is a score from 16-21; and "Severe" is a score of at least 22.

Network Elicitation

I elicit networks by asking participants to report the names of each person with whom they engage in a series of economic and social activities. For each network link, I ask respondents details about their relationship. The first time that they report a given network tie, I ask a series of objective questions about each link that are likely to remain dynamically stable: their ethnicity, religion, household roof material, marital status, and proximity in walking time. In each survey round where a respondent names someone in the network, I ask her to report two objective facts about the relationship—how often they see each other, and if they most typically see each other in groups or alone—and five subjective questions, which I use to define "strength of ties".

Strength of Ties: For each network link that I elicit, I collect data about the respondent's subjective experience in that relationship. I use these variables to measure "strength of ties". There are five variables that enter the "strength of ties" index, capturing trust,

humor, closeness, and reciprocity (questions described in Appendix Section A.4).

I conduct a PCA analysis of these five variables across all ties (at the dyad level) in the Listing Sample. I use the first principal component to construct weights that I then apply to these five variables across all following surveys. This method applies the baseline levels and correlation structure of these five variables to analyze strength of ties in following survey rounds, so that the definition of "strong ties" remains fixed. Furthermore, this method allows me to conduct a levels comparison in strength-of-ties over time.

In order to identify the ties that are the strongest, and most clearly represent something like a "best friend", I define "strong ties" as ties whose strength-of-ties index is equal to the maximum (6.2% of baseline ties; 13% of midline ties; and 17% of endline ties). I also analyze ties both above and below the baseline median on the strength-of-ties index ("weak ties" and "mid-strength ties" for simplicity of exposition).

3.4 Estimating Equations

3.4.1 First Stage

First I evaluate Inviters' decisions about who to invite using the following specification:

$$I_i = \beta_0 + \beta_1 T_i^{R,V} + \beta_2 T^{H,NV} + \beta_3 T^{H,V} + \beta_4 T^{L,NV} + \beta_5 T^{L,V} + X_i + \delta_i + \gamma_i + \epsilon_i \quad (1)$$

where I_i takes the value of 1 if Inviter i sends an invitation to any Guest, and 0 otherwise; T_i indicates treatment groups (Random Guest List with Voucher, High-SES Guest List without Voucher, High-SES Guest List with Voucher, Low-SES Guest List without Voucher, and Low-SES Guest List with Voucher); X_i is a vector of unbalanced baseline controls and lasso-selected controls, δ_i are wealth-quintile fixed effects, and γ_i are geographic cluster fixed effects (my two randomization strata). I use heteroskedasticity-robust standard errors. For regressions concerning invitation-sending, Inviters with the Random Guest List without the Voucher are the omitted group.

Next, I conduct several analyses to understand the composition of the Guests that Inviters select. The following specification is restricted to the sample of Inviters with the High-SES Guest List or the Random Guest List:

$$I_i^H = \beta_0^H + \beta_1^H T_i^{R,V} + \beta_2^H T_i^{H,NV} + \beta_3^H T_i^{H,V} + X_i + \delta_i + \gamma_i + \epsilon_i \quad (2)$$

where I_i^H takes the value of 1 if Inviter i sends an invitation to a High-SES Guest, and 0 otherwise. The next specification is restricted to Inviters with the Low-SES Guest List

or the Random Guest List:

$$I_i^L = \beta_0^L + \beta_1^L T_i^{R,V} + \beta_2^L T_i^{L,NV} + \beta_3^L T_i^{L,V} + X_i + \delta_i + \gamma_i + \epsilon_i \tag{3}$$

where I_i^L takes the value of 1 if Inviter i sends an invitation to a Low-SES Guest, and 0 otherwise.[21] Finally, using only Inviters with the Random Guest List, I estimate:

$$I_i^{H\&L} = \beta_0^{H\&L} + \beta_1^{H\&L} T_i^{R,V} + X_i + \delta_i + \gamma_i + \epsilon_i \tag{4}$$

where $I_i^{H\&L}$ takes value 1 for Inviters who send an invitation to at least one High-SES Guest and at least one Low-SES Guest.

3.4.2 Second Stage: Intent-to-Treat Results

To estimate the intent-to-treat treatment effects of the intervention on second-stage outcomes, I conduct the following analysis:

$$Y_i = \beta_0 + \beta_1 T_i^{R,NV} + \beta_2 T_i^{R,V} + \beta_3 T_i^{H,NV} + \beta_4 T_i^{H,V} + \beta_5 T^{L,NV} + \beta_6 T^{L,V} + Y_{BL,i} + X_i + \bar{S}_i + \delta_i + \gamma_i + \epsilon_i \tag{5}$$

where Y_i is an outcome variable, T_i denotes the treatment group, $Y_{BL,i}$ is the baseline measure of the dependent variable (following Cilliers et al. (2024)),[22] X_i is a matrix of lasso-selected controls and unbalanced baseline variables, δ_i are wealth-quantile fixed effects, and γ_i are geographic cluster fixed effects. When testing second-stage outcomes, the omitted group is always the Control group.

There is the potential for SUTVA concerns with this experiment. When an Inviter shares a meal with a Guest, the Inviter or Guest might reduce meal-sharing with their other neighbors, including Control participants, which bias treatment effects to be larger than against an unaffected control. Conversely, when an Inviter shares a meal with a Guest, Control participants who are friends with the Guest or the Inviter might be included in these meals and new friendships, attenuating treatment effects.

In Equation 5, $\bar{S}_i$ is a vector of four control variables that account for exposure to spillovers: the total number of surrounding Inviters and Guests; the total number of Inviters with the Low-SES Guest List plus Low-SES Guests; the total number of Inviters with the High-SES Guest List; and the total number of High-SES Guests. The first measure establishes the effect of the treatment as a whole. The second measure establishes the

[21] Mechanically, for Inviters with the Low-SES Guest List, $I_i^H = 0$, and $I_i^L = I_i$. Mechanically, for Inviters with the High-SES Guest List, $I_i^L = 0$, and $I_i^H = I_i$.

[22] In cases where the baseline measure is missing, I use the most similar baseline measure. Of my main outcomes, I have baseline measures of food security and mild-to-severe depressive symptom, but I do not have baseline farm yields. I have a baseline measure of the number of months that a maize harvest will usually last before depleted, and use this measure as my baseline proxy for farm yields instead.

effect of being exogenously surrounded by within-SES relationships. The third and fourth measures establish the effect of be exogenously surrounded by cross-SES relationships, separately by proximity to the low-SES and high-SES sides of the link. I include each of these measures since the direction of potential bias on the treatment effects depends on the nature of the relationship. For each spillover measure, I estimate $\bar{S}_i$ by counting the number of participants of interest within fifty meters, and then recentering the variables to account for endogenous spatial variation following Borusyak and Hull (2023).

4 Results: First Stage

4.1 Invitation Sending Across Treatment Sub-Groups

Table 1: Treatment Effects on Invitation-Sending to Low-SES and High-SES Guests

	(1) Invites Any Guest		(2) Invites Low-SES Guest		(3) Invites High-SES Guest		(4) Invites Both	
Random with Voucher	-0.017	(0.039)	0.006	(0.044)	0.079*	(0.043)	0.106**	(0.042)
High-SES without Voucher	0.035	(0.039)			0.319***	(0.042)		
High-SES with Voucher	-0.026	(0.039)			0.257***	(0.042)		
Low-SES without Voucher	-0.045	(0.039)	0.261***	(0.044)				
Low-SES with Voucher	-0.048	(0.039)	0.261***	(0.045)				
Observations	1200		800		800		400	
Random without Voucher Mean	0.830		0.530		0.535		0.235	
P-values:								
High-SES: Voucher = No Voucher	0.119				0.151			
Low-SES: Voucher = No Voucher	0.933		0.999					
Test: *HSES without V = HSES with V = LSES without V = LSES with V*					[p= 0.358]			
DID Estimates [P-values]								
High-SES without Voucher – Low-SES without Voucher					0.058 [p= 0.349]			
High-SES with Voucher – Low-SES with Voucher					-0.003 [p= 0.957]			
Random with Voucher:								
(1) Invites High-SES – Invites Low-SES					0.073 [p= 0.274]			
(2) Invites Both – Invites Either Type			0.100 [p= 0.012]		0.027 [p= 0.521]			

Standard errors in parentheses
All specifications include baseline unbalanced control variables (number of garden plots and listed husband in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Any Guest (column 1) indicates if the Inviter sent an invitation to at least one Guest. Low-SES Guest (column 2) indicates if the Inviter sent an invitation to at least one low-SES Guest (Inviters with the High-SES Guest List did not see any low-SES names, so this outcomes is mechanically zero for this group). High-SES Guest (column 3) indicates if the Inviter sent an invitation to at least one high-SES Guest (Inviters with the Low-SES Guest List did not see any high-SES names, so this outcome is mechanically zero for this group). Invites Both (column 4) indicates if the Inviter sent at least one invitation to a High-SES Guest, and at least one invitation to a Low-SES Guest.
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

First, I analyze Inviter's invitation-sending decisions in the experiment. Control participants did not have the opportunity to send any invitations, so I use Inviters with the Random Guest List without the Voucher as a comparison group for all other subtreatments. Eighty-one percent of Inviters invite at least one Guest, and the rate is not different across treatment arms (column (1) of Table 1). Inviters invite 1.4 Guests on

average (1.8 Guests among Inviters who invite anyone) (Appendix Table C.1). The vast majority of these invitations are to women who are not in participants' baseline networks, indicating that effort costs of social linking are prohibitively high, yet easy to overcome with a simple intervention.

In all of the Inviter treatment arms, there are no differences in the probability of sending an invitation with or without the voucher, suggesting that serving a low-quality meal to either a high-SES or low-SES woman is still better than nothing. However, evaluating Inviters with the Random Guest List elucidates that the Voucher does have a treatment effect on the *composition* of invitations.

Inviters with the Random Guest List invite Low-SES Guests and High-SES Guests at the same rate without the voucher (53% and 53.5% of the time, respectively), suggesting that there is heterogeneity in who Inviters prefer to interact with. Furthermore, inviting *both* is the most popular option. Among Inviters with the Random Guest List without the Voucher, 23.5% of respondents invite both, more than twice as common as selecting two-or-more low-SES Guests or high-SES Guests (selected 9.5% and 10.5% of the time, respectively, shown in Appendix Table C.3). This already is highly suggestive that most women value interacting with both high-SES and low-SES women.

The Voucher induces an 7.9 percentage point increase in inviting High-SES Guests (Table 1). This comes without a reduction in the probability of inviting Low-SES Guests; I interpret this to imply that women who would have only invited Low-SES Guests without the voucher are induced by the voucher to *also* invite a High-SES Guest.[23] This pattern reveals that high prices constrain cross-SES relationships because of substitution with within-SES relationships. By revealed preference, cross-SES relationships are *relatively* more valuable at higher levels of financial investment, but not at the expense of within-SES relationships.

4.2 Meal-Sharing Across Treatment Sub-Groups

While 81% of Inviters invite at least one Guest, 45% of Inviters share a meal with a Guest within one month, and 62% share a meal within one year. Meal-sharing rates are statistically indistinguishable across Inviter groups with High-SES and Low-SES Guests (Appendix Table C.4), suggesting that the effort costs of meal-sharing itself, like the effort costs of sending an invitation, are not systematically higher for cross-SES matches.

The rate of meal-sharing is higher for Inviters with the Random Guest List with the voucher. This is largely driven by increases in meals with high-SES Guests, but I cannot

[23] Women who otherwise would have invited one low-SES Guest are more likely to invite one Low-SES Guest and one High-SES Guest; and women who otherwise would have invited multiple Low-SES Guests are more likely to substitute one of those invitations for a High-SES Guest (Appendix Table C.3).

Table 2: Text Response Analysis: Keyword Counts

Theme	Keywords	N: Inviters	N: Guests
Positive Emotions	trust, happy, love, warm, welcome, respect	253 (22%)	187 (19%)
Information Diffusion	info, learn, idea, teach, taught	299 (26%)	207 (21%)
Mutual Support	encourage, support, lend, borrow, help	279 (24%)	235 (24%)
Share	share, sharing	501 (44%)	450 (47%)

Notes: In the endline survey, enumerators ask Inviters about their experience with each Guest who they sent an invitation to; and ask Guests about their experience with each Inviter who they received an invitation from. Enumerators also ask all respondents about their overall opinions of the program. I conduct a simple keyword search from these text responses. Column (3) ("N: Inviters") is the number of Inviters who ever use one of the keywords in column (2) ("Keywords") across any of these text response questions. Column (4) ("N: Guests") is the number of Guests who ever use one of the keywords in column (2) ("Keywords") across any of these text response questions.

rule out that this increase is no different from a small increase in meals with low-SES Guests. This increase occurs only when agents can choose across types and afford higher-quality meals, suggesting a complementarity between selection and investment.

4.3 Long-Term Relationships

While the extent of the intervention is to directly encourage women to send invitations for a shared meal, it is crucial to understand the implications of these invitations beyond the first shared meal to interpret the effects of the intervention. At endline, the majority of Inviters report that they saw at least one Guest in the past month, and these are often intentional interactions rather than chance encounters. Twenty percent of Inviters and 16% of Guests report that they have a strong or deep relationship with at least one Guest or Inviter.[24] In open-ended responses describing experiences with the intervention, 65% percent of Inviters and 65% of Guests mention a keyword describing positive emotions, learning opportunities, and support (Table 2).[25] The intervention creates positive interpersonal experiences, where women share information and express happiness to spend time with each other. Furthermore, a non-trivial portion of these experiences lead to persistent and mutually meaningful friendships.

Are these relationships marginal, or do they crowd out relationships that would have formed without the intervention? While the number of people that participants report doing activities with is not affected by the Inviter treatment, *who* women report doing these activities with does change. Among all Inviters there is a significant increase in "network churn", defined as the sum of baseline ties that are absent from the follow-up network and ties in the follow-up network that were not present in the baseline network

[24] At endline, participants describe their experiences with each invitation that they sent or received in open-ended responses. In Appendix Section A.5, I list the key phrases I use to determine what a close relationship is, such as "we talk often" and "we are still friends".

[25] If I omit the keywords "share" and "sharing", to remove mechanical discussion of meal-sharing, 52% percent of Inviters and 47% of Guests mention any of the other positive keywords.

(Appendix Table D.13).[26]

This network churn is driven by Inviters dropping weak ties present in the baseline network and replacing them with other weak ties (Appendix Table D.14 and D.15). There is a small but precise increase in new strong ties, without any change in the probability of dropping baseline strong ties. I find suggestive evidence that networks change so as to increase the presence and strength of women's own friends in the network (Appendix Table D.16). While there are small and insignificant reductions in the presence and strength of relatives in the network overall, there are significant reductions in engaging with the husband's side of the family across specific activities (Appendix Tables D.17). Recall that women's networks are dominated by their husband's relatives, but the intervention specifically creates a new non-familial relationship. Women shift away from interacting with their husband's relatives, and towards their own friends, in lending, secret-sharing, and advice-seeking (Appendix Tables D.18, D.19 and D.20).[27]

5 Results: Second Stage Outcomes

I evaluate the effect of the intervention on three main outcomes: food security, farm yields, and mild-to-severe depressive symptoms. These outcomes capture both the instrumental return that relationships can provide and the intrinsic benefits of companionship.

Pooling all Inviters, the Inviter treatment leads to a 0.140 standard deviation increase in food security ($p < 0.05$); an 8.3 percentage point reduction in depressive symptoms (23%, $p < 0.01$); and no distinguishable impact on farm yields (Table 3). The farm yields measure is extremely noisy, with the 95% confidence interval spanning a 15% reduction to an 18% increase in the value of farm yields, but I highlight the outcome because farm yields are the most important income-earning opportunity in the year. Other income-earning opportunities primarily serve as a means to liquidity outside of the harvest season and protection from shocks.[28]

The effects on food security are larger among Inviters with the High-SES Guest List than Inviters with the Low-SES Guest List (DID: 0.153 SD, $p = 0.025$), while the treatment effects on depressive symptom reductions are larger among Inviters with the Low-SES

[26] There is also network churn specifically among meal-sharing ties (Appendix Tables D.21).

[27] This set of results helps clarify why my main treatment effects are inconsistent with the "rotten kin" literature, which shows that social pressure to share income can distort investment and labor supply decisions (Carranza et al., 2022; Dillon et al., 2021; Jakiela and Ozier, 2016; Swanson, 2025). Contrarily, I find that relationships built specifically by sharing resources can improve investment, savings, and business decisions. The shift in activities (including lending) away from extended family networks and towards own friends suggests that the intervention may allow participants to rely *less* on "rotten kin", and that relationships built on resource-sharing need not imply extractive obligations.

[28] Eighty percent of participants report cultivating agriculture and 25% report earning income through self-employment. Among Control participants who earn through agriculture and self-employment, the value of their harvest yields are more than four times the self-employment profits in the past year.

Table 3: Treatment Effects on Main Outcomes

	(1)	(2)	(3)	(4)	(5)
	Food Security Index		Farm Yields USD Value	Mild-to-Severe Depressive Symptoms	
	1 Month	1 Year	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects					
All Inviters	0.012	0.140**	5.496	0.009	-0.083***
	(0.056)	(0.056)	(26.984)	(0.026)	(0.027)
Panel B: Pooled Inviter by Guest List Treatment Effects					
Inviter with High-SES Guest List	0.057	0.226***	31.281	-0.000	-0.045
	(0.069)	(0.071)	(35.432)	(0.031)	(0.033)
Inviter with Low-SES Guest List	-0.086	0.073	-13.383	0.018	-0.104***
	(0.067)	(0.065)	(32.099)	(0.032)	(0.032)
Inviter with Random Guest List	0.055	0.105	-4.031	0.019	-0.092***
	(0.069)	(0.068)	(31.126)	(0.032)	(0.032)
Panel C: Inviter Sub-Group Treatment Effects					
High-SES without Voucher	0.047	0.223**	-8.503	-0.026	-0.047
	(0.083)	(0.089)	(40.877)	(0.038)	(0.039)
High-SES with Voucher	0.071	0.219**	29.133	0.027	-0.049
	(0.087)	(0.089)	(41.701)	(0.040)	(0.041)
Low-SES without Voucher	-0.089	0.012	-42.277	0.034	-0.111***
	(0.079)	(0.074)	(36.942)	(0.039)	(0.038)
Low-SES with Voucher	-0.086	0.145*	7.755	-0.009	-0.115***
	(0.085)	(0.086)	(43.092)	(0.040)	(0.040)
Random without Voucher	0.015	0.106	13.750	-0.022	-0.090**
	(0.082)	(0.082)	(38.323)	(0.040)	(0.041)
Random with Voucher	0.091	0.131	-32.925	0.047	-0.111***
	(0.089)	(0.088)	(37.517)	(0.039)	(0.039)
Observations	1522	1528	1528	1522	1528
Control Mean	-0.000	-0.000	324.783	0.297	0.361
P-values:					
Pooled: High-SES vs. Low-SES Guest List	0.034**	0.025**	0.191	0.569	0.060*
Pooled: Voucher vs. No Voucher	0.652	0.375	0.528	0.260	0.784
High-SES: Voucher vs. No Voucher	0.803	0.968	0.436	0.232	0.969
Low-SES: Voucher vs. No Voucher	0.973	0.136	0.266	0.333	0.938
Random: Voucher vs. No Voucher	0.453	0.810	0.273	0.132	0.630

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The Food Security Index (columns 1 and 2) is a PCA index in standard deviations from the Control mean. The Food Security Index at 1 Month is a PCA index of: the typical number of meals consumed per day in the past month, and the number of meals with fish or eggs consumed in the past month. The Food Security Index at 1 Year is a PCA index of: the typical number of meals consumed per day in the past month, and the number of meals with meat, fish or eggs consumed in the past month, the typical number of meals consumed per day during the lean season, and the number of meals with meat, fish or eggs consumed during the lean season. To calculate the Farm Yields in USD value (column 3), households estimate their harvest yields for each crop. I multiply market rates in agricultural reports by yields for the most high-yield crops: maize, soy beans, groundnuts, beans, tobacco, sugar cane, sunflower seeds, sweet potatoes, and pumpkin. Depression is calculated using the CESDR-10, a 10-item revised version of the Center of Epidemiologic Studies Depression Scale. Depression (columns 4 and 5) indicates a score greater than or equal to 10.

Guest List than Inviters with the High-SES Guest List (DID: -0.059 percentage points, $p = 0.060$). Treatment effects on almost every question in the CESDR-10 questionnaire move in the direction of "less depression" by the one year mark, and the changes are almost always largest among Inviters with the Low-SES Guest List (Appendix Table D.2). Treatment effect estimates among Inviters with the Random Guest List lie between the other two groups on both outcomes.

Many agricultural economies experience seasonal variation in food supply and prices, creating a phenomenon where agents consume less food during the high-price "lean season". The endline increase in food security is driven by increases in animal protein consumption in any time period, and increases in the typical number of daily meals consumed during the lean season (Appendix Table D.1). The endline survey is conducted towards the end of the harvest season, so the fact that there is no detectable change in farm yields is consistent with the fact that there is likewise no change in the typical number of daily meals consumed in the past month.[29]

While the depressive symptom reductions may be related to food security increases, food security cannot be the primary mechanism through which the Inviter treatment improves mental health. The group that experiences the largest depressive symptoms reductions experiences the smallest food security gains, and vice versa. Benchmarking the magnitudes of the effects on depressive symptoms against cash and asset transfer programs, which intervene on income rather than on social connection, my impacts are almost twice as large (Ridley et al., 2020).[30]

These large effects are not necessarily surprising, and rather speak to the immense value of social relationships for psychological well-being specifically. My estimates are comparable to or lower than the correlational coefficient between measures of social support and the probability of depression that descriptive studies find across varied contexts (Choi et al., 2023; Hoang et al., 2024; Thompson et al., 2021), and my treatment effects are driven by a reduction in mild depressive symptoms (Appendix Table D.3). Experiencing mild depressive symptoms is significantly more common than moderate or severe depressive symptoms; while still clinically relevant, it may not require professional intervention and is plausibly treatable with social support. Furthermore, clinical psychology specifically recommends peer support as a treatment plan even for severe mental health conditions

[29]There is a development economics literature evaluating ways to smooth seasonal variation in food security, which has largely focused on heavy-handed financial interventions (Dillon, 2021; Stephens and Barrett, 2011; Fink et al., 2020; Basu and Wong, 2015; Burke et al., 2019). Consistent with Augenblick et al. (2024), which finds that a simple psychological budgeting exercise can increase lean-season liquidity, I find evidence that little more than a nudge (peer-to-peer information-sharing and relationship-building) can generate new lean-season income streams, and that high-cost programs may not be necessary.

[30]Ridley et al. (2020), which averages the effects of twenty-eight cash transfer and multifaceted antipoverty programs, find a 0.094 SD improvement in mental health, whereas I find an improvement from 0.17 SD (pooled estimate) to 0.22 SD (Inviters with the Low-SES Guest List).

because of its immense psychological value (Chinman et al., 2014; Cooper et al., 2024; Gunderson and Links, 2014; Yim et al., 2023).

Taken together, these results demonstrate that high-SES and low-SES links offer different returns that are likely non-substitutable, and offer a compelling explanation for why Inviters with the Random Guest List are most likely to prefer to interact with both low-SES and high-SES women when they face the least constraints. Inviters with the Random Guest List experience intermediate treatment effects in both food security gains and depressive symptom reductions, which may be because they balance both rather than maximize one at the expense of the other.[31]

5.1 Treatment Effect Channels

In this section, I evaluate potential explanations for how these social relationships lead to such large economic and psychological improvements, and why different types of relationships are more impactful across different outcomes. While I present results pooled and for all sub-groups, I rarely analyze or interpret differential Voucher effects, since the first-stage is largely equivalent across the Voucher sub-treatments. I correct for multiple hypothesis testing across thirty hypotheses in Appendix Table E.3, and every statistically significant result from the main analysis remains significant at the 90% confidence level using False Discovery Rate-sharpened q-values (Benjamini et al., 2006).

5.1.1 Instrumental Utility

What allows women who create new high-SES relationships to consume more meals per day during the lean season, and more protein throughout the year? I explore several potential channels, and identify two persuasive explanations: improved productive time use and reduced lending increase liquidity, while risk mitigation increases the option value of consumption during the lean season. I find increases in borrowing or intra-household bargaining power (Appendix Table D.11 and Table D.12). In Appendix Section D.1.3, I present evidence for channels for which I find suggestive evidence, but do not have the data to make conclusive statements.

Improved Productive Time Use

Aside from agriculture, households earn income through piecework and self-employment. Piecework, which 63% of Control households participate in at some point during the study period, constitutes supplying labor on a task-by-task basis to farmers with larger

[31] While this may be in part due to the fact that Inviters with the Random Guest List represent a combination of the other two treatment arms, recall that the most common choice is to invite both high-SES and low-SES Guests, suggesting a non-trivial component of these treatment effects is explained by women who interact with both types.

land-holdings. Self-employment, which only 27% of Control households earn income from in the study period, constitutes selling goods and services around the village or in a market.[32] Self-employment provides an alternate stream of liquidity, has the potential for much higher earnings than piecework, and requires less physical exertion.

There is no increase in piecework employment in any treatment group (Appendix Table D.5), but there is evidence of a reduction in piecework search frictions. Control respondents report searching for piecework without finding it on average three days per month. Inviters spend fewer days searching unsuccessfully for piecework, which may free up time or energy for other productive activities (Table 4).

One such productive activity is self-employment. At endline, Inviters with the High-SES Guest List, who experience the largest improvements in food security, are 5.5 percentage points (20%, $p < 0.10$) more likely to earn income through self-employment in the past year than the Control group. Inviters with the High-SES Guest List enter self-employment immediately after the experiment begins. In the midline survey, only one month after enumerators send meal-sharing invitations to High-SES Guests, these Inviters are 9.2 percentage points (51%, $p < 0.01$) more likely to earn income through self-employment than Control participants. There are modest increases among Inviters with the Random Guest List, commensurate with smaller but still positive food security effects. These effects are largely driven by women entering self-employment for the first time, and many of them directly copy the self-employment activities of a high-SES Guest (Table D.4).[33]

I find evidence that High-SES women share valuable information about entering self-employment in these meals. This explanation is consistent with the "strength of weak ties" theory (Granovetter, 1973), which argues that weak non-homophilic ties are precisely those that bring in novel information (recall that the experiment leads to more churn among weak links in the network).[34] Most of the self-employment activities that participants engage in do not require formalization nor a physical business space (for example, selling baked goods to other villagers). Consequently, the fixed capital costs of entering self-employment are low, and information may be enough to help women overcome barriers to entry. For participants whose household enters into any new self-employment activity in the past month by midline, I ask them why they entered into this activity. New information emerges as a key reason: The most common reason respondents give for earning from a new self-employment activity is that "somebody told me

[32] The most common self-employment activities involve selling simple consumables (most frequently, "selling prepared foods"), and 70% of self-employed women sell only to customers around the village.

[33] I define copying self-employment activities as an Inviter who sells a good or service for the first time that a Guest on the Inviter's list reports selling at *baseline*.

[34] This explanation is also consistent with a literature on peer effects in entrepreneurship, which finds an important role for social networks in the diffusion of information and business practices (Asiedu et al., 2025; Cai and Szeidl, 2017; Fafchamps and Quinn, 2018; Hardy and McCasland, 2021; Vega-Redondo et al., 2019).

Table 4: Treatment Effects on Productive Time Use

	(1)	(2)	(3)	(4)	(5)
	Idle Search: Growing Season	Idle Search: Harvest Season	Any Income from Self-Employment This Year	Any Income from Self-Employment This Month	
	1 Year	1 Year	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects					
All Inviters	-0.193	-0.483*	0.022	0.053**	0.017
	(0.274)	(0.266)	(0.026)	(0.022)	(0.025)
Panel B: Pooled Inviter by Guest List Treatment Effects					
Inviter with High-SES Guest List	-0.279	-0.141	0.055*	0.092***	0.049
	(0.334)	(0.335)	(0.032)	(0.028)	(0.031)
Inviter with Low-SES Guest List	-0.208	-0.759**	-0.005	0.013	-0.007
	(0.331)	(0.304)	(0.031)	(0.027)	(0.030)
Inviter with Random Guest List	-0.032	-0.553*	0.013	0.039	0.006
	(0.335)	(0.310)	(0.032)	(0.028)	(0.030)
Panel C: Inviter Sub-Group Treatment Effects					
High-SES without Voucher	0.175	-0.189	0.021	0.091**	0.030
	(0.423)	(0.389)	(0.038)	(0.036)	(0.037)
High-SES with Voucher	-0.685*	-0.164	0.096**	0.094***	0.071*
	(0.399)	(0.437)	(0.041)	(0.036)	(0.040)
Low-SES without Voucher	-0.250	-0.622*	-0.014	-0.001	-0.029
	(0.402)	(0.373)	(0.037)	(0.032)	(0.036)
Low-SES with Voucher	-0.055	-0.834**	0.012	0.022	0.014
	(0.425)	(0.367)	(0.038)	(0.034)	(0.037)
Random without Voucher	-0.156	-0.497	-0.007	0.042	-0.012
	(0.405)	(0.366)	(0.038)	(0.035)	(0.036)
Random with Voucher	0.092	-0.578	0.030	0.028	0.012
	(0.428)	(0.393)	(0.040)	(0.033)	(0.037)
Observations	1515	1506	1528	1522	1528
Control Mean	3.072	2.872	0.277	0.180	0.237
P-values:					
Pooled: High-SES vs. Low-SES Guest List	0.831	0.048**	0.056*	0.005***	0.062*
Pooled: Voucher vs. No Voucher	0.481	0.650	0.078*	0.952	0.210
High-SES: Voucher vs. No Voucher	0.064*	0.959	0.105	0.943	0.356
Low-SES: Voucher vs. No Voucher	0.684	0.596	0.534	0.530	0.293
Random: Voucher vs. No Voucher	0.609	0.845	0.417	0.723	0.576

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Idle Search Days (columns 2 and 3) is the average number of days the participant reports typically searching for piecework in a month, but not finding it. Column (2) is the average number of idle search days per month during the growing season (the agricultural season largely overlapping with the lean season), and Column (3) is the average number of idle search days per month during the harvest season. Any Income from Self-Employment (columns (4)-(6)) takes the value of 1 if the participant engaged in self-employment, i.e. the sale of goods or services, to earn income. Column (4) indicates earning income from self-employment any time in the past year, whereas columns (5) and (6) indicate earning income from self-employment in just the past month, measured at different times.

this is profitable" (45%), and five percent of women say that they received training.[35]

I also investigate the possibility that high-SES women provide low-SES women with credit or market linkages, but find no evidence consistent with either channel.[36]

Reduced Lending

Inviters across all sub-treatments report lending money or goods to fewer people in the past year than Control, particularly husband's relatives, which may be in part responsible for having the liquidity for more food consumption (Appendix Table D.23). See Appendix Section D.1.1 for more details.

Reduced Risk of Lean Season Consumption

Inviters Diversify Their Crop Portfolios

Inviters undertake a striking transformation in their agricultural production function by diversifying their crop portfolios, reducing the risk they face during the lean season associated with the upcoming harvest. All Inviters are more likely to grow fruits and vegetables, while Inviters with the High-SES Guest List are more likely to grow cash crops (Appendix Table D.6). See Appendix Section D.1.2 for more details.

Inviters Report Financially Reliable Network Relationships

If Inviters have networks ties that they are confident they can rely on in case of a negative shock, they may be more comfortable consuming rather than saving. Inviters are 28% more likely to join "Welfare Groups" ($p < 0.10$, Table D.9), home-grown self-help groups that are not operated by any formal institution. There are two narrow groups of Inviters who say they would be very comfortable borrowing and lending with more people in their networks: Inviters with the High-SES Guest List and the Voucher, and Inviters with the Random Guest List and the Voucher. Though not definitive, when households have exogenous access to high-SES links, they might use the Voucher to mitigate their vulnerability to shocks by establishing relationships with financially reliable ties. See Appendix Section D.1.2 for more details.

[35]There is also evidence that seeking out information was intentional. When asked to explain why they chose to invite the Guests they selected, Inviters with the High-SES Guest List are more likely than either of the other groups to say that it was because their guests are knowledgeable (Appendix Table C.7). Furthermore, at midline, Inviters with the High-SES Guest List are more likely to say that they discussed ways to find business resources in their shared meals (Appendix Table C.8), while Inviters with the Low-SES Guest List are more likely to discuss ways to find piecework.

[36]Twenty-one percent of respondents report that they enter into a new self-employment activity because somebody helped them to find capital, suggesting that fixed costs are an important constraint. However, Inviter borrowing rates are not different from Control, so information is the more binding constraint that high-SES social ties effectively resolve. Similarly, only 8% of participants who ever engage in self-employment report having any stable business relationships (for example, regular suppliers or customers), with no differences between Inviters and Control, so Guest referrals to business-relevant contacts cannot fully account for the large and persistent increase in self-employment activity.

5.1.2 Intrinsic Utility

Peer Relationships Serve as a Mental Health Resource Themselves

Literature across the social sciences hypothesizes that interactions with similar-type people are more effective at providing emotional and psychological support (Jacobson, 1987; Lomnitz, 1988; Steffens et al., 2019; Umberson and Montez, 2010), and clinical psychology specifically recommends *peer* support for treating mental health conditions. Evaluating the treatment effects on depression heterogeneously by baseline income *within* low-SES women, I find further evidence that "peer" status—or, in this case, people who are proximate to one another in social-class—is a key feature of these relationships that leads to mental health improvements.

I evaluate effects heterogeneously by the SES index, considering women in the first two quintiles as "poorest of the poor" and women in the third and fourth quintiles as "richest of the poor" (recall that women in the fifth quintile are "high-SES"). Among Inviters with the Low-SES Guest List, the poorest and richest of the poor both experience meaningful reductions in depression (Table 5). Among Inviters with the High-SES Guest List, who only have the choice to interact with women in the top quintile of the SES index, the richest of the poor experience depression reductions that are equal in magnitude to the reductions among Inviters with the Low-SES Guest List, while the poorest of the poor experience *no* depression reduction. The benefit of relationships for psychological well-being is *symmetrically* increasing in social-class proximity, implying that low-SES women do not offer some specific resource that is useful for mental health. Rather, the evidence suggests that socially proximate relationships are in themselves a mental health resource.

Consistent with this hypothesis, I find modest evidence that socially proximate relationships improve participants' subjective experience of social inclusion. Using my primary pre-specified measure of loneliness (the UCLA-3 score), magnitudes follow the same directions across groups as the CESDR-10 score but with much less variation that is not statistically distinguishable across groups (Appendix Table D.25). However, Inviters with the Low-SES Guest List are significantly less likely to say they experienced loneliness ever or often in the past two weeks when I pose the question directly.[37]

Why do relationships with people who are proximate in the distribution of socio-economic status serve as a mental health resource? Inviters with the Low-SES Guest List are far more likely than Control or Inviters with the High-SES Guest List to share secrets with their own friends rather than with relatives (Appendix D.20). While I do not collect the

[37]Measures of subjective experience of social inclusion are likely context-specific, and are either minimally validated or not validated at all in settings such as rural Malawi. Following Banerjee et al. (2023), which uses a plain-language question to measure loneliness to address context and validation issues, I convert a question from the CESDR-10 scale – "How often in the past two weeks did you experience loneliness?" – into two binary measures of loneliness depending on the frequency of loneliness experience.

Table 5: Treatment Effects on Mild-to-Severe Depressive Symptoms: by Baseline SES Index

	(1)	(2)	(3)	(4)
	Mild-to-Severe Depressive Symptoms: Poorest of the Poor		Mild-to-Severe Depressive Symptoms: Richest of the Poor	
	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects				
All Inviters	-0.021	-0.089**	0.032	-0.055
	(0.038)	(0.038)	(0.041)	(0.043)
Panel B: Pooled Inviter by Guest List Treatment Effects				
Inviter with High-SES Guest List	-0.026	-0.014	-0.008	-0.035
	(0.047)	(0.047)	(0.048)	(0.051)
Inviter with Low-SES Guest List	-0.019	-0.127***	0.070	-0.059
	(0.046)	(0.046)	(0.050)	(0.051)
Inviter with Random Guest List	-0.020	-0.135***	0.028	-0.043
	(0.048)	(0.047)	(0.050)	(0.051)
Panel C: Inviter Sub-Group Treatment Effects				
High-SES without Voucher	-0.084	-0.004	-0.002	-0.026
	(0.055)	(0.057)	(0.060)	(0.061)
High-SES with Voucher	0.029	-0.022	-0.015	-0.025
	(0.059)	(0.056)	(0.058)	(0.063)
Low-SES without Voucher	-0.018	-0.139**	0.105	-0.054
	(0.055)	(0.056)	(0.064)	(0.062)
Low-SES with Voucher	-0.031	-0.127**	0.048	-0.088
	(0.055)	(0.055)	(0.060)	(0.060)
Random without Voucher	-0.013	-0.162***	-0.027	-0.036
	(0.062)	(0.061)	(0.060)	(0.063)
Random with Voucher	-0.011	-0.154***	0.070	-0.049
	(0.060)	(0.057)	(0.062)	(0.061)
Observations	764	768	755	757
Control Mean	0.314	0.359	0.279	0.364
P-values:				
Pooled: High-SES vs. Low-SES Guest List	0.873	0.015**	0.099*	0.626
Pooled: Voucher vs. No Voucher	0.391	0.682	0.694	0.770
High-SES: Voucher vs. No Voucher	0.088*	0.786	0.839	0.990
Low-SES: Voucher vs. No Voucher	0.835	0.842	0.422	0.609
Random: Voucher vs. No Voucher	0.971	0.903	0.158	0.852

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Depression is measured using the CESDR-10, a 10-item revised version of the Center of Epidemiologic Studies Depression Scale. For each question, participants self-report the frequency with with they experienced the feelings that the question describes in the past two weeks. Each question is answered on a scale of 0-3: (0) Never, (1) Rarely, (2) Sometimes, (3) Always. The CESDR-10 score is the aggregate of these scores across all ten questions. Mild-to-severe depression is a score >=10. The regressions in columns (1)-(2) limit the sample to women in the first and second quintiles of the baseline SES index. The regressions in columns (3)-(4) limit the sample to women in the third and fourth quintiles of the baseline SES index. Low-SES women are in the first four quintiles of the SES index, and High-SES women are all in the top quintile of the SES index.

data to explain why sharing secrets with a friend might improve mental health relative to sharing secrets with extended family members, it is conceivable that women's own friends are able to provide more empathy for sensitive issues, and that talking to a non-relative can widen the range of sensitive topics they can discuss.

5.2 Discussion of Results

In this section, I present descriptive and exploratory evidence to address several questions that follow from the results.

5.2.1 What social frictions does the Inviter treatment resolve?

I argue that the Inviter treatment reduces the costs of initiating social relationships, and resolves information asymmetries about others' willingness to interact. In this section, I evaluate the specific costs and information asymmetries that women face in social interactions. In Appendix Section D.3, I discuss three alternate explanations, for which I find no supportive evidence: lack of mutual consent and reciprocation, low-SES women under-estimate the returns to relationships, and social norms mitigate negative externalities.

Evidence of High Face-to-Face Social Costs of Initiating Relationships

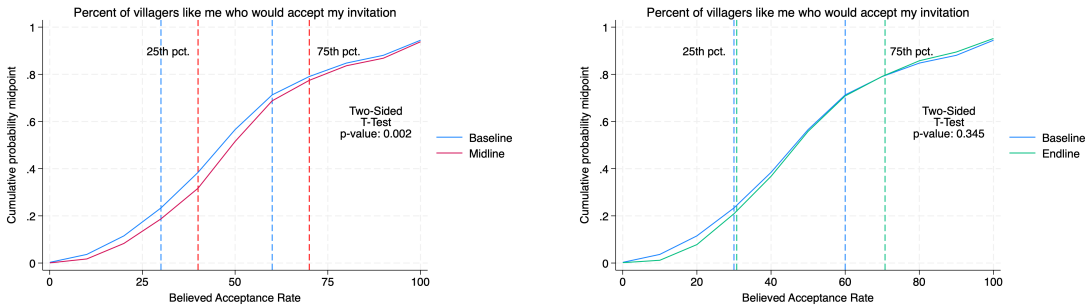
At baseline, I implement vignettes to understand social norms and barriers to meal-sharing that women face. Respondents describe how the fact that people share food both for companionship and to assist the needy leads to deep concerns about burdening others, stigmatizing others, or stigmatizing oneself. The primary concerns with eating food at a neighbor's home are monetary, such as burdening the neighbor financially or appearing needy, while concerns with inviting a neighbor to share one's own food are both monetary and social, such as worry that the neighbor does not like her.

Only 34% of Inviters, less than half of the women who send any invitation, accompany the enumerator to deliver the invitations to Guests. This is consistent with a revealed preference not to send invitations in person, and a high cost of face-to-face interaction.

Evidence of Information Asymmetries

I ask women at baseline: What percent of women similar to you in the village do you believe would accept an invitation for a shared meal from you? On average, participants believe that 50% of women similar to her in the village would accept her invitation (Figure A.6). One year after the experiment, participants report that 63% of invitations resulted in a shared meal, an invitation acceptance rate that is 26% higher than the mean, and higher than the interquartile range, of the baseline believed acceptance rate.

Figure 3: Belief Updating



Notes: In each survey, I ask women: "Out of 10 women in the village who are similar to you, how many would accept an invitation from you for a shared meal?" Both graphs plot the cumulative distribution function of responses to this question (multiplied by 10 to represent percentages for readability), across different survey rounds, and the inter-quartile range. The blue line represents the baseline distribution, the red line represents the midline distribution, and the green line represents the endline distribution. Using a two-sided t-test, the baseline and midline distributions are statistically different from each other ($p = 0.002$). Using a two-sided t-test, we can not rule out that the baseline and endline distributions are equal ($p = 0.345$).

This evidence indicates that there is a significant information asymmetry whereby women systematically and pluralistically underestimate others' willingness to share meals.

At midline, the time when participants observe high rates of meal-sharing in the community, they positively update their beliefs about others' willingness to interact (Figure 3, left-side panel). By endline, the distribution of beliefs is no longer distinguishable from the baseline distribution, suggesting that correcting beliefs about aggregate willingness to interact is not a sufficient intervention (Figure 3, right-side panel). Without knowing who specifically is willing to interact, or without a third party bearing the face-to-face costs of the initial interaction, meal-sharing activity slows and beliefs revert to their initial distribution.[38]

5.2.2 Does the price of meat contribute to homophily in durable networks?

In Section 4, I find evidence that the price of meat constrains low-SES women from initiating social interaction with high-SES women. To what extent can the price of one shared meal explain income-based homophily in durable social networks?

First, I evaluate the probability that Inviters obtain and save the phone number of the women on the Guest List. Patterns are consistent with first-stage results: Inviters with the High-SES Guest List obtain the phone numbers of their high-SES Guests at the same rate that Inviters with the Low-SES Guest List obtain the phone numbers of their low-SES Guests, with or without the Voucher (Appendix Table D.32). Inviters with the Random Guest List, on the other hand, are more likely to obtain the phone numbers of

[38] At midline, Inviters with the *most* pessimistic baseline beliefs share more meals in the past week than Control (Figure A.7). The gap closes among participants who thought at baseline that 45% of women in the village would accept her invitation—the women whose beliefs should *not* update at midline, since 45% of Inviters shared at least one meal with Guests within one month.

low-SES Guests than high-SES Guests unless they have the Voucher.[39]

By endline, the Inviter treatment does not affect the composition of the network by roof material (Table D.33). While much of the churn in the network is accounted for by network ties with thatched roofs, for Inviters with the Random Guest List and the Voucher, the treatment induces more economic diversity in network churn while, for every other group, the intervention induces network churn within the same income groups who enter and exit the network absent the treatment.[40] Notably, the importance of changes to the diversity of the network churn is minute relative to changes in the total volume of network churn that the treatment induces, implying that the first-and-foremost impact of the Inviter treatment on networks in the long run is to spur any amount of movement in and out of the network.

5.2.3 Does Guest SES or a correlated variable drive heterogeneous results?

SES is correlated with other characteristics, so it is possible that average differences in a characteristic *other* than resources or social status generate heterogeneous treatment effects. Aside from SES, age differs systematically between the invited Guests from the High-SES and Low-SES Guest Lists, while High-SES Guests' networks are descriptively larger (Appendix Table C.5). Inviter treatment effects do not consistently depend on Guest age or the number of friends that women on the Guest List have, suggesting that it is SES itself that drives the treatment effects (Appendix Tables D.34, D.35, and D.36).[41]

5.2.4 Are there repercussions to weakening extended family networks?

Inviters are less likely to interact with their husbands' extended family in secret-sharing, advice-seeking, and lending, and instead engage with their own friends. I do not find any evidence of negative repercussions. If anything, marriages appear to improve, especially among Inviters with the Low-SES Guest List, who are the most likely to discuss marriage and romance with their Guests (Table C.8).[42] Furthermore, it is the women with the

[39]Homophily in networks is often accounted for by a combination of having more opportunities to meet similar people (meeting bias) and, conditional on meeting someone, preferring to befriend similar people (friending bias). If we consider randomization to the Inviter treatment as an exogenous meeting opportunity, and the choice to send an invitation or save a phone number as a friending decision, then 'friending bias' only appears in one circumstance: when bias is permitted in choosing meeting opportunities, and the price of meat is high. These results suggest that descriptive measures of friending bias may conflate underlying preferences with price-constrained selection into meeting opportunities.

[40]See Appendix Section C.1 for detailed results on economic-diversity-weighted network churn.

[41]Among Inviters with the Random Guest List, a younger Guest List leads to greater depressive symptom reductions, suggesting that age-based homophily may be beneficial for mental health (Appendix Table D.37). However, the effects go in the opposite direction for Inviters with the Low-SES Guest List, implying that age cannot drive the heterogeneous results by Guest SES, and that age-based homophily may only be relevant for mental health in the absence of class-based homophily.

[42]Inviters with the Low-SES Guest List are 3.9 percentage points less likely than Control to divorce during the trial (43%, $p < 0.10$, not statistically different from Inviters with the High-SES Guest List)

least baseline household decision-making power, who are ostensibly the most vulnerable to family backlash, who experience the greatest depressive symptom reductions—a 59% decrease across all Inviter groups (Appendix Table D.26). These women also experience high rates of network churn among their husband's family members, suggesting that the intervention enables disempowered women to find new people within the family network to interact with (Appendix Table D.27). Inviters who are ever pregnant during the trial, another group who is vulnerable and likely depends on the extended family, also experience much more network churn in husband's relatives (Appendix Table D.30).

5.2.5 Are the results externally valid?

External Validity of Network Formation Frictions

I conduct a survey of sixty-eight people in Central Malawi that does not restrict the sample on the basis of gender or migrant status. Participants' beliefs about others' willingness to accept an invitation are even more pessimistic than in the experimental sample, and male and female beliefs are not different. The majority of both male and female respondents say they would prefer to share more meals per week than they currently do. While it is impossible to say if the experiment would be effective among men or non-migrants, the survey evidence suggests that pessimism and fear of rejection may lead to inaction in a broader sample.

External Validity of Downstream Treatment Effects

External Validity to Other Contexts

While the specific channels through which higher-income peers offer economic advantages to their lower-income friends likely differs across settings, my results offer external validity to a large literature based in high-income countries showing the economic value of information, access, or resources that higher-income and higher-achieving peers offer. My results on mental health symptoms are consistent with a wealth of descriptive evidence identifying a powerful correlational relationship between social relationships and psychological well-being across countries, age groups, gender, and income status. Collins et al. (2022) finds that "relational diversity" in social relationships—that is, more variety and evenness in the types of relationships that people engage in—is predictive of well-being across eight countries, which is consistent with my respondents' revealed preference for economic diversity in their networks.

External Validity to Stable Networks

(Appendix Table D.28). Divorce is a feasible route for women in this setting (9% of Control divorce during the study). There are no differences in the strength of relationships or comfort risk-sharing with husbands (Appendix Table D.29), suggesting that the treatment likely does not induce low-agency women to stay in harmful marriages. Ninety-six percent of participants say that there is somebody who they can talk to in case of a quarrel with their husband, and there is no treatment effect on this outcome.

Because my sample is selected on the basis of having recently moved, women's social networks are likely in disequilibrium. If networks adjust with time, the experiment speeds up the process of social integration. Alternatively, if disadvantage on the basis of income, migrant status, and gender mean these women are perpetually excluded, the treatment is a shock to all women in the sample, including those whose networks are in equilibrium.

I test for treatment effect heterogeneity by log-years in the village. While some treatment effects are noisily larger for women who are new to the village, there are no variables that are clearly and robustly heterogeneous by time in the village. This suggests that the treatment serves as a shock for everyone, rather than as a device that speeds up network formation that otherwise would happen with time.

6 Conceptual Model of Social Network Formation

What set of preferences and constraints that women face can explain the patterns that I observe in the experiment? I present a conceptual model of social network formation that serves as a non-formal organizing framework with which to interpret the experimental results. A key feature of the model is, first, to consider relationships with different types of people as different goods with different benefits, implying that the trade-off across types is a central tension in network formation. Second, I allow the benefits of relationships to increase in financial investment. When the returns to relationships with higher- and lower-SES people are differentially sensitive to financial investment, the model generates predictions for how the price of high-value social activities can act as a determinant of income-based homophily.

I consider two ways in which individuals obtain utility from social capital, paralleling the Grossman (1972) theoretical formulation of health capital. First, individuals get utility from social interaction itself—the non-market value of being in the company of others—which I refer to as "intrinsic utility". Secondly, there is a financial return to social interactions, which allows people to consume other utility-generating goods. I refer to the utility of the consumption that individuals can afford because of the financial return on social interaction as the "instrumental utility" of social relationships.

6.1 Model Set-up

Low-SES women decide if they want to initiate relationships with low-SES or high-SES women, and then choose whether to invest financially in these social engagements. First, they select across a set of relationship-initiation bundles (S_H, S_L), choosing to initiate relationships with high-SES women, low-SES women, none, or both, where they can

initiate a maximum of two relationships.[43] Initiating relationships with high-SES and low-SES women bear fixed effort costs c_H and c_L, respectively. Conditional on initiating at least one relationship, agents choose whether to invest financially to improve the social experience, $I \in \{0,1\}$, which costs p^I per relationship. Agents spend their remaining income on consumption Z.

Women select S_H, S_L, and I to maximize:

$$\max_{S_H \in \{0,1,2\},\, S_L \in \{0,1,2\},\, I \in \{0,1\}} U\big(Z, V(S_H, S_L, I)\big)$$

$$\begin{aligned} \text{s.t.} \quad & Y(S_H, S_L, I) = Z + p^I \cdot I \cdot (S_H + S_L) \\ & Y(S_H, S_L, I) = \bar{y} + \mathbb{Y}(S_H, S_L, I) \\ & V(S_H, S_L, I) = \bar{v} + \mathbb{V}(S_H, S_L, I) - S_H \cdot c_H - S_L \cdot c_L \\ & I = 1 \quad \Rightarrow \quad S_H + S_L > 0 \\ & S_H + S_L \leq 2 \end{aligned}$$

where $\mathbb{Y}(S_H, S_L, I)$ is the financial return function for these social interactions, and $\mathbb{V}(S_H, S_L, I)$ is the intrinsic utility return function; $\bar{y}$ is the existing income, and $\bar{v}$ is the intrinsic utility the agent gets from her pre-existing network. All functions are increasing in S_H, S_L, and I. $U = \alpha u(Z) + \beta h(V)$ is strictly concave, increasing, and additively separable in Z and V, where u and h are both concave functions with u' bounded below by some $\underline{u}' > 0$, and α and β are both non-negative scalars weighting an individual's relative preference for consumption versus intrinsic utility.

Assumption 6.1 (Each type has an absolute advantage in producing one type of utility)**.** *Without loss of generality, H is more productive in generating a financial return, and L is more productive in generating intrinsic utility.*[44] *Complementarities are sufficiently weak so that these advantages cannot be overturned in any pairing.*[45]

Assumption 6.2 (One type is more responsive to investment)**.** *Replacing an L-type with an H-type in any bundle strictly raises the utility gain from investing.*[46]

Lemma 6.1 (Definition of a preference space where diversity is optimal)**.** *For every sufficiently concave pair of $(u(Z), h(V))$, there is a non-empty preference space $\mathscr{P}$ such that, for all $(\alpha, \beta) \in \mathscr{P}$ the choice set $(S_H, S_L) = (1,1)$ generates more utility than the*

[43] i.e., $(S_H, S_L) \in \{(0,0); (1,0); (0,1); (2,0); (0,2); (1,1)\}$.

[44] The predictions are unchanged if L is better at generating a financial return, and H is better at generating intrinsic utility. Label assignments are chosen to be consistent with empirical results.

[45] Mathematical statement is Assumption F.1 in Appendix Section F.2.

[46] Mathematical statement is Assumption F.2 in Appendix Section F.2. Note that the model conditions are symmetric, so the predictions are symmetric if L is the type that is more responsive to investment. Label assignments are chosen to be consistent with empirical results.

choice sets $(S_H, S_L) \in \{(2,0),(0,2)\}$*, for both* $I \in \{0,1\}$.[47]

6.1.1 Investment Price Subsidy

Let τ be a subsidy to the price of I, so that $p(\tau) = p^I - \tau$. Following the empirical results, assume that the price of investment p^I is too expensive unsubsidized for investment to be worthwhile.[48] Let $I^*(\tau)$ denote the optimal investment level at subsidy τ; then the optimal investment level $I_g^*(0) = 0$. $S_H^*(0)$ and $S_L^*(0)$ denote the optimal number of high- and low-SES initiations that the agent chooses without the subsidy, and $S_H^*(\tau)$ and $S_L^*(\tau)$ denote the optimal number of initiations under subsidy τ.

Lemma 6.2 (Subsidies induce diversification)**.** *Assume that* u *and* h *satisfy the concavity conditions in Lemma 6.1 so that* $(\alpha, \beta) \in \mathscr{P}$*. Then, if* $S_H^*(0) + S_L^*(0) < 2$*, there exists a subsidy large enough to induce an increase in the number of initiations; and, given the assumed preferences, this subsidy will induce a diverse set of initiations if it is large enough to induce two invitations.*

6.1.2 Preference Heterogeneity

I introduce a set of groups $g \in G_1, G_2$, each with a pair of weights (α^g, β^g), implying a different set of preferences over financial return and intrinsic utility. I assume that groups have opposite but symmetric tastes: let $\psi = \alpha_1 = \beta_2$, and $\eta = \alpha_2 = \beta_1$. All other model primitives remain constant across groups. Let

$$G_1 := \left\{g : \ (S_H^*(0), S_L^*(0)) = (0,1)\right\}, \qquad G_2 := \left\{g : \ (S_H^*(0), S_L^*(0)) = (1,0)\right\}.$$

That is, everyone either prefers interaction with L or H, and the second relationship of either type fails to repay its effort cost absent investment. I assume both sets are non-empty and that each group's tastes are symmetric when $\tau = 0$; that is, G_1 is as far from adding or switching to an H as G_2 is from adding or switching to a L.[49] Furthermore, preference weight heterogeneity is moderate enough that the investment advantage of H cannot be overturned by a preference for L, and the diversity premium guaranteed by $\alpha/\beta \in \mathscr{P}$ cannot be overturned.[50]

6.2 Comparative Statics

Proposition 6.1 (Substitution across types)**.** *Let Assumptions 6.1 and 6.2 hold, and assume preferences lie in* $\mathscr{P}$*.*

[47] Non-emptiness of $\mathscr{P}$ is proven formally in Lemma F.1 in Appendix F.2.

[48] This assumption corresponds to the empirical result that meat-sharing almost never happens without a meat subsidy.

[49] The mathematical statement is Assumption F.3 in Appendix F.2.

[50] The mathematical statements are Assumption F.4 and F.5.

(a) ***(Substitution is available)*** *There is a non-empty interval of subsidies where it is only profitable to invest in the same bundle that is selected without the subsidy, or to invest in the diverse bundle* $(S_H, S_L) = (1, 1)$*. Then, for any* τ *in the interval:*

- *a positive mass of people who prefer a low-SES relationship absent the subsidy (people in* G_1*) initiate a high-SES relationship, choosing* $(S_H, S_L) = (1, 1)$ *or* $(S_H, S_L) = (1, 0)$*, with investment;*
- *every group in* G_2 *continues to choose* $(S_H, S_L) = (1, 0)$*, and initiates no additional low-SES relationship;*
- *no group initiates two relationships of the same type.*

Thus, across all agents, high-SES initiations strictly increase and low-SES initiations weakly decrease.

(b) ***(Substitution is unavailable)*** *Suppose one type of relationship is completely unavailable (e.g., due to a prohibitively high initiation cost), restricting the feasible set so the diverse bundle* $(1, 1)$ *is unavailable. Evaluate behavior at the same subsidy* τ *as in part (a). Then:*

- *if each agent still initiates a relationship of her feasible type absent the subsidy (she values her less-preferred type sufficiently), the subsidy does not change initiation choices.*

All model proofs are in Appendix Section F.2.

By assumption, a sufficiently large subsidy induces diversity in initiations. Proposition 6.1(a) shows that a fixed subsidy is more likely to induce diversification among people who, without a subsidy, prefer to initiate relationships with the type that is less responsive to investment. The experiment enables me to measure if participants perceive the returns to social interaction with high- or low-SES women to be more sensitive to financial investment, and the model generates implications about what this means for network formation patterns. Then, if:

1. Invitations to H are more responsive to an investment subsidy than invitations to L: high prices of social interaction constrain cross-SES linking and contribute to income-based homophily
2. Invitations to L are more responsive to an investment subsidy than invitations to H: high prices of social interaction constrain within-SES linking and enable more economic connectedness
3. Neither type are sensitive to an investment subsidy: high prices of social interaction do not contribute to the economic composition of networks

Proposition 6.1(b) highlights an important counterfactual: even if marginal returns to investment are higher with one type, a subsidy may fail to shift behavior when the margin of substitution is shut down because *all* relationships may be beneficial even without investment. Someone who prefers an L may elect to interact with an H, even without investment, when an L is not available. Evaluating this counterfactual can reveal if prices are a *feasibility* constraint, or if, as the model proposes, the relative returns to initiating relationships with high-SES and low-SES links changes across the price domain, driving differences in network composition across prices.

6.2.1 Mapping Results to the Model

The invitation response patterns show that high- and low-SES relationships are distinct goods, where each type provides unique benefits that are both valuable, implying that economic diversity in social interactions is most-preferred in an environment with the least constraints (Assumption 6.1).[51] The Voucher makes multiple interactions with a diverse set of people even more attractive. Because this shift comes from an increase towards high-SES links, I interpret the perceived benefits of high-SES relationships as more sensitive to financial investment. The model tests also clarify the nature of constraints that inhibit cross-SES linking. High-SES invitations increase in response to the voucher only when Inviters can choose between high- and low-SES Guests, but not when they are restricted to high-SES Guests alone, indicating that high-SES relationships are not infeasible at high prices. Instead, they are relatively less attractive than low-SES alternatives due to their greater sensitivity to investment.

7 Conclusion

My results add to our understanding of social network formation in three important ways. First, I show, while social capital is broadly beneficial, the types of benefits that it provides vary by the nature of the tie. In my setting, within-SES links yield large mental health benefits, while cross-SES links yield large benefits for food security. This implies that people face very consequential trade-offs in linking when social network formation is constrained. Furthermore, if people value both food security and mental health, it implies that people also value economic *diversity* in their networks.

Second, I find that information frictions and effort costs of the activities that form or strengthen network links are prohibitive. Resolving these frictions is remarkably easy, and leads to large positive treatment effects for both cross-SES and within-SES links, without any evidence of displacing the returns to social relationships from anyone else in

[51] Appendix Section F.1 details the relationship between the model predictions and empirical tests, and Appendix Table F.1 presents the first-stage results as a set of empirical tests of the model predictions.

the sample. These results suggest that there is likely a significant market failure in the creation of new social relationships.

Third, I show that the price of valuable social interaction serves as an important barrier to social connection, and that, without intervening on price specifically, links may form or strengthen in ways that can even reinforce existing inequities. This insight into social network formation has important implications for development policy. While evidence shows that group-based development programs—a common tool to execute development policy—have important exogenous economic impacts (Annan et al., 2013; Beaman et al., 2014; Karlan et al., 2017; Ksoll et al., 2016), these are often reported as average impacts and may be concentrated among the relatively higher-income households in a village. Indeed, there is evidence suggesting that development policy relying on self-selection into groups does not reach the poorest households (Lønborg and Rasmussen, 2014). My results show that this exclusion may be at least in part driven by social capital deficits, which is not a determinant that need be strictly relevant for the efficacy of the intervention. Without measuring the distributional impacts of group-based interventions and without being deliberately inclusive, development programs may ignore or deepen social and economic vulnerability of the most marginalized populations. A promising avenue for future research is to test, when introducing interventions in which women form groups themselves, if providing lower-income women with the opportunities and financial means to engage in valuable social activities can help them to be included.

Furthermore, a review of group-based interventions among women found that the primary mechanism through which group-based interventions are successful is their ability to reach many women at once, rather than the social interactions that they foster (Díaz-Martin et al., 2023). My experiment, which generates large positive treatment effects by helping loose acquaintances spend time together in a meaningful way, suggests that the absence of social interaction effects in many group-based interventions may not be surprising since these interventions often do little more socially than bring people to the same place. Without understanding and removing specific barriers to *meaningful* social interaction, group-based interventions may be ineffective at generating social connection, especially for people who are systematically left out. At the same time, including the most marginalized groups can induce a virtuous cycle: social connection alone generates large positive treatment effects on economic and psychological well-being, which, when paired with other development initiatives, has theoretical grounds to significantly improve their efficiency. *Inclusive* development policy may require specific efforts to help marginalized community members build and access social capital.

References

M. H. Algren, O. Ekholm, L. Nielsen, A. K. Ersbøll, C. Kronborg Bak, and P. Tanggaard Andersen. Social isolation, loneliness, socioeconomic status, and health-risk behaviour in deprived neighbourhoods in denmark: A cross-sectional study. *SSM-population health*, 10, 2020.

M. Alsan, O. Garrick, and G. Graziani. Does diversity matter for health? experimental evidence from oakland. *American Economic Review*, 109:4071–4111, 2019.

J. Anders and A. Pallais. Inefficient social disconnection. *Working Paper*, 2025.

M. L. Anderson. Multiple inference and gender differences in the effects of early intervention: A reevaluation of the abecedarian, perry preschool, and early training projects. *Journal of the American Statistical Association*, 103(484):1481–1495, 2008.

M. Angeluci and D. Bennett. The economic impact of depression treatment in india: Evidence from community-based provision of pharmacotherapy. *The American Economic Review*, 114(1):169–198, 2024.

F. Annan and B. Archibong. The value of communication for mental health. *NBER Working Paper 31638*, 2023.

J. Annan, T. Bundervoet, J. Seban, and J. Costigan. A randomized impact evaluation of village savings and loans associations and family-based interventions in burundi. *USAID (Retrieved from http://www.rescue.org/sites/default/files/resource-file/New$_G$eneration$_F$inal$_R$eport$_0$5312013.pdf)*, 2013.

E. Asiedu, M. Lambon-Quayefio, F. Truffa, and A. Wong. Female entrepreneurship and professional networks. *Working Paper*, 2025.

O. Attanasio, A. Barr, J. C. Cardenas, G. Genicot, and C. Meghir. Risk pooling, risk preferences, and social networks. *American Economic Journal: Applied Economics*, 4: 134–167, 2012.

N. Augenblick, K. B. Jack, S. Kaur, F. Masiye, and N. Swanson. Retrieval failures and consumption smoothing: A field experiment on seasonal poverty. *Working Paper*, 2024.

A. Banerjee, A. G. Chandrasekhar, E. Duflo, and M. O. Jackson. The diffusion of microfinance. *Science*, 341:363, 2013.

A. Banerjee, A. Chandrasekhar, E. Duflo, and M. O. Jackson. Using gossips to spread information. *The Review of Economic Studies*, 86:2453–2490, 2019.

A. Banerjee, E. Duflo, E. Grela, M. McKelway, F. Schilbach, G. Sharma, and G. Vaidyanathan. Depression and loneliness among the elderly in low- and middle-income countries. *The Journal of Economic Perspectives*, 37(2):179–202, 2023.

A. Banerjee, E. Breza, A. Chandrasekhar, E. Duflo, M. O. Jackson, and C. Kinnan. Changes in social network structure in response to exposure to formal credit markets.

The Review of Economic Studies, 91:1331–1372, 2024a.

A. Banerjee, E. Breza, A. Chandrasekhar, and B. Golub. When less is more: Experimental evidence on information delivery during india's demonization. *Review of Economic Studies*, 91:1884–1992, 2024b.

K. Basu and M. Wong. Evaluating seasonal food storage and credit programs in east indonesia. *Journal of Development Economics*, 115:200–215, 2015.

L. Beaman. Social networks and the dynamics of labour market outcomes: Evidence from refugees resettled in the u.s. *Review of Economic Studies*, 79:128–161, 2012.

L. Beaman, D. Karlan, and B. Thuysbaert. Saving for a (not so) rainy day: A randomized evaluation of savings groups in mali. *NBER Working Paper 20600*, 2014.

L. Beaman, A. BenYishay, J. Magruder, and A. M. Mobarak. Can network theory-based targeting increase technology adoption? *American Economic Review*, 111(6), 2021.

L. A. Beaman and J. Magruder. Who gets the job referral? evidence from a social networks experiment. *American Economic Review*, 102(7):3574–3593, 2012. ISSN 0002-8282. doi: 10.1257/aer.102.7.3574. URL https://www.aeaweb.org/articles?id=10.1257/aer.102.7.3574.

G. S. Becker. A theory of social interactions. *Journal of Political Economy*, 82:1063–1093, 1974.

Y. Benjamini, A. M. Krieger, and D. Yekutieli. Adaptive linear step-up procedures that control the false discovery rate. *Biometrica*, 93:491–507, 2006.

J. Bentham. *The principles of morals and legislation*. Oxford: Clarendon, 1789.

P. Bergman, R. Chetty, S. DeLuca, N. Hendren, L. F. Katz, and C. Palmer. Creating moves to opportunity: Experimental evidence on barriers to neighborhood choice. *American Economic Review*, 114:1281–1337, 2024.

B. Bhat, J. de Quidt, J. Haushofer, V. H. Patel, G. Rao, F. Schilbach, and P.-L. P. Vautrey. The long-run effects of psychotherapy on depression, beliefs, and economic outcomes. *NBER Working Paper 30011*, 2022.

C. Blattman, S. Chaskel, J. C. Jamison, and M. Sheridan. Cognitive behavioral therapy reduces crime and violence over ten years: Experimental evidence. *American Economic Review: Insights*, 5(4):527–545, 2023.

P. M. Blau. *Inequality and Heterogeneity: A Primitive Theory of Social Structure*. Free Press, New York, 1977.

K. Borusyak and P. Hull. Nonrandom exposure to exogenous shocks. *Econometrica*, 91: 2155–2185, 2023.

P. Bourdieu and J.-C. Passeron. Reproduction in education, society and culture. 1977.

E. Breza and A. Chandrasekhar. Social networks, reputation, and commitment: Evidence

from a savings monitors experiment. *Econometrica*, 87(1):175–216, 2019.

E. Breza, S. Kaur, and N. Krishnaswamy. Social norms as a determinant of aggregate labor supply: Evidence from decentralized spot markets. *NBER Working Paper 25880*, 2024.

K. B. Burchardi and T. A. Hassan. The economic impact of social ties: Evidence from german reunification. *The Quarterly Journal of Economics*, 128:1219–1271, 2013.

M. Burke, L. F. Bergquist, and E. Miguel. Sell low and buy high: Arbitrage and local price effects in kenyan markets. *The Quarterly Journal of Economics*, 134:785–842, 2019.

J. Cai and A. Szeidl. Interfirm relationships and business performance. *The Quarterly Journal of Economics*, 133:1229–1282, 2017.

B. Cannon, D. Hirshleifer, and J. Thornton. Friends with benefits: Social capital and household financial behavior. *NBER Working Paper 32186*, 2024.

E. Carranza, A. Donald, F. Grosset, and S. Kaur. The social tax: Redistributive pressure and labor supply. *Revise and Resubmit, Econometrica*, 2022.

A. G. Chandrasekhar, C. Kinnan, and H. Larreguy. Social networks as contract enforcement: Evidence from a lab experiment in the field. *American Economic Journal: Applied Economics*, 10(4):43–78, 2018. doi: 10.1257/app.20150057.

A. G. Chandrasekhar, E. Dufo, M. Kremer, J. F. Pugliese, J. Robinson, and F. Schilbach. Blue spoons: Sparking communication about appropriate technology use. *NBER Working Paper 30423*, 2022.

T. Chaney. The network structure of international trade. *American Economic Review*, 11:3600–3634, 2014.

R. Chetty, N. Hendren, and L. F. Katz. The effects of exposure to better neighborhoods on children: New evidence from the moving to opportunity experiment. *American Economic Review*, 106:855–902, 2016.

R. Chetty, M. O. Jackson, T. Kuchler, J. Stroebel, N. Hendren, R. B. Fluegge, S. Gong, F. Gonzalez, A. Grondin, M. Jacob, D. Johnston, M. Koenen, E. Laguna-Muggenburg, F. Mudekereza, T. Rutter, N. Thor, W. Townsend, R. Zhang, M. Bailey, P. Barberá, M. Bhole, and N. Wernerfelt. Social capital ii: determinants of economic connectedness. 608, 2022a.

R. Chetty, M. O. Jackson, T. Kuchler, J. Stroebel, N. Hendren, R. B. Fluegge, S. Gong, F. Gonzalez, A. Grondin, M. Jacob, D. Johnston, M. Koenen, E. Laguna-Muggenburg, F. Mudekereza, T. Rutter, N. Thor, W. Townsend, R. Zhang, M. Bailey, P. Barberá, M. Bhole, and N. Wernerfelt. Social capital i: measurement and associations with economic mobility. *Nature*, 608:108–121, 2022b.

M. Chinman, P. George, R. H. Dougherty, A. S. Daniels, S. S. Ghose, A. Swift, and

M. E. Delphin-Rittmon. Peer support services for individuals with serious mental illnesses: Assessing the evidence. *Psychiatric Services*, 65(4):429–441, 2014. doi: 10.1176/appi.ps.201300244. URL https://ps.psychiatryonline.org/doi/10.1176/appi.ps.201300244.

K. W. Choi et al. Social support and depression during a global crisis. *Nature Mental Health*, 2023. doi: 10.1038/s44220-023-00078-0. URL https://www.nature.com/articles/s44220-023-00078-0.

J. Cilliers, N. Elashmawy, and D. McKenzie. Using post-double selection lasso in field experiments. *Working Paper*, 2024.

S. Coate and M. Ravallion. Reciprocity without commitment: Characterization and performance of informal insurance arrangements. *Journal of Development Economics*, 40:1–24, 1993.

J. S. Coleman. Social capital in the creation of human capital. *American Journal of Sociology*, 94(Supplement):S95–S120, 1988.

H. K. Collins, S. F. Hagerty, J. Quoidbach, M. I. Norton, and A. Wood Brooks. Relational diversity in social portfolios predicts well-being. *PNAS*, 119(43):1–9, 2022.

T. G. Conley and C. Udry. Learning about a new technology: Pineapple in ghana. *The American Economic Review*, 100(1):35–69, 2010.

R. E. Cooper et al. The effectiveness, implementation, and experiences of peer support approaches for mental health: A systematic umbrella review. *BMC Medicine*, 22:72, 2024. doi: 10.1186/s12916-024-03260-y. URL https://bmcmedicine.biomedcentral.com/articles/10.1186/s12916-024-03260-y.

P. Dasgupta and D. Ray. Inequality as a determinant of malnutrition and unemployment: Theory. *The Economic Journal*, 86:1011–1034, 1986.

L. Derksen and P. C. Souza. Who knows? information access and endogenous network formation. *Revise and Resubmit, Amerian Economic Journal: Applied*, 2024.

B. Dillon. Selling crops early to pay for school: A large-scale natural experiment in malawi. *Journal of Human Resources*, 56(4):1296–1325, 2021.

B. Dillon, J. De Weerdt, and T. O'Donoghue. Paying more for less: Why don't households in tanzania take advantage of bulk discounts? *The World Bank Economic Review*, 35: 148–179, 2021.

E. Duflo, D. Keniston, T. Suri, and C. Zipfel. Chat over coffee? diffusion of agronomic practices and market spillovers in rwanda. *NBER Working Paper 31368*, 2023.

L. Díaz-Martin, A. Gopalan, E. Guarnieri, and S. Jayachandran. Greater than the sum of the parts? evidence on mechanisms operating in women's groups. *The World Bank Research Observer*, 38:1–35, 2023.

N. Epley and J. Schroeder. Mistakenly seeking solitude. *Journal of Experimental Psychology: General*, 143(5):1980–1999, 2014.

N. Epley, M. Kardas, X. Zhao, S. Atir, and J. Schroeder. *Trends in Cognitive Sciences*, 26(5), 2022.

M. Fafchamps and S. Lund. Risk-sharing networks in rural philippines. *Journal of Development Economics*, 71(2):261–287, 2003. doi: 10.1016/S0304-3878(03)00029-4.

M. Fafchamps and S. Quinn. Networks and manufacturing firms in africa: Results from a randomized field experiment. *The World Bank Economic Review*, 32:656–675, 2018.

B. Feigenberg, E. Field, and R. Pande. The economic returns to social interaction: Experimental evidence from microfinance. *Review of Economic Studies*, 80:1459–1483, 2013.

G. Fink, K. B. Jack, and F. Masiye. Seasonal liquidity, rural labor markets, and agricultural production. *American Economic Review*, 110:3351–3392, 2020.

R. Fisman, D. Paravisini, and V. Vig. Cultural proximity and loan outcomes. *American Economic Review*, 107:457–492, 2017.

A. D. Foster and M. R. Rosenzweig. Learning by doing and learning from others: Human capital and technical change in agriculture. *The Journal of Political Economy*, 103(6): 1176–1209, 1995.

M. S. Granovetter. The strength of weak ties. *American Journal of Sociology*, 78(6), 1973.

M. Grossman. On the concept of health capital and the demand for health. *Journal of Political Economy*, 80:223–255, 1972.

J. G. Gunderson and P. S. Links. *Handbook of Good Psychiatric Management for Borderline Personality Disorder*. American Psychiatric Publishing, 2014.

M. Hardy and J. McCasland. It takes two: Experimental evidence on the determinants of technology diffusion. *Journal of Development Economics*, 149, 2021.

J. Haushofer and D. Salicath. The psychology of poverty: Where do we stand? *Social Philosophy and Policy*, 40:150–184, 2023.

V. M. Hoang et al. Social support and symptoms of antenatal depression among women screened for gestational diabetes mellitus: A cross-sectional study in northern vietnam (the valid ii study). *PLOS Global Public Health*, 2024. URL https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11620646/.

G. Iacobelli and J. Singh. Endogenous institutions: a network experiment in nepal. *Working Paper*, 2020.

W. Jack and T. Suri. Risk sharing and transactions costs: Evidence from kenya's mobile money revolution. *American Economic Review*, 104:183–223, 2014.

M. O. Jackson. Inequality's economic and social roots: the role of social networks and homophily. *arXiv preprint*, arXiv:2506.13016, 2025. doi: 10.48550/arXiv.2506.13016. URL https://arxiv.org/abs/2506.13016.

D. Jacobson. The cultural context of social support and support networks. *Medical Anthropology Quarterly*, 1(2):130–146, 1987. doi: 10.1525/maq.1987.1.2.02a00020. URL https://www.jstor.org/stable/648770.

P. Jakiela and O. Ozier. Does africa need a rotten kin theorem? experimental evidence from village economies. *The Review of Economic Studies*, 83:231–268, 2016.

R. Jensen. The digital provide: Information (technology), market performance, and welfare in the south indian fisheries sector. *The Quarterly Journal of Economics*, 122 (3):879–924, 2007. doi: 10.1162/qjec.122.3.879.

R. Jensen and N. H. Miller. Giffen behavior and subsistence consumption. *American Economic Review*, 98:1553–1577, 2008.

D. Karlan, B. Savonittob, B. Thuysbaertb, and C. Udry. Impact of savings groups on the lives of the poor. *PNAS*, 114(12):3079–3084, 2017.

C. Kinnan and R. M. Townsend. Kinship and financial networks, formal financial access, and risk reduction. *American Economic Review*, 102(3):289–293, 2012. doi: 10.1257/aer.102.3.289.

C. Kinnan, K. Samphantharak, R. M. Townsend, and D. Vera-Cossio. Propagation and insurance in village networks. *American Economic Review*, 114(1):252–284, 2024. doi: 10.1257/aer.20220892.

F. Kondylis, J. A. Loeser, M. Mobarak, M. R. Jones, and D. K. Stein. Learning from self and learning from others : Experimental evidence from bangladesh (english). *Policy Research working paper ; no. WPS 10545; Impact Evaluation series Washington, D.C. : World Bank Group.*, 2023.

C. Ksoll, H. B. Lilleør, J. H. Lønborg, and O. D. Rasmussen. Impact of village savings and loan associations: Evidence from a cluster randomized trial. *Journal of Development Economics*, 120:70–85, 2016.

E. La Ferrara. Kin groups and reciprocity: A model of credit transactions in ghana. *American Economic Review*, 93:1730–1751, 2003.

R. A. Laschever. The doughboys network: Social interactions and the employment of world war i veterans. *Available at SSRN: https://ssrn.com/abstract=1205543*, 2013.

E. Ligon. Risk sharing and information in village economies. *The Review of Economic Studies*, 65:847–864, 1998.

E. Ligon and L. Schechter. Measuring vulnerability. *The Economic Journal*, 113(486): C95–C102, 2003.

E. Ligon and L. Schechter. Motives for sharing in social networks. *Journal of Development Economics*, 99:13–26, 2012.

E. Ligon, J. P. Thomas, and T. Worrall. Informal insurance arrangements with limited commitment: Theory and evidence from village economies. *The Review of Economic Studies*, 69:209–244, 2002.

N. Lin. *Social Capital: A Theory of Social Structure and Action*. Cambridge University Press, Cambridge, 2001.

S. Linde and L. E. Egede. Community social capital and population health outcomes. *JAMA Netw Open*, 6, 2023.

L. A. Lomnitz. Informal exchange networks in formal systems: A theoretical model. *American Anthropologist*, 90(1):42–55, 1988. doi: 10.1525/aa.1988.90.1.02a00030. URL https://www.jstor.org/stable/678801.

J. Ludwig, G. J. Duncan, L. A. Gennetian, L. F. Katz, R. C. Kessler, J. R. Kling, and L. Sanbonmatsu. Long-term neighborhood effects on low-income families: Evidence from moving to opportunity. *American Economic Review*, (3):226–231, 2013.

J. H. Lønborg and O. D. Rasmussen. Can microfinance reach the poorest: evidence from community-managed microfinance intervention. *World Development*, 64:460–472, 2014.

F. Mallah. Cross-income exposure in schools and long-run student outcomes. *Working Paper*, 2026.

M. McKelway, A. Banerjee, E. Grela, F. Schilbach, M. Sequeira, G. Sharma, G. Vaidyanathan, and E. Duflo. Effects of cognitive behavioral therapy and cash transfers on older persons living alone in india: A randomized trial. *Annals of Internal Medicine*, 176(5), 2023.

M. McPherson, L. Smith-Lovin, and J. M. Cook. Birds of a feather: Homophily in social networks. *Annual Review of Sociology*, 27:415–444, 2001.

A. M. Mobarak and M. R. Rosenzweig. Informal risk sharing, index insurance, and risk taking in developing countries. *American Economic Review*, 103:375–380, 2013.

N. Morrish, R. Mujica-Mota, and A. Medina-Lara. Understanding the effect of loneliness on unemployment: propensity score matching. *BMC Public Health*, 22(740), 2022.

K. Munshi. Community networks and the process of development. *Journal of Economic Perspectives*, (4):49–76, 2014.

G. P. Murdock. Ethnographic atlas. *Pittsburgh: University of Pittsburgh Press*, 1967.

P. C. Patel and M. T. Wolfe. Friends for richer or poorer: Economic connectedness, regional social capital, and county entrepreneurial activity. *Journal of Small Business Management*, 62, 2024.

V. Patel, R. Araya, S. Chatterjee, D. Chisholm, A. Cohen, M. De Silva, C. Hosman, H. McGuire, G. Rojas, and M. van Ommeren. Treatment and prevention of mental disorders in low-income and middle-income countries. *The Lancet*, 370:991–1005, 2007.

B. A. Pescosolido. The public stigma of mental illness: What do we think; what do we know; what can we prove? *Journal of Health and Social Behavior*, 4, 2013.

M. Poll. Micro-enterprise saturation and poverty graduation at high frequency, 2024.

A. Portes. Social capital: Its origins and applications in modern sociology. *Annual Review of Sociology*, 24:1–24, 1998.

R. D. Putnam. *Bowling alone: The collapse and revival of American community.* Simon Schuster, 2000.

G. Rao. Familiarity does not breed contempt: Generosity, discrimination, and diversity in delhi schools. *American Economic Review*, 109:774–809, 2019.

L. A. Rico-Uribe, F. F. Caballero, N. Martín-María, M. Cabello, J. L. Ayuso-Mateos, and M. Miret. Association of loneliness with all-cause mortality: A meta-analysis. *PLOS ONE*, 13(1), 2018.

M. Ridley. Mental illness discrimination. *Working Paper*, 2025.

M. Ridley, G. Rao, F. Schilbach, and V. Patel. Poverty, depression, and anxiety: Causal evidence and mechanisms. *Science*, 370(6522), 2020.

E. Riley. Mobile money and risk sharing against village shocks. *Journal of Development Economics*, 135:43–58, 2018.

M. Rozensweig and Stark. Consumption smoothing, migration, and marriage: Evidence from rural india. *Journal of Political Economy*, 97:905–926, 1989.

B. Sacerdote. Peer effects in education: How might they work, how big are they and how much do we know thus far? *Handbook of the Economics of Education, Volume 3*, 2011.

P. Santos and C. B. Barrett. Persistent poverty and informal credit. *Journal of Development Economics*, 96:337–347, 2011.

L. Schechter and S. Vasudevan. Value of social connection: Evidence from randomized introduction of peer contact. *Working Paper*, 2024.

M. K. Shenk, R. O. Begley, D. A. Nolin, and A. Swiatek. When does matriliny fail? the frequencies and causes of transitions to and from matriliny estimated from a de novo coding of a cross-cultural sample. *Philosophical Transactions of the Royal Society B*, 1780:20190006, 2019.

E. Smith. Stigma and social cover: A mental health care experiment in refugee network. *Working Paper*, 2024.

N. K. Steffens, C. Haslam, J. Jetten, and S. A. Haslam. Social identification-building interventions to improve health: A systematic review and meta-analysis. *Health Psy-*

chology Review, 13(2):110–130, 2019. doi: 10.1080/17437199.2018.1483239. URL https://doi.org/10.1080/17437199.2018.1483239.

E. C. Stephens and C. B. Barrett. Incomplete credit markets and commodity marketing behavior. *Journal of Agricultural Economics*, 62(1):1–24, 2011.

S. Subramanian and A. Deaton. The demand for food and calories. *Journal of Political Economy*, 104:133–162, 1996.

D. L. Surkalim, M. Luo, R. Eres, K. Gebel, J. van Buskirk, and D. Bauman, Adrian Baumanand Ding. The prevalence of loneliness across 113 countries: systematic review and meta-analysis. *British Medical Journal*, 376, 2022.

N. Swanson. All in the family: Kinship pressure and firm-worker matching. *Working Paper*, 2025.

Q. Thompson, K. Dobson, and K. G. Card. How many friends are enough? optimizing number of close friendships for mental health benefit. *PsyArXiv*, 2021. doi: 10.31234/osf.io/9p8g7. URL https://psyarxiv.com/9p8g7/.

C. Tilly. *Durable Inequality*. University of California Press, Berkeley, 1998.

R. M. Townsend. Risk and insurance in village india. *Econometrica*, 62:539–591, 1994.

C. Udry. Risk and insurance in a rural credit market: An empirical investigation in northern nigeria. *The Review of Economic Studies*, 61:495–526, 1994.

D. Umberson and J. K. Montez. Social relationships and health: A flashpoint for health policy. *Journal of Health and Social Behavior*, 51(1_suppl):S54–S66, 2010. doi: 10.1177/0022146510383501. URL https://doi.org/10.1177/0022146510383501.

UNICEF. Humanitarian situation report no. 5, malawi. *UNICEF.org*, 2025.

K. Vasilaky. Female social networks and farmer training: Can randomized information exchange improve outcomes? *American Journal of Agricultural Economics*, 95:376–383, 2013.

K. N. Vasilaky and K. L. Leonard. As good as the networks they keep? improving outcomes through weak ties in rural uganda. *Economic Development and Cultural Change*, 66:755–792, 2018.

F. Vega-Redondo, C. Brummit, P. Pin, G. Rubera, D. Ubfal, D. Hovy, C. Benedetti-Fasil, and T. Fornaciari. Peer networks and entrepreneurship: A pan-african rct. *IZA Discussion Paper No. 12848*, 2019.

L. M. Verbrugge. The structure of adult friendship choices. *Social Forces*, 56:576–597, 1977.

C. S. T. Yim, J. H. L. Chieng, X. R. Tang, J. X. Tan, V. K. F. Kwok, and S. M. Tan. Umbrella review on peer support in mental disorders. *International Journal of Mental Health*, 52(4):379–400, 2023. doi: 10.1080/00207411.2023.2166444. URL

https://www.tandfonline.com/doi/abs/10.1080/00207411.2023.2166444.

Online Supplementary Material

Appendix A Background and Cultural Context

Figure A.1: Percent of Women Who Never Left Their Home Village (Malawi DHS, 2015)

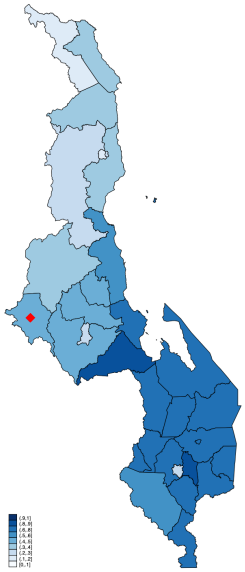


Table A.1: Malawi DHS 2015: Determinants of Moving Villages

	(1) Ever Moved Villages	(2) Ever Moved Villages	(3) Ever Moved Villages	(4) Ever Moved Villages
Property Owner	-0.287***	-0.222***	-0.277***	-0.220***
	(0.005)	(0.005)	(0.005)	(0.005)
Observations	67987	67987	67987	67987
Mean of Women Without Owned Property	0.677	0.677	0.677	0.677
District Fixed Effects		X		X
Tribe Fixed Effects			X	X

Standard errors in parentheses
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Figure A.2: Social Inclusion – by SES and Years in the Village

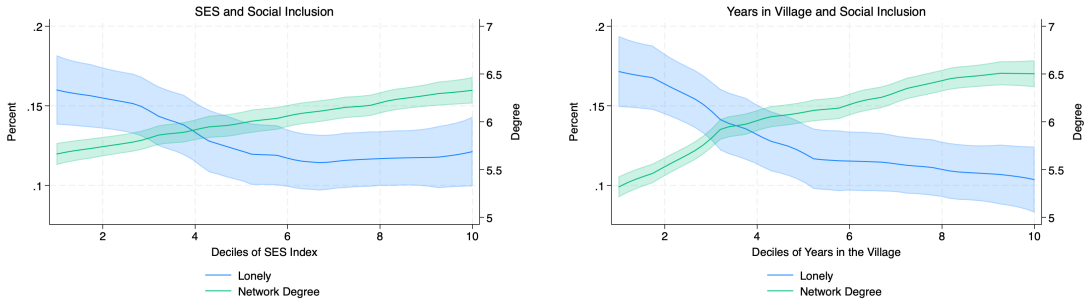
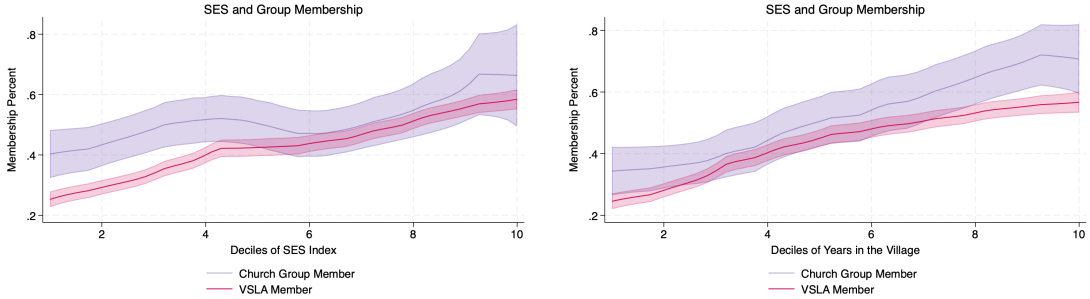


Figure A.3: Group Membership – by SES and Years in the Village



Notes: I only measured church group membership at endline, so I only use the Control group to analyze underlying correlations between church group membership and other demographic characteristics.

Figure A.4: Mental Health – by Age and Distance from Home Village

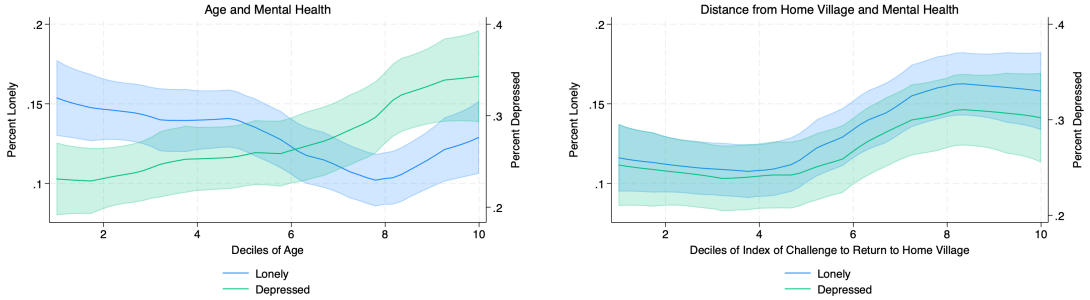


Table A.2: Summary Statistics by Socio-Economic Status

	Low-SES	**Thatched**	**High-SES**	**Iron Sheets**
Iron Sheets Roof	0.26		0.84	
Any HH Business	.39	.43	.76	.53
Acres of Owned Farmland	1.7	1.8	3.0	2.3
Meals in Past 2 Days	4.5	4.5	4.9	4.7
Months Consuming Harvest	4.5	4.3	5.5	5.2
Acres of Rented Farmland	0.8	0.7	1.1	1.2
Any HH Sec.-School Grad.	0.13	0.10	0.20	0.22

Table A.3: Closeness of a Conversation Across Settings

	(1) Full Sample	(2) Thinking About a Meal-Sharing Friend	(3) Thinking About a Work Friend	(4) Female Sample	(5) Male Sample
Conversation Over a Meal	1.282***	1.276***	1.303***	1.249***	1.452**
	(0.150)	(0.212)	(0.161)	(0.159)	(0.443)
Friend from Work	0.020			0.023	-0.000
	(0.119)			(0.134)	(0.256)
Observations	504	252	252	432	72
Mean in Other Settings	2.045	2.042	2.048	2.123	1.574

Standard errors in parentheses
The outcome is how personal a conversation could become in a given setting, and with a given person, on a scale from 1 to 5 (where higher numbers indicate more personal conversations). Each survey participant was asked to consider a conversation across four settings: over a meal, while working, while doing chores, or while at work. They were also asked to consider two types of people: a person with whom they often share meals, and a person with whom they often work together. The order of questions was randomized. Each observation is at the question level (with eight observations per participatn). All regressions include individual fixed effects and fixed effects for the order in which they were asked the question. Standard errors are clustered at the individual level.
* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Figure A.5: Preferred versus Actual Meal-Sharing

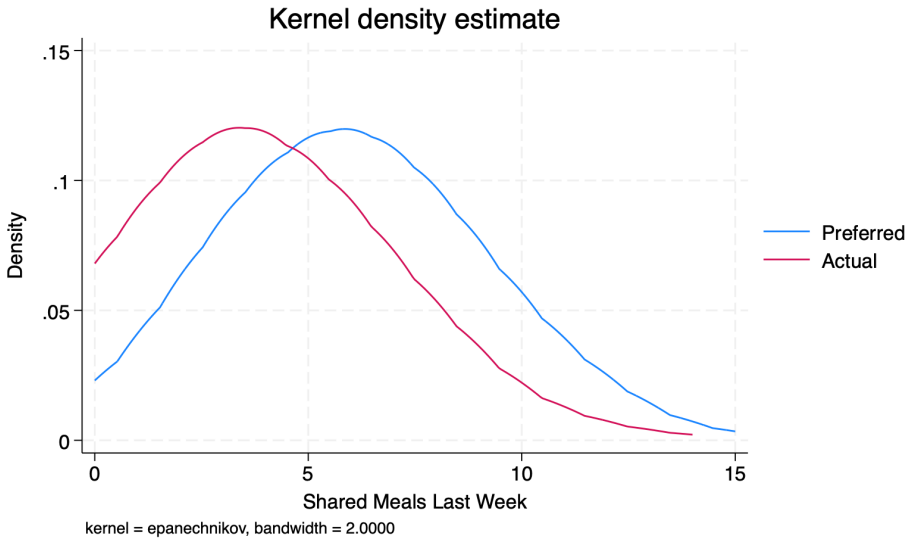


Figure A.6: Pluralistic Ignorance

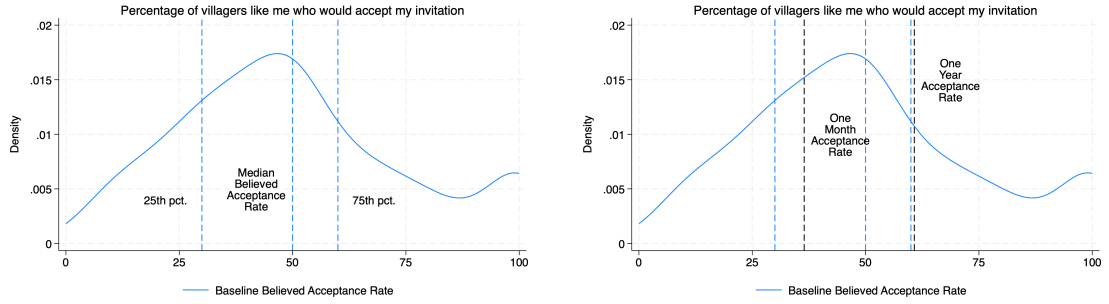
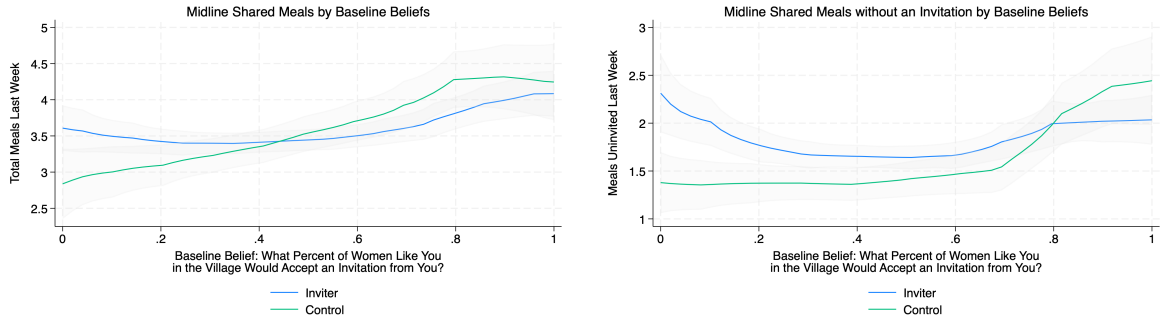


Figure A.7: Baseline Beliefs and Midline Meal-Sharing



A.1 Business Capital Index Construction

I create a proxy for business capital in the following way: each enumerator estimates the fixed costs of establishing each business type. I regress the within-enumerator rank of fixed costs on an indicator for the business type, and use the coefficient as a measure of the capital required to run that business, controlling for the enumerators' stated confidence in their response. I add these coefficients across every household business, plus the mean rank of the omitted group in the regression, so that people without any household businesses have no business capital, and people with more capital-intensive or a greater number of household businesses have more business capital. Enumerators come from within these communities, and several have family members or friends engaged in these types of businesses. Furthermore, they have a fairly high degree of agreement about the relative rankings of different business types.

A.2 Cluster-level Program Saturation

I randomize the saturation of the Inviters and Guests at a geographic cluster level. In a "high-saturation" cluster, 87% of eligible women are randomized to be Inviters or Guests, and 13% of women are randomized to Control. In a "low-saturation" cluster, 35% of eligible women are randomized to be Inviters or Guests, and 65% of women are randomized to Control. First, I define geographic clusters. Then, I randomize each cluster to high or low saturation. Finally, I randomize individuals within clusters to treatment assignments, using the ratios as defined by their cluster-saturation-type.

I define geographic clusters using k-means clustering following Poll (2024). This unsupervised learning technique creates groups based on similarities in latitude and longitude. I specify that clusters must have at least 20 people in them, and should have 36 people per cluster on average. I choose 36 as the average size so that I attain around 100 clusters, in order to ensure sufficient statistical power to detect differences across Control respondents in high- and low-saturation areas. In order to ensure that this is a relevant geographic unit for social networks, I link the names of social network connections with people in my listing dataset and calculate how many relationships with someone in my sample are within the same cluster. I am able to link 7% of network links. Seventy percent of these links are within the same cluster, indicating that this is a network-relevant geographic unit. It is important to note that the relationships between respondents in my sample (the names who I can link) are not representative relationships. They are on average 4.4 minutes (58%) walking-distance closer than other relationships, and are 5 percentage points (16%) less likely to be relatives. However, these relationships are likely good approximations for the types of relationships that my intervention creates.

A.3 Sharing List Construction Details

There are 68 participants (52 Inviters) who are randomized into the Intervention Sample but who attrit from the Treatment Allocation Survey. These participants are randomly replaced with women who originally are randomized to Control and who live in the same geographic cluster as the attriter. In all but 19 cases, I also am able to randomly select a woman from the same wealth quintile, to hold strata constant. In all analyses, these women are analytically handled as "treatment", while the women originally assigned to treatment, who attrited in the treatment allocation survey, are handled as "control". I check for balance on the treatment assignment after replacements.

In the 19 cases where there is not a Control household available from the same geographic cluster and the same wealth quintile, I select a woman from the same geographic cluster and the proceeding or succeeding wealth quintile. Because the names from the Sharing List that each Inviter sees is dependent on *other* Inviters' Sharing Lists, I do not reconstruct the Sharing List for these replacement households. That is, "replacement" Inviters see the Sharing List originally constructed for a different woman, and the Guests available for them to invite are not necessarily within 800 meters.

A.4 Variables Construction Details

Food Security Index

For each index, I take the first component of the PCA of the variables entering the index. Each index is then normalized into standard deviation units from the Control mean.

Midline Food Security Index Variables:

1. How many days in the past month did you eat fish? + How many days in the past month did you eat eggs?
2. On a typical day in the past month, how many meals did you eat?

Endline Food Security Index Variables:

1. How many days in the past month did you eat fish? + How many days in the past month did you eat eggs? + How many days in the past month did you eat meat?
2. On a typical day in the past month, how many meals did you eat?
3. How many days during the most challenging period this past growing season did you eat fish? + How many days during the most challenging period this past growing season did you eat eggs? + How many days during the most challenging period this past growing season did you eat meat?

4. On a typical day during the most challenging period this past growing season, how many meals did you eat?

I define the lean season as the time during the past growing season where participants report facing the most financial hardship. On average, households report that this time period lasted eleven weeks. In the Control group, 64% report that they ate one meal per day during this time period, and only 6% report that they ate three meals per day. The endline survey is conducted right after the harvest, so food security measured over the past month represents the theoretical best time for food security. Consistent with extreme swings in consumption following the agricultural seasons, in the Control group, only 5% report typically eating one meal per day in the past month, while 36% report eating three meals per day. Note that there are no differences across groups in the length of the lean season, so question (3) of the endline food security index should not confound length of the lean season with diet quality.

Strength of Ties

These five variables entering the strength of ties measure are the following:

1. How comfortable would you feel to trust [NAME] with your thoughts and feelings? *(Five-point Likert scale).*
2. How easy is it for you to laugh with [NAME] when you are together? *(Five-point Likert scale).*
3. "Who do you share secrets with?" *(Binary, =1 if selects name of given tie).*
4. Inclusion-of-Self in Others Score *(Visual measure of relationship "closeness" on a 7 point scale).*
5. Confidante reciprocity status *(0 if the tie is not a confidante; 1 if the tie is a confidante, but the relationship is certainly not reciprocated; 2 if the tie is a confidante, and the relationship is probably not reciprocated; 3 if the tie is a confidante and the relationship is most likely reciprocated; 4 if the tie is a confidante and the relationship is certainly reciprocated).*

Crop Classes

I classify crops into five crop classes: maize, other starches, fruits and vegetables, legumes, and cash crops.

Other starches include: cassava, sorghum, rice, Irish potatoes, and sweet potatoes.

Fruits and vegetables include: bananas, leafy greens, tomatoes, onions and pumpkins.

Legumes include: soy beans, plain beans, groundnuts, pigeon peas, and cowpeas.

Cash crops include: sugar cane, tobacco, and sunflower seeds.

A.5 Text Analysis

In the endline survey, enumerators ask Inviters about their experience with each Guest who they sent an invitation to; and ask Guests about their experience with each Inviter who they received an invitation from. Enumerators also ask all respondents about their overall opinions of the program. I conduct a text analysis from these text responses. Across all of the possible text response variables, I compute the cosine similarity of the text response to a set of key-words. I code each row as falling within a category if the cosine similarly is ≥ 0.6.

If text is similar to multiple key phrases, they are assigned a category based on the following priority ordering: (1) strong relationships, (2) any other relationships, (3) no relationship, (4) other.

The breakdown of text analysis categorizations is the following:

Table A.4: Text Analysis

	Strong Relationship	Any Relationship	No Contact	Other
Inviters	229	308	265	125
Guests	153	251	275	73

Phrases indicating "strengthened relationships":

"closer now", "grew closer", "became close", "we talk often", "grew close", "closer together", "we are close now", "stronger friendship", "stronger relationship", "closer than before", "friendship grew", "our bond", "bond grew", "bond stronger", "love", "relationship improved", "friendship improved", "we are still friends", "we are good friends", "we continue", "we keep", "we still", "persist", "more open", "united", "support each other", "encourage each other", "help each other", "sharing continues", "sharing food regularly", "continued sharing", "always together", "keep meeting", "continue sharing" , "we still see each other", "we still meet", "we visit each other", "we visit often", "we still visit", "we are in touch", "we keep in touch", "we remain friends", "we are friends", "we are close", "we became closer", "our relationship is good", "relationship continues", "she helps me", "he helps me", "we help one another", "we support one another", "she encourages me", "he encourages me", "we encourage one another", "we are still together", "always together", "still together", "friendship continues", "friendship persists", "we remain close", "we became united", "we are still united", "relationship is strong", "friendship is strong", "friendship remains", "relationship remains", "bonds much stronger", "encouraging each other", "increases our relationship", "we are friends", "strengthen our relationship", "friendship very strong", "share meals sometimes"

Phrases indicating "any relationship":

"became friends", "made a friend", "started talking", "she visits", "he visits", "now friends", "friends now", "our bond", "we get along", "invited to my home", "we met and chat", "visited me", "went to her house", "we know each other now", "we chat", "we chatted", "chatting", "talked", "discussed", "greeted each other", "we still chat", "conversation", "just chatted", "just chatting", "chat here and there", "i forgot what we discussed", "share food", "share meals", "first time we met", "she came for the first time", "she introduced herself", "he introduced himself", "introduced me", "we started as friends", "began visiting", "share things", "shared with me", "sharing my meal", "we talked about", "we ate together", "discussing about", "we do share meals", "it has made them to know each other", "we talked more", "we really talked"

Phrases indicating "nothing happened":

"nothing happened", "nothing", "didn't happen", "did not happen", "she never came", "he never came", "we didn't meet", "we did not meet", "didn't share", "did not share", "never visited", "no friendship", "didn't invite", "not invited", "never invited", "did not receive invitation", "she refused", "he refused", "we haven't met", "never met", "don't know her", "don't know him", "don't know her", "don't know him", "moved away", "she moved", "he moved", "we have never meet", "can't recall being invited", "can't remember being invited", "i wasn't invited", "no invitation", "invited but didn't go", "couldn't make it", "wasn't available", "missed the meeting", "can't recall being invited", "did hear I was invited", "don't remember the person", "name is new", "moved to another village", "moved away", "at a funeral", "did not manage to go", "wasn't invited by her", "never received any invitation", "did not receive invitation", "never discussed anything concerning this program", "never turned up", "never showed up", "didn't show up", "she refused to come", "he refused to come", "she denied", "he denied", "she ignored" , "he ignored" , "not available" , "we never interacted", "we never connected", "did not show up", "we have never eaten", "we did not talk","meeting was canceled", "never met", "she didnt find her", "did not invite me", "never invites me", "never invited me"

Appendix B Balance and Attrition

B.1 Baseline Balance

Across the following balance tables, all outcomes are presented in standard deviations from the full sample mean. All outcomes are from the Listing Survey. Thus, both the Control and Inviter samples can be above or below the sample mean, since it includes High-SES women, Guests, and Spillover Control participants.

Table B.1: Baseline Balance: Comparing Control with Inviters

Variable	(1) Control Mean/SE	(2) Inviter Mean/SE	T-test Difference (1)-(2)
Age	-0.029	-0.042	0.013
	(0.056)	(0.029)	
Years in Village	-0.197	-0.099	-0.097
	(0.046)	(0.028)	
Chewa	0.047	-0.026	0.073
	(0.046)	(0.030)	
Number of Children	-0.108	-0.057	-0.051
	(0.049)	(0.028)	
SES Index (Z-Score)	-0.372	-0.359	-0.014
	(0.039)	(0.022)	
Consuming Own Maize	-0.020	-0.085	0.065
	(0.050)	(0.030)	
VSLA	-0.053	-0.053	0.000
	(0.050)	(0.029)	
Rented Acres of Farmland	-0.093	-0.079	-0.014
	(0.039)	(0.027)	
Rented Gardens	-0.009	0.024	-0.033
	(0.047)	(0.030)	
Moved for Marriage	-0.063	-0.089	0.026
	(0.053)	(0.031)	
Yearly Home Visits	0.016	-0.003	0.019
	(0.053)	(0.030)	
Social Network Size	0.008	-0.033	0.041
	(0.051)	(0.029)	
Husband in Social Network Size	0.055	0.037	0.018
	(0.052)	(0.030)	
Listed in Others' Networks	-0.044	-0.015	-0.029
	(0.048)	(0.028)	
% Network High-SES	-0.051	-0.012	-0.039
	(0.050)	(0.030)	
% Network Same Roof Material	0.082	0.080	0.002
	(0.049)	(0.029)	
UCLA-3 Score	0.031	0.071	-0.040
	(0.048)	(0.030)	
Average Strength of Network Links	-0.069	-0.014	-0.054
	(0.054)	(0.028)	
N	400	1200	
F-test of joint significance (F-stat)			0.661
F-test, number of observations			1600

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table B.2: Baseline Balance: Comparing Inviter Sub-groups

Variable	(1) Inviter: Random SES Mean/SE	(2) Inviter: High-SES Mean/SE	(3) Inviter Low-SES Mean/SE	T-test Difference (1)-(2)	(1)-(3)	(2)-(3)
Age	0.005	-0.098	-0.033	0.103	0.038	-0.065
	(0.052)	(0.048)	(0.052)			
Years in Village	-0.061	-0.088	-0.149	0.027	0.088	0.061
	(0.049)	(0.049)	(0.048)			
Chewa	-0.062	0.010	-0.026	-0.073	-0.036	0.036
	(0.054)	(0.049)	(0.052)			
Number of Children	0.007	-0.070	-0.110	0.077	0.117*	0.040
	(0.051)	(0.049)	(0.047)			
SES Index (Z-Score)	-0.340	-0.362	-0.374	0.022	0.035	0.012
	(0.038)	(0.039)	(0.038)			
Consuming Own Maize	-0.078	-0.046	-0.130	-0.031	0.052	0.084
	(0.051)	(0.051)	(0.052)			
VSLA	-0.134	0.003	-0.028	-0.137*	-0.106	0.030
	(0.049)	(0.050)	(0.050)			
Rented Acres of Farmland	-0.133	-0.011	-0.093	-0.122	-0.040	0.082
	(0.043)	(0.057)	(0.040)			
Rented Gardens	0.012	-0.054	0.115	0.066*	-0.104	-0.170***
	(0.054)	(0.047)	(0.055)			
Moved for Marriage	0.008	-0.105	-0.169	0.113	0.177**	0.064
	(0.050)	(0.055)	(0.057)			
Yearly Home Visits	-0.002	0.091	-0.099	-0.093	0.097	0.190
	(0.051)	(0.063)	(0.041)			
Social Network Size	-0.048	-0.032	-0.018	-0.016	-0.029	-0.013
	(0.049)	(0.049)	(0.051)			
Husband in Social Network Size	0.028	0.076	0.008	-0.047	0.020	0.068*
	(0.051)	(0.053)	(0.050)			
Listed in Others' Networks	-0.027	-0.011	-0.005	-0.017	-0.022	-0.006
	(0.047)	(0.050)	(0.049)			
% Network High-SES	-0.036	0.063	-0.062	-0.099	0.026	0.125
	(0.050)	(0.052)	(0.054)			
% Network Same Roof Material	0.039	0.106	0.096	-0.067	-0.058	0.009
	(0.050)	(0.049)	(0.053)			
UCLA-3 Score	0.106	0.085	0.023	0.021	0.083	0.062
	(0.055)	(0.053)	(0.049)			
Average Strength of Network Links	-0.017	-0.018	-0.009	0.001	-0.007	-0.009
	(0.052)	(0.046)	(0.050)			
N	400	400	400			
F-test of joint significance (F-stat)				1.169	1.520*	2.105***
F-test, number of observations				800	800	800

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table B.3: Baseline Balance: Comparing Voucher Sub-groups

Variable	(1) Random Mean/SE	(2) Random + Voucher Mean/SE	T-test Difference (1)-(2)
Age	0.024	-0.014	0.038
	(0.072)	(0.075)	
Years in Village	-0.098	-0.024	-0.074
	(0.068)	(0.070)	
Chewa	-0.080	-0.044	-0.036
	(0.079)	(0.075)	
Number of Children	0.014	0.001	0.012
	(0.074)	(0.072)	
SES Index (Z-Score)	-0.340	-0.340	-0.000
	(0.053)	(0.055)	
Consuming Own Maize	-0.088	-0.067	-0.021
	(0.073)	(0.072)	
VSLA	-0.119	-0.149	0.030
	(0.069)	(0.068)	
Rented Acres of Farmland	-0.180	-0.087	-0.093
	(0.062)	(0.060)	
Rented Gardens	-0.061	0.084	-0.145
	(0.062)	(0.089)	
Moved for Marriage	0.072	-0.056	0.128
	(0.065)	(0.075)	
Yearly Home Visits	0.001	-0.006	0.007
	(0.062)	(0.082)	
Social Network Size	0.002	-0.097	0.100
	(0.068)	(0.070)	
Husband in Social Network Size	0.062	-0.006	0.068
	(0.075)	(0.071)	
Listed in Others' Networks	-0.022	-0.033	0.011
	(0.073)	(0.060)	
% Network High-SES	-0.092	0.019	-0.111
	(0.068)	(0.074)	
% Network Same Roof Material	0.122	-0.045	0.166
	(0.066)	(0.075)	
UCLA-3 Score	0.134	0.078	0.056
	(0.080)	(0.077)	
Average Strength of Network Links	-0.029	-0.004	-0.025
	(0.073)	(0.074)	
N	200	200	
F-test of joint significance (F-stat)			0.961
F-test, number of observations			400

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table B.4: Baseline Balance: Comparing Voucher Sub-groups

Variable	(1) Low-SES Mean/SE	(2) High-SES Mean/SE	(3) High-SES + Voucher Mean/SE	(4) Low-SES + Voucher Mean/SE	T-test Difference (1)-(2)	(1)-(3)	(1)-(4)
Age	0.036	-0.115	-0.080	-0.101	0.151	0.116	0.137
	(0.076)	(0.070)	(0.065)	(0.071)			
Years in Village	-0.065	-0.137	-0.038	-0.233	0.073	-0.027	0.168
	(0.068)	(0.070)	(0.069)	(0.067)			
Chewa	0.010	0.029	-0.008	-0.062	-0.018	0.018	0.073
	(0.070)	(0.068)	(0.072)	(0.077)			
Number of Children	-0.051	-0.045	-0.094	-0.169	-0.006	0.043	0.117
	(0.071)	(0.076)	(0.062)	(0.063)			
SES Index (Z-Score)	-0.328	-0.344	-0.380	-0.421	0.017	0.052	0.094
	(0.051)	(0.055)	(0.056)	(0.055)			
Consuming Own Maize	-0.141	-0.099	0.006	-0.120	-0.042	-0.147	-0.021
	(0.073)	(0.073)	(0.071)	(0.073)			
VSLA	0.084	-0.007	0.013	-0.139	0.091	0.071	0.223*
	(0.072)	(0.071)	(0.071)	(0.069)			
Rented Acres of Farmland	-0.101	-0.112	0.090	-0.085	0.012	-0.191	-0.015
	(0.058)	(0.058)	(0.097)	(0.056)			
Rented Gardens	0.012	-0.071	-0.037	0.219	0.083	0.049	-0.207
	(0.073)	(0.065)	(0.068)	(0.082)			
Moved for Marriage	-0.127	-0.098	-0.112	-0.212	-0.028	-0.014	0.085
	(0.079)	(0.077)	(0.078)	(0.083)			
Yearly Home Visits	-0.029	0.068	0.115	-0.169	-0.097	-0.144	0.140
	(0.064)	(0.084)	(0.094)	(0.051)			
Social Network Size	0.010	-0.015	-0.049	-0.046	0.024	0.058	0.056
	(0.071)	(0.073)	(0.066)	(0.074)			
Husband in Social Network Size	0.008	0.103	0.049	0.008	-0.095	-0.041	0.000
	(0.071)	(0.077)	(0.074)	(0.071)			
Listed in Others' Networks	-0.022	0.011	-0.033	0.011	-0.033	0.011	-0.033
	(0.064)	(0.080)	(0.059)	(0.076)			
% Network High-SES	-0.049	0.041	0.084	-0.075	-0.090	-0.134	0.026
	(0.077)	(0.073)	(0.074)	(0.077)			
% Network Same Roof Material	0.049	0.059	0.152	0.144	-0.011	-0.104	-0.096
	(0.077)	(0.071)	(0.068)	(0.074)			
UCLA-3 Score	0.070	0.076	0.094	-0.024	-0.006	-0.024	0.094
	(0.071)	(0.072)	(0.078)	(0.069)			
Average Strength of Network Links	0.030	-0.067	0.032	-0.048	0.097	-0.001	0.079
	(0.068)	(0.064)	(0.065)	(0.073)			
N	200	200	200	200			
F-test of joint significance (F-stat)					1.036	1.120	1.235
F-test, number of observations					400	400	400

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

B.2 Complier Balance

Table B.5: Complier Balance (Sent an Invitation): Comparing Inviter Sub-groups

Variable	(1) Inviter: Random SES Mean/SE	(2) Inviter: High-SES Mean/SE	(3) Inviter Low-SES Mean/SE	T-test Difference (1)-(2)	T-test Difference (1)-(3)	T-test Difference (2)-(3)
Age	-0.022	-0.094	-0.048	0.072	0.026	-0.046
	(0.055)	(0.053)	(0.058)			
Years in Village	-0.054	-0.075	-0.115	0.021	0.061	0.040
	(0.053)	(0.054)	(0.054)			
Chewa	-0.085	0.049	-0.034	-0.134	-0.051	0.083
	(0.062)	(0.051)	(0.059)			
Number of Children	0.005	-0.077	-0.115	0.082	0.120*	0.038
	(0.057)	(0.054)	(0.053)			
SES Index (Z-Score)	-0.320	-0.339	-0.333	0.019	0.013	-0.006
	(0.041)	(0.044)	(0.040)			
Consuming Own Maize	-0.112	-0.031	-0.126	-0.081	0.013	0.095
	(0.057)	(0.056)	(0.058)			
VSLA	-0.125	0.060	0.045	-0.185**	-0.170**	0.015
	(0.054)	(0.055)	(0.057)			
Rented Acres of Farmland	-0.150	0.007	-0.076	-0.156	-0.073	0.083
	(0.047)	(0.065)	(0.045)			
Moved for Marriage	0.067	-0.068	-0.139	0.135	0.206*	0.071
	(0.051)	(0.059)	(0.063)			
Yearly Home Visits	-0.002	0.118	-0.074	-0.121	0.071	0.192
	(0.056)	(0.071)	(0.050)			
Social Network Size	-0.020	-0.027	0.045	0.008	-0.064	-0.072*
	(0.056)	(0.055)	(0.056)			
Listed in Others' Networks	-0.057	-0.057	0.027	0.000	-0.083	-0.083
	(0.051)	(0.050)	(0.058)			
% Network High-SES	-0.051	0.038	-0.090	-0.089	0.039	0.128
	(0.056)	(0.057)	(0.061)			
% Network Same Roof Material	0.028	0.076	0.084	-0.048	-0.056	-0.008
	(0.056)	(0.054)	(0.061)			
UCLA-3 Score	0.124	0.124	0.051	0.001	0.073	0.072
	(0.062)	(0.059)	(0.056)			
Average Strength of Network Links	-0.024	0.020	-0.016	-0.044	-0.008	0.036
	(0.058)	(0.052)	(0.056)			
N	329	331	314			
F-test of joint significance (F-stat)				0.872	1.411	1.150
F-test, number of observations				660	643	645

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. The covariate variables told_about_husband_z, and rented_garden_z are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table B.6: Complier Balance (Shared a Meal by Midline): Comparing Inviter Sub-groups

Variable	(1) Inviter: Random SES Mean/SE	(2) Inviter: High-SES Mean/SE	(3) Inviter Low-SES Mean/SE	T-test Difference (1)-(2)	(1)-(3)	(2)-(3)
Age	-0.049	-0.106	-0.001	0.056	-0.048	-0.104
	(0.073)	(0.074)	(0.081)			
Years in Village	-0.039	-0.032	-0.080	-0.007	0.041	0.048
	(0.069)	(0.078)	(0.076)			
Chewa	-0.020	0.027	-0.051	-0.047	0.031	0.078
	(0.077)	(0.076)	(0.084)			
Number of Children	-0.027	-0.012	-0.039	-0.015	0.012	0.027
	(0.075)	(0.077)	(0.075)			
SES Index (Z-Score)	-0.303	-0.302	-0.319	-0.001	0.016	0.017
	(0.057)	(0.064)	(0.053)			
Consuming Own Maize	0.058	-0.009	-0.059	0.067	0.116	0.049
	(0.073)	(0.080)	(0.079)			
VSLA	-0.086	0.097	0.136	-0.183	-0.222	-0.039
	(0.073)	(0.080)	(0.079)			
Rented Acres of Farmland	-0.118	0.011	-0.053	-0.128	-0.065	0.064
	(0.066)	(0.072)	(0.065)			
Moved for Marriage	0.052	-0.105	-0.172	0.157	0.224	0.067
	(0.070)	(0.087)	(0.090)			
Yearly Home Visits	-0.046	0.147	-0.045	-0.192	-0.001	0.191
	(0.062)	(0.096)	(0.073)			
Social Network Size	-0.011	0.017	0.192	-0.028	-0.203	-0.175
	(0.071)	(0.081)	(0.078)			
Listed in Others' Networks	-0.050	-0.145	0.072	0.094	-0.123	-0.217
	(0.070)	(0.073)	(0.083)			
% Network High-SES	-0.087	0.126	-0.032	-0.213	-0.055	0.158
	(0.071)	(0.080)	(0.084)			
% Network Same Roof Material	-0.011	0.029	-0.033	-0.040	0.022	0.062
	(0.073)	(0.077)	(0.086)			
UCLA-3 Score	0.174	0.176	0.025	-0.002	0.149	0.151
	(0.092)	(0.093)	(0.078)			
Average Strength of Network Links	-0.021	-0.018	-0.100	-0.003	0.079	0.082
	(0.076)	(0.071)	(0.076)			
N	181	159	165			
F-test of joint significance (F-stat)				0.435	0.495	0.492
F-test, number of observations				340	346	324

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. The covariate variables told_about_husband_z, and rented_garden_z are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table B.7: Complier Balance (Shared a Meal by Endline): Comparing Inviter Sub-groups

Variable	(1) Inviter: Random SES Mean/SE	(2) Inviter: High-SES Mean/SE	(3) Inviter Low-SES Mean/SE	T-test Difference (1)-(2)	(1)-(3)	(2)-(3)
Age	-0.007	-0.084	-0.045	0.078	0.039	-0.039
	(0.063)	(0.060)	(0.066)			
Years in Village	-0.003	-0.045	-0.096	0.043	0.093	0.050
	(0.061)	(0.062)	(0.063)			
Chewa	-0.036	0.065	-0.036	-0.101	0.000	0.101
	(0.066)	(0.055)	(0.067)			
Number of Children	0.019	-0.065	-0.114	0.083	0.133*	0.050
	(0.063)	(0.061)	(0.061)			
SES Index (Z-Score)	-0.303	-0.320	-0.331	0.017	0.029	0.012
	(0.048)	(0.049)	(0.045)			
Consuming Own Maize	-0.007	-0.047	-0.116	0.040	0.109	0.070
	(0.063)	(0.063)	(0.066)			
VSLA	-0.094	0.068	0.063	-0.162	-0.157	0.005
	(0.061)	(0.063)	(0.064)			
Rented Acres of Farmland	-0.113	0.020	-0.106	-0.133	-0.008	0.126
	(0.055)	(0.077)	(0.051)			
Moved for Marriage	0.040	-0.086	-0.137	0.126	0.177	0.051
	(0.060)	(0.067)	(0.071)			
Yearly Home Visits	0.018	0.200	-0.068	-0.183	0.085	0.268
	(0.066)	(0.088)	(0.056)			
Social Network Size	-0.007	0.020	0.049	-0.027	-0.055	-0.028
	(0.062)	(0.062)	(0.064)			
Listed in Others' Networks	-0.036	-0.077	0.071	0.041	-0.107	-0.147
	(0.061)	(0.056)	(0.069)			
% Network High-SES	-0.023	-0.001	-0.083	-0.022	0.060	0.083
	(0.063)	(0.062)	(0.070)			
% Network Same Roof Material	0.011	0.078	0.049	-0.067	-0.037	0.029
	(0.063)	(0.060)	(0.071)			
UCLA-3 Score	0.106	0.106	0.084	0.000	0.022	0.022
	(0.072)	(0.067)	(0.066)			
Average Strength of Network Links	-0.047	-0.002	-0.028	-0.044	-0.018	0.026
	(0.064)	(0.057)	(0.064)			
N	258	261	247			
F-test of joint significance (F-stat)				0.593	1.090	0.678
F-test, number of observations				519	505	508

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. The covariate variables told_about_husband_z, and rented_garden_z are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table B.8 sample: Control sample only includes women who send an outside-experiment invitation between Baseline and Midline that has not yet resulted in a shared meal; Inviter sample only includes women who send a within-experiment invitation between Baseline and Midline that has not yet resulted in a shared meal.

Table B.8: Treatment Balance Among Women Whose Invitations are Rejected

Variable	(1) Control Mean/SE	(2) Inviters Mean/SE	T-test Difference (1)-(2)
Age	-0.059	-0.038	-0.021
	(0.140)	(0.048)	
Years in Village	-0.007	-0.086	0.079
	(0.138)	(0.047)	
Chewa	0.059	-0.040	0.099
	(0.118)	(0.050)	
Number of Children	0.072	-0.088	0.160
	(0.127)	(0.047)	
SES Index (Z-Score)	-0.397	-0.349	-0.048
	(0.101)	(0.036)	
Consuming Own Maize	0.069	-0.177	0.246
	(0.127)	(0.049)	
VSLA	0.157	-0.040	0.197
	(0.132)	(0.047)	
Rented Acres of Farmland	0.006	-0.103	0.108
	(0.097)	(0.049)	
Rented Gardens	0.158	-0.055	0.213
	(0.142)	(0.043)	
Moved for Marriage	0.035	-0.018	0.053
	(0.125)	(0.048)	
Yearly Home Visits	-0.025	0.041	-0.067
	(0.118)	(0.056)	
Social Network Size	0.379	-0.066	0.445
	(0.130)	(0.047)	
Husband in Social Network Size	-0.032	0.065	-0.097
	(0.126)	(0.050)	
Listed in Others' Networks	-0.089	-0.022	-0.067
	(0.127)	(0.044)	
% Network High-SES	0.061	-0.053	0.115
	(0.120)	(0.050)	
% Network Same Roof Material	-0.088	0.132	-0.219
	(0.121)	(0.049)	
UCLA-3 Score	-0.076	0.071	-0.147
	(0.120)	(0.046)	
Average Strength of Network Links	-0.294	0.022	-0.316
	(0.113)	(0.048)	
N	60	447	
F-test of joint significance (F-stat)			0.564
F-test, number of observations			507

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

B.3 Attrition

Table B.9: Attrition

	(1)	(2)	(3)	(4)	(5)	(6)
	Participated in Survey		Took Survey in Person		Took Survey by Phone	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.002	0.007	-0.008	-0.010	0.009	0.016
	(0.006)	(0.012)	(0.009)	(0.021)	(0.007)	(0.018)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	-0.004	0.010	-0.004	-0.009	-0.000	0.019
	(0.008)	(0.014)	(0.011)	(0.025)	(0.008)	(0.022)
Inviter with Low-SES Guest List	0.009	0.017	-0.011	0.004	0.019*	0.011
	(0.006)	(0.015)	(0.012)	(0.025)	(0.011)	(0.022)
Inviter with Random Guest List	0.002	-0.006	-0.008	-0.025	0.009	0.019
	(0.007)	(0.015)	(0.011)	(0.026)	(0.009)	(0.022)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	-0.003	0.021	-0.008	0.004	0.004	0.016
	(0.010)	(0.017)	(0.015)	(0.030)	(0.011)	(0.027)
High-SES with Voucher	-0.005	-0.001	0.001	-0.027	-0.006	0.027
	(0.010)	(0.018)	(0.013)	(0.032)	(0.009)	(0.028)
Low-SES without Voucher	0.011*	0.010	-0.006	0.005	0.017	0.006
	(0.006)	(0.018)	(0.015)	(0.031)	(0.014)	(0.026)
Low-SES with Voucher	0.006	0.024	-0.013	0.001	0.018	0.019
	(0.007)	(0.017)	(0.016)	(0.031)	(0.015)	(0.028)
Random without Voucher	-0.006	0.011	-0.013	0.004	0.006	0.009
	(0.010)	(0.018)	(0.015)	(0.031)	(0.011)	(0.027)
Random with Voucher	0.010*	-0.024	-0.002	-0.054	0.011	0.031
	(0.005)	(0.021)	(0.013)	(0.033)	(0.011)	(0.028)
Observations	1599	1599	1599	1599	1599	1599
Control Mean	0.990	0.950	0.977	0.850	0.013	0.100
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.048**	0.604	0.604	0.593	0.076*	0.693
Pooled: Voucher vs. No Voucher	0.284	0.293	0.507	0.222	0.753	0.307
High-SES: Voucher vs. No Voucher	0.918	0.267	0.592	0.381	0.394	0.717
Low-SES: Voucher vs. No Voucher	0.351	0.468	0.712	0.916	0.935	0.667
Random: Voucher vs. No Voucher	0.066*	0.118	0.518	0.119	0.761	0.496

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for column (1) takes value 1 if the respondent participated in the Midline Survey, and 0 otherwise. The outcome for column (2) takes the value 1 if the respondent participated in the Endline Survey, and 0 otherwise. The outcome for column (3) takes the value 1 if the respondent was interviewed in the Midline Survey in person, and 0 otherwise (includes both attriters and participants who were interviewed over the phone). The outcome for column (4) takes the value 1 if the respondent was interviewed in the Endline Survey in person, and 0 otherwise (includes both attriters and participants who were interviewed over the phone). The outcome for column (5) takes the value 1 if the respondent was interviewed in the Midline Survey over the phone, and 0 otherwise (includes both attriters and participants who were interviewed in person). The outcome for column (6) takes the value 1 if the respondent was interviewed in the Endline Survey over the phone, and 0 otherwise (includes both attriters and participants who were interviewed in person).

Appendix C First Stage Results: Additional Tables and Figures

Table C.1: Treatment Effects on Number of Invitations Sent Across Treatment Arms

	(1) Any	(2) Any	(3) Any	(4) Low-SES	(5) Low-SES	(6) High-SES	(7) High-SES
Random with Voucher	0.074	0.091	0.079	-0.041	-0.031	0.103	0.112
	(0.105)	(0.104)	(0.107)	(0.080)	(0.076)	(0.083)	(0.078)
High-SES without Voucher	0.049	0.135	0.176*	0.000	0.000	0.779***	0.231**
	(0.098)	(0.098)	(0.106)	(.)	(.)	(0.088)	(0.105)
High-SES with Voucher	-0.099	-0.021	0.017	0.000	0.000	0.620***	0.068
	(0.102)	(0.102)	(0.111)	(.)	(.)	(0.092)	(0.107)
Low-SES without Voucher	-0.160	-0.059	-0.013	0.595***	0.085	0.000	0.000
	(0.100)	(0.101)	(0.115)	(0.093)	(0.107)	(.)	(.)
Low-SES with Voucher	-0.183*	-0.083	-0.092	0.577***	0.061	0.000	0.000
	(0.105)	(0.106)	(0.118)	(0.097)	(0.120)	(.)	(.)
Observations	1200	1200	994	800	800	800	800
Random without Voucher Mean	1.505	1.505	1.468	0.750	0.750	0.755	0.755
Controls for Number of Type on List	NO	YES	NA	NO	YES	NO	YES
P-values:							
High-SES: Voucher = No Voucher	0.142	0.118	0.160			0.112	0.098
Low-SES: Voucher = No Voucher	0.827	0.825	0.536	0.865	0.816		
Random + Voucher = High-SES + Voucher	0.105	0.291	0.585			0.000	0.701
Random + Voucher = Low-SES + Voucher	0.019	0.114	0.163	0.000	0.438		

Standard errors in parentheses
All specifications include baseline unbalanced control variables (number of garden plots and listed husband in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Column 3 drops Inviter participants who saw fewer than six names on the Sharing List (206 participants, or 17 percent).
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Table C.2: Treatment Effects on Invitation-Sending to Guest "Bundles"

	(1) Mixed Bundle		(2) Low-SES Only		(3) High-SES Only		(4) >= 2 Invitations	
Random with Voucher	0.106**	(0.042)	-0.099**	(0.042)	-0.024	(0.042)	0.056	(0.047)
Low-SES without Voucher			0.499***	(0.042)			0.022	(0.068)
Low-SES with Voucher			0.490***	(0.042)			-0.015	(0.068)
High-SES without Voucher					0.548***	(0.042)	-0.048	(0.060)
High-SES with Voucher					0.493***	(0.042)	-0.083	(0.060)
Observations	400		800		800		1200	
Random without Voucher Mean	0.235		0.295		0.300		0.435	
P-values:								
High-SES: Voucher = No Voucher					0.191		0.454	
Low-SES: Voucher = No Voucher			0.831				0.425	
DID Estimates [P-values]								
Random with Voucher:								
(1) Chooses High-SES Only – Chooses Low-SES Only					0.075 [p= 0.258]			
(2) Chooses a Mixed Bundle – Chooses a Homogeneous Bundle			0.205 [p= 0.001]		0.130 [p= 0.049]			

Standard errors in parentheses
All specifications include baseline unbalanced control variables (number of garden plots and listed husband in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. (column 1) indicates if the Inviter sent an invitation to at least one low-SES Guest and at least one high-SES Guest (Inviters with the High-SES Guest List or the Low-SES Guest List only saw one type of Guest, so the outcome is mechanically zero). (column 2) indicates if the Inviter sent an invitation to at least one low-SES Guest, and sent no invitations to any high-SES Guests (Inviters with the High-SES Guest List did not see any low-SES names, so this outcomes is mechanically zero for this group). (column 3) indicates if the Inviter sent an invitation to at least one high-SES Guest, and no low-SES Guests (Inviters with the Low-SES Guest List did not see any high-SES names, so this outcome is mechanically zero for this group).
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Table C.3: Treatment Effects on Invitation-Sending to Low-SES and High-SES Guests

	(1) Mixed Bundle		(2) 1 Low-SES Only		(3) >= 2 Low-SES Only		(4) 1 High-SES Only		(5) >= 2 High-SES Only	
Random with Voucher	0.106**	(0.042)	-0.047	(0.043)	-0.054	(0.038)	-0.021	(0.045)	-0.001	(0.041)
Low-SES without Voucher			0.212***	(0.043)	0.288***	(0.038)				
Low-SES with Voucher			0.240***	(0.044)	0.250***	(0.039)				
High-SES without Voucher							0.221***	(0.045)	0.179***	(0.054)
High-SES with Voucher							0.190***	(0.044)	0.156***	(0.054)
Observations	400		800		800		800		800	
Random without Voucher Mean	0.235		0.200		0.095		0.195		0.105	
P-values:										
High-SES: Voucher = No Voucher							0.494		0.566	
Low-SES: Voucher = No Voucher			0.520		0.322					

Standard errors in parentheses

All specifications include baseline unbalanced control variables (number of garden plots and listed husband in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. (column 1) indicates if the Inviter sent an invitation to at least one low-SES Guest and at least one high-SES Guest (Inviters with the High-SES Guest List or the Low-SES Guest List only saw one type of Guest, so the outcome is mechanically zero). (column 2) indicates if the Inviter sent an invitation to exactly one low-SES Guest, and sent no invitations to any high-SES Guests; (column 3) indicates if the Inviter sent invitations to at least two low-SES Guests, and sent no invitations to any high-SES Guests (Inviters with the High-SES Guest List did not see any low-SES names, so these outcomes are mechanically zero for this group). (column 4) indicates if the Inviter sent an invitation to exactly one high-SES Guest, and no low-SES Guests; (column 5) indicates if the Inviter sent invitations to at least two high-SES Guests, and sent no invitations to any low-SES Guests (Inviters with the Low-SES Guest List did not see any high-SES names, so these outcomes is mechanically zero for this group).

* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Table C.4: Treatment Effects on Meal-Sharing with Guests within 1 Month

	(1) Any Guest		(2) Low-SES Guest		(3) High-SES Guest	
Random with Voucher	0.094*	(0.049)	0.043	(0.046)	0.101**	(0.046)
High-SES without Voucher	0.015	(0.049)			0.189***	(0.046)
High-SES with Voucher	-0.010	(0.049)			0.164***	(0.046)
Low-SES without Voucher	-0.006	(0.049)	0.175***	(0.046)		
Low-SES with Voucher	0.012	(0.049)	0.205***	(0.046)		
Observations	1190		800		800	
Random without Voucher Mean	0.401		0.215		0.220	
P-values:						
High-SES: Voucher = No Voucher	0.615				0.593	
Low-SES: Voucher = No Voucher	0.719		0.507			
Test: *HSES without V = HSES with V = LSES without V = LSES with V*					[p= 0.848]	
DID Estimates [P-values]						
High-SES without Voucher – Low-SES without Voucher					0.014 [p= 0.819]	
High-SES with Voucher – Low-SES with Voucher					-0.042 [p= 0.499]	
Random with Voucher:						
Shares Meal with High-SES – Shares Meal with Low-SES					0.057 [p= 0.334]	

Standard errors in parentheses
All specifications include baseline unbalanced control variables (number of garden plots and listed husband in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. (column 1) indicates if the Inviter shared a meal with at least one Guest within one month. (column 2) indicates if the Inviter shared a meal with at least one low-SES Guest within one month (Inviters with the High-SES Guest List did not see any low-SES names, so this outcomes is mechanically zero for this group). (column 3) indicates if the Inviter shared a meal with at least one high-SES Guest within one month (Inviters with the Low-SES Guest List did not see any high-SES names, so this outcome is mechanically zero for this group).
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Table C.5: Chosen Guest Characteristics

	(1) Age	(2) Networks	(3) SES	(4) UCLA-3	(5) Roof	(6) Distance	(7) HH Power
High-SES	1.985***	0.395***	0.740***	-0.011	0.367***	0.009	-0.033
	(0.703)	(0.131)	(0.053)	(0.097)	(0.061)	(0.013)	(0.077)
Low-SES	-0.948	0.219	-0.799***	0.028	-0.470***	0.009	0.090
	(0.613)	(0.143)	(0.062)	(0.100)	(0.052)	(0.015)	(0.082)
Observations	974	974	974	967	1200	974	974
Controls for Average on List	NO	NO	NO	NO	NO	NO	NO
Random-SES Mean	29.608	6.238	0.600	3.698	0.820	0.198	0.012
P-values							
Low-SES = High-SES	0.000	0.268	0.000	0.728	0.000	0.998	0.183

Standard errors in parentheses
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Table C.6: Reasons Chose First-Pick Guest

	(1) Generous	(2) Humble	(3) Warm	(4) Knowledgeable	(5) Connected	(6) Business	(7) Mother	(8) Hard-working	(9) Trust-worthy	(10) Respected	(11) Prayerful	(12) Marriage	(13) Needy
Voucher	0.020**	0.010	-0.012	0.007	-0.020**	0.001	0.003	0.001	-0.011	-0.009	-0.006	0.007	0.002
	(0.009)	(0.008)	(0.009)	(0.006)	(0.009)	(0.005)	(0.006)	(0.007)	(0.007)	(0.006)	(0.006)	(0.007)	(0.006)
Observations	974	974	974	974	974	974	974	974	974	974	974	974	974
No Voucher Mean	0.104	0.152	0.183	0.053	0.230	0.026	0.051	0.061	0.072	0.055	0.063	0.059	0.037

Standard errors in parentheses
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Table C.7: Reasons Chose Guest (Average Across Guest)

	(1) Generous	(2) Humble	(3) Warm	(4) Knowledgeable	(5) Connected	(6) Business	(7) Mother	(8) Hard-working	(9) Trust-worthy	(10) Respected	(11) Prayerful	(12) Marriage	(13) Needy
Inviter with High-SES Guests	-0.003	0.021**	0.025***	0.015*	-0.013	0.004	-0.006	-0.010	-0.023**	-0.005	0.017**	-0.014*	-0.015**
	(0.011)	(0.011)	(0.009)	(0.008)	(0.014)	(0.006)	(0.007)	(0.009)	(0.010)	(0.009)	(0.008)	(0.008)	(0.008)
Inviter with Low-SES Guests	0.012	0.027**	0.030***	-0.005	-0.003	-0.003	-0.005	-0.017*	-0.017*	-0.012	0.012	-0.014*	-0.008
	(0.010)	(0.011)	(0.010)	(0.007)	(0.014)	(0.006)	(0.007)	(0.009)	(0.009)	(0.007)	(0.008)	(0.007)	(0.008)
Observations	974	974	974	974	974	974	974	974	974	974	974	974	974
Inviter with Random-SES Guests Mean	0.121	0.156	0.177	0.061	0.243	0.033	0.069	0.080	0.089	0.067	0.063	0.081	0.053
P-values													
High-SES = Low-SES	0.164	0.608	0.670	0.032	0.438	0.249	0.976	0.406	0.520	0.378	0.652	0.951	0.278

Standard errors in parentheses
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Figure C.1: First-Stage Meal Sharing

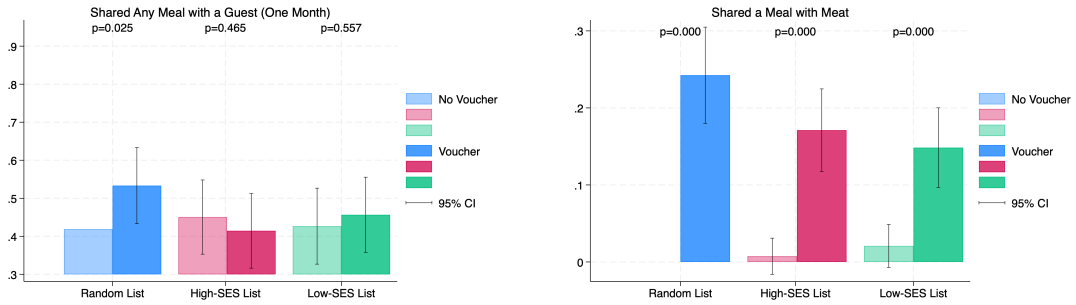


Figure C.2: First-Stage Meal Sharing Reciprocation

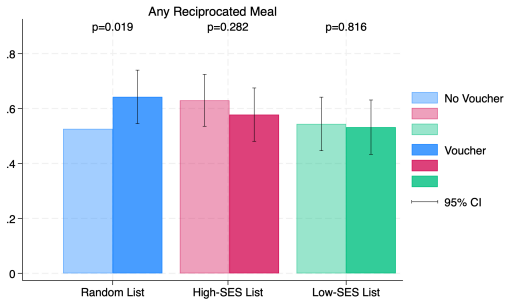


Figure C.3: Vignette: Returns to Relationships

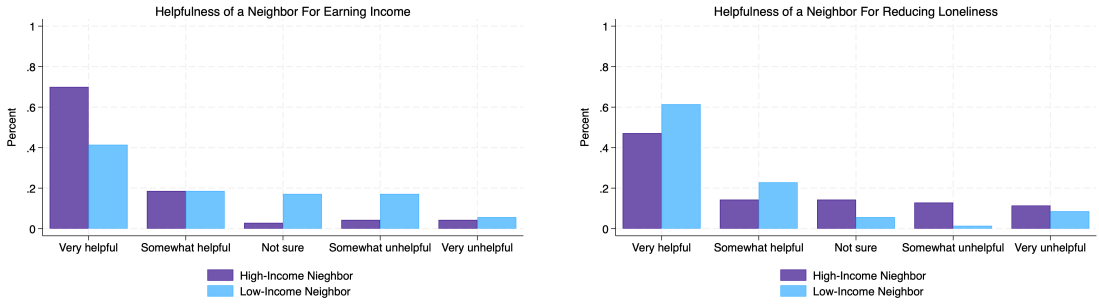


Table C.8: Meal Conversation Topics

Variable	(1) Inviter: Random SES Mean/SE	(2) Inviter: High-SES Mean/SE	(3) Inviter Low-SES Mean/SE	T-test Difference (1)-(2)	(1)-(3)	(2)-(3)
Discussed with an Invitee: Marriage/Romance	0.134	0.137	0.145	-0.004	-0.012**	-0.008
	(0.017)	(0.017)	(0.018)			
Discussed with an Invitee: Childcare	0.108	0.109	0.095	-0.001	0.013	0.014
	(0.016)	(0.016)	(0.015)			
Discussed with an Invitee: Other Family	0.101	0.099	0.080	0.002	0.021	0.019
	(0.015)	(0.015)	(0.014)			
Discussed with an Invitee: Social Relationships	0.134	0.096	0.133	0.037*	0.001	-0.036
	(0.017)	(0.015)	(0.017)			
Discussed with an Invitee: Professional Relationship	0.028	0.046	0.015	-0.018	0.013	0.031
	(0.008)	(0.011)	(0.006)			
Discussed with an Invitee: Business Resources	0.043	0.033	0.008	0.010	0.035*	0.025**
	(0.010)	(0.009)	(0.004)			
Discussed with an Invitee: Finding Piecework	0.033	0.046	0.070	-0.013	-0.037**	-0.024*
	(0.009)	(0.011)	(0.013)			
Discussed with an Invitee: Growing Businesses	0.040	0.036	0.025	0.005	0.015	0.010
	(0.010)	(0.009)	(0.008)			
Discussed with an Invitee: Finding Loans	0.015	0.008	0.018	0.007	-0.002	-0.010
	(0.006)	(0.004)	(0.007)			
Discussed with an Invitee: Agriculture	0.116	0.086	0.080	0.030	0.036	0.006
	(0.016)	(0.014)	(0.014)			
Discussed with an Invitee: Stories from our Past	0.038	0.033	0.023	0.005	0.015	0.010
	(0.010)	(0.009)	(0.007)			
Discussed with an Invitee: Stories from the Village	0.058	0.048	0.055	0.010	0.003	-0.007
	(0.012)	(0.011)	(0.011)			
Discussed with an Invitee: Discussions about Norms	0.030	0.020	0.028	0.010	0.003	-0.007
	(0.009)	(0.007)	(0.008)			
Discussed with an Invitee: Health	0.045	0.018	0.020	0.028	0.025**	-0.002
	(0.010)	(0.007)	(0.007)			
Discussed with an Invitee: Religion	0.020	0.008	0.023	0.013	-0.002	-0.015
	(0.007)	(0.004)	(0.007)			
N	397	394	399			
F-test of joint significance (F-stat)				0.970	2.437***	1.835**
F-test, number of observations				791	796	793

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. Standard errors are robust. Fixed effects using variable strata are included in all estimation regressions. The covariate variables rented_garden_z, vsla_z, children_z, and move_why_1_z are included in all estimation regressions. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

C.1 Economic Diversity-Weighted Network Churn

My measure of economic diversity-weighted network churn is:

$$Churn^{Diverse} = (Dropped + New) \cdot \left(\frac{(Dropped_{IS} + New_{IS}) \cdot (Dropped_{Th} + New_{Th})}{(Dropped + New)^2} \right)^{\alpha}$$

where $Dropped$ is the number of dropped links, New is the number of new links, IS subscripts ties that have iron-sheets roofs, and Th subscripts ties that have thatched roofs, $Dropped = Dropped_{IS} + Dropped_{Th}$ and $New = New_{IS} + New_{Th}$. Finally, α is a parameter that weights the importance of diversity relative to volume. Note that, if $\alpha > 0$, the total measure is maximized when network churn among ties with iron-sheets roofs is equal to network churn among ties with thatched roofs.

I regress $Churn^{Diverse}$ on my treatment indicators across a range of values for α. If the treatment effect is maximized when $\alpha = 0$, then the intervention induces a greater volume of network churn, among the same types of ties (by income) that we would expect in the absence of the treatment. Conversely, if $\alpha > 0$, then the intervention induces both a greater volume of network churn and more diversity (by income) in who enters and exits the network. For almost every group, $\alpha^* = 0$ maximizes treatment effects of that group, indicating that the intervention induces a larger volume of network churn within the same groups who would enter and exit the network to a lesser extent in the absence of the treatment. However, consistent with the first-stage treatment effects on invitation-sending, this is not the case among Inviters with the Random Guest List and the Voucher. The α that maximizes the treatment-effect is $\alpha^* = 0.016$, indicating that Inviters with the Random Guest List and the Voucher have a more economically-diverse set of ties enter and exit their networks.

Because the intervention increases total network churn, applying a penalty for homogeneous ties (an $\alpha > 0$) naturally penalizes the treatment group more heavily. Therefore, the fact that a positive weight ($\alpha^* = 0.016$) actually maximizes the treatment effect for the Random Guest List with Voucher arm is a demanding test. It indicates that the churn in this group became sufficiently more economically diverse to overcome the penalty on higher volume, directly aligning with the first-stage finding that this arm actively sought diverse interactions.

While $\alpha^* > 0$ provides evidence that patterns in invitation-sending carry through to actual changes in the network, the difference between the magnitude of the treatment effect among Inviters with the Random Guest List and the Voucher when $\alpha = 0.016$ and when $\alpha = 0$ is minute. When $\alpha = 0.016$, the Inviter treatment with the Random Guest List and the Voucher leads to a 0.698-person increase in network churn. When $\alpha = 0$, the same group leads to a 0.696-person increase in network churn.

Table C.9: $\alpha^* := \alpha$ that Maximizes Treatment Effects on Diversity-Weighted Network Churn

Inviter Arm	α^*
High-SES Guest List without the Voucher	0
High-SES Guest List and the Voucher	0
Low-SES Guest List without the Voucher	0
Low-SES Guest List and the Voucher	0
Random Guest List without the Voucher	0
Random Guest List and the Voucher	0.016

I construct a measure of network churn that is weighted by economic diversity of individuals entering and existing networks: $(Dropped + New) \cdot (p^{IS} \cdot (1 - p^{IS}))^{\alpha}$, where p^{IS} is the percent of individuals who enter or exit the network that have an iron sheets roof. I regress this measure on my treatment indicators across a range of α. In this table I report α^*, the α that maximizes the treatment effects for each treatment indicator.

Appendix D Second Stage Results: Additional Figures and Tables

Table D.1: Treatment Effects on Food Security Index Components

	(1)	(2)	(3)	(4)	(5)	(6)
	Lean Season Protein Consumption	Last Month Protein Consumption		Lean Season Daily Meals	Last Month Daily Meals	
	1 Year	1 Month	1 Year	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.515*	0.069	0.621**	0.078**	-0.008	0.074
	(0.297)	(0.178)	(0.288)	(0.031)	(0.032)	(0.098)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	0.778**	0.129	1.054***	0.094**	0.026	0.254
	(0.372)	(0.224)	(0.361)	(0.038)	(0.039)	(0.246)
Inviter with Low-SES Guest List	0.259	-0.251	0.390	0.089**	-0.042	-0.037
	(0.337)	(0.206)	(0.351)	(0.040)	(0.039)	(0.049)
Inviter with Random Guest List	0.544	0.312	0.283	0.032	-0.017	-0.009
	(0.356)	(0.233)	(0.370)	(0.038)	(0.038)	(0.051)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	0.649	0.047	1.061**	0.118**	0.028	0.482
	(0.417)	(0.270)	(0.465)	(0.048)	(0.048)	(0.506)
High-SES with Voucher	0.947*	0.193	1.019**	0.062	0.022	0.034
	(0.506)	(0.285)	(0.445)	(0.047)	(0.048)	(0.062)
Low-SES without Voucher	0.134	-0.243	-0.104	0.088*	-0.046	-0.105*
	(0.372)	(0.250)	(0.412)	(0.047)	(0.047)	(0.059)
Low-SES with Voucher	0.396	-0.204	1.013**	0.075	-0.045	0.032
	(0.438)	(0.254)	(0.469)	(0.052)	(0.050)	(0.061)
Random without Voucher	0.602	0.313	0.264	0.045	-0.044	-0.005
	(0.418)	(0.279)	(0.477)	(0.047)	(0.047)	(0.063)
Random with Voucher	0.613	0.329	0.432	0.030	0.018	-0.007
	(0.467)	(0.311)	(0.474)	(0.047)	(0.046)	(0.061)
Observations	1528	1522	1528	1528	1522	1528
Control Mean	3.185	3.117	6.135	1.420	2.247	2.309
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.126	0.076*	0.072*	0.898	0.079*	0.231
Pooled: Voucher vs. No Voucher	0.521	0.903	0.235	0.471	0.587	0.547
High-SES: Voucher vs. No Voucher	0.583	0.655	0.938	0.313	0.919	0.386
Low-SES: Voucher vs. No Voucher	0.534	0.891	0.030**	0.823	0.995	0.044**
Random: Voucher vs. No Voucher	0.983	0.964	0.770	0.784	0.238	0.970

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Protein meals consumed during the lean season and during the last month at the 1 Year follow-up include meals with meat, fish, or eggs (columns 1 and 3). Meals consumed with protein during the last at the 1 Month follow-up include meals with fish or eggs (column 2). Daily meals in the lean season (column 4) and daily meals in the past month (columns 5 and 6) are the number of meals consumed on a typical day during those time periods.

Table D.2: Treatment Effects on CESDR-10 Score (Depression Score) Components

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
	I was unusually bothered		I was unfocused		I felt depressed		Everything was effortful		I felt hopeful		I felt fearful		My sleep was restless		I was happy		I felt lonely		I could not get going	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects																				
All Inviters	0.018	-0.121**	0.021	-0.061	-0.053	-0.028	0.092**	-0.076	0.126**	-0.011	0.013	-0.091**	0.047	-0.077	0.027	0.068	0.079**	-0.077*	-0.049	-0.043
	(0.052)	(0.048)	(0.049)	(0.048)	(0.052)	(0.050)	(0.041)	(0.047)	(0.055)	(0.053)	(0.046)	(0.046)	(0.049)	(0.049)	(0.049)	(0.048)	(0.039)	(0.042)	(0.050)	(0.051)
Panel B: Pooled Inviter by Guest List Treatment Effects																				
Inviter with High-SES Guest List	0.009	-0.081	0.015	0.001	-0.065	0.037	0.086*	0.000	0.146**	-0.021	-0.033	-0.045	0.028	-0.041	0.000	0.077	0.042	-0.001	-0.084	0.018
	(0.063)	(0.058)	(0.058)	(0.059)	(0.061)	(0.060)	(0.050)	(0.058)	(0.065)	(0.066)	(0.054)	(0.056)	(0.058)	(0.060)	(0.058)	(0.057)	(0.047)	(0.052)	(0.060)	(0.062)
Inviter with Low-SES Guest List	0.022	-0.150**	-0.017	-0.091	-0.056	-0.116*	0.087	-0.144***	0.131**	-0.002	0.085	-0.148***	0.072	-0.100*	0.064	0.089	0.109**	-0.119**	-0.000	-0.091
	(0.064)	(0.059)	(0.060)	(0.059)	(0.063)	(0.060)	(0.053)	(0.055)	(0.066)	(0.066)	(0.057)	(0.055)	(0.061)	(0.057)	(0.057)	(0.059)	(0.048)	(0.050)	(0.059)	(0.061)
Inviter with Random Guest List	0.026	-0.112*	0.071	-0.075	-0.040	-0.001	0.112**	-0.071	0.111*	-0.008	-0.007	-0.073	0.065	-0.076	0.017	0.048	0.088*	-0.086*	-0.068	-0.053
	(0.062)	(0.058)	(0.061)	(0.059)	(0.061)	(0.063)	(0.052)	(0.056)	(0.065)	(0.064)	(0.055)	(0.057)	(0.058)	(0.060)	(0.059)	(0.058)	(0.048)	(0.051)	(0.059)	(0.063)
Observations	1521	1525	1520	1524	1522	1528	1522	1527	1522	1527	1522	1527	1521	1527	1521	1527	1522	1528	1522	1527
Control Mean	1.751	1.651	1.615	1.615	1.854	1.879	1.393	1.594	2.875	3.119	1.488	1.608	1.570	1.681	2.931	3.008	1.340	1.480	1.708	1.728
P-values:																				
Pooled: High-SES vs. Low-SES Guest List	0.837	0.242	0.593	0.112	0.871	0.011**	0.983	0.010**	0.821	0.780	0.033**	0.060*	0.457	0.308	0.243	0.833	0.176	0.019**	0.137	0.065*
Pooled: Voucher vs. No Voucher	0.497	0.680	0.307	0.680	0.595	0.926	0.469	0.606	0.352	0.502	0.289	0.573	0.087*	0.682	0.637	0.509	0.933	0.737	0.315	0.626

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Each column represents a question from the CESDR-10, a 10-item revised version of the Center of Epidemiologic Studies Depression Scale. For each question, participants self-report the frequency with with they experienced the feelings that the question describes in the past two weeks. Each question is answered on a scale of 0-3: (0) Never, (1) Rarely, (2) Sometimes, (3) Always.

Table D.3: Treatment Effects on Depression

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	CESDR-10 Score		Any Depression		Mild Depression		Moderate Depression		Severe Depression	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects										
All Inviters	0.077	-0.646**	0.009	-0.083***	0.010	-0.071***	0.018	-0.014	-0.022**	-0.001
	(0.311)	(0.328)	(0.026)	(0.027)	(0.024)	(0.026)	(0.016)	(0.017)	(0.010)	(0.006)
Panel B: Pooled Inviter by Guest List Treatment Effects										
Inviter with High-SES Guest List	-0.098	-0.137	-0.000	-0.045	0.017	-0.049	-0.003	-0.004	-0.024**	0.003
	(0.367)	(0.396)	(0.031)	(0.033)	(0.029)	(0.031)	(0.018)	(0.021)	(0.011)	(0.008)
Inviter with Low-SES Guest List	0.252	-0.962**	0.018	-0.104***	0.009	-0.094***	0.029	-0.014	-0.020*	-0.006
	(0.384)	(0.393)	(0.032)	(0.032)	(0.029)	(0.030)	(0.020)	(0.020)	(0.012)	(0.006)
Inviter with Random Guest List	0.117	-0.768**	0.019	-0.092***	0.008	-0.066**	0.032	-0.023	-0.022**	0.000
	(0.379)	(0.388)	(0.032)	(0.032)	(0.029)	(0.031)	(0.020)	(0.019)	(0.011)	(0.008)
Observations	1522	1528	1522	1528	1522	1528	1522	1528	1522	1528
Control Mean	7.103	7.303	0.297	0.361	0.194	0.261	0.066	0.090	0.037	0.011
P-values:										
Pooled: High-SES vs. Low-SES Guest List	0.344	0.029**	0.569	0.060*	0.762	0.111	0.091*	0.637	0.568	0.221
Pooled: Voucher vs. No Voucher	0.618	0.645	0.260	0.784	0.203	0.102	0.840	0.066*	0.882	0.794

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Depression is measured using the CESDR-10, a 10-item revised version of the Center of Epidemiologic Studies Depression Scale. For each question, participants self-report the frequency with with they experienced the feelings that the question describes in the past two weeks. Each question is answered on a scale of 0-3: (0) Never, (1) Rarely, (2) Sometimes, (3) Always. The CESDR-10 score (column (1)) is the aggregate of these scores across all ten questions. Any depression (column (2)) is a score >=10; Mild depression (column (3)) is a score from 10-15; Moderate depression (column (4)) is a score from 16-21; and Severe depression (column (5)) is a score >=22.

D.1 Treatment Effect Channels: Details

D.1.1 Reduced Lending

Inviters across all sub-treatments report lending money or goods to fewer people in the past year than Control, particularly husband's relatives, which may be in part responsible for having the liquidity for more food consumption (Appendix Table D.23). See Appendix Section D.1.1 for more details. This is driven entirely by a reduction in lending to baseline network members and husband's relatives (Appendix Table D.19). Inviters lend less to the same people who they lent to the previous year, and are less likely to give loans for the first time to members of the baseline network. There is actually an *increase* in the number of borrowers who are new to the network, but this is more than offset by the reduction in the number of baseline network-members to whom they lend, leading to an overall decrease in the number of borrowers.

The reduction in borrowers is explained by weak- and medium-strength ties, while the number of strong ties who Inviters lend to is unaffected (Appendix Table D.22).[52] Consistent with a reshaping of risk-sharing networks, Inviters borrow from more people who are new to the network, and from fewer people who they listed in their baseline networks, even though the total number of people or amount of borrowing does not change (Appendix Table D.24).

D.1.2 Reduced Risk of Lean Season Consumption

Inviters Diversify Their Crop Portfolios

Inviters undertake a striking transformation in their agricultural production function by diversifying their crop portfolios, reducing the risk they face during the lean season associated with the upcoming harvest. Inviters' primary method of crop diversification is to diversify the *types* of crops that they cultivate. I consider five classes of crops: maize (the staple grain), other starches, fruits and vegetables, legumes, and cash crops.[53] The probability that Inviters grow crops from at least three of these classes increases by 4.7 percentage points (54%, $p < 0.01$) (Table D.8). This implies that Inviters are less vulnerable to environmental or market shocks that affect any single crop *or* any single class of crops.

All Inviters are more likely to grow fruits and vegetables, while Inviters with the High-

[52] Inviters report that they have more network members who they would feel very comfortable to lend to, indicating that the reduction in lending is not driven by a decrease in willingness to lend broadly speaking, but instead a decrease in lending to people who they are uncomfortable lending to (Appendix Table D.7). Compared to Control, Inviters reduce lending to their husbands' relatives but not their own friends, suggesting that this pattern is more likely explained by Inviters foregoing "kinship taxes" altogether, rather than Inviters replacing friends who need loans with friends who do not need loans.

[53] More details on how I classify crops are in Appendix Section A.4.

Table D.4: Business Details

	(1)	(2)	(3)	(4)	(5)	(6)
	Any Self-Employment Activity		New Household Self-Employment Activity		Copied Self-Employment Activity	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.053**	0.017	0.034	-0.025	0.014	-0.007
	(0.022)	(0.025)	(0.026)	(0.028)	(0.013)	(0.019)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	0.092***	0.049	0.082**	-0.033	0.038**	0.011
	(0.028)	(0.031)	(0.032)	(0.034)	(0.018)	(0.024)
Inviter with Low-SES Guest List	0.013	-0.007	-0.017	-0.037	-0.018	-0.045**
	(0.027)	(0.030)	(0.031)	(0.033)	(0.015)	(0.021)
Inviter with Random Guest List	0.039	0.006	0.023	-0.014	0.017	0.010
	(0.028)	(0.030)	(0.032)	(0.034)	(0.017)	(0.023)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	0.091**	0.031	0.075*	-0.052	0.044*	-0.002
	(0.036)	(0.037)	(0.040)	(0.041)	(0.024)	(0.029)
High-SES with Voucher	0.094***	0.064	0.076*	-0.003	0.034	0.028
	(0.036)	(0.040)	(0.041)	(0.042)	(0.023)	(0.030)
Low-SES without Voucher	-0.001	-0.029	-0.022	-0.053	-0.024	-0.075***
	(0.032)	(0.036)	(0.039)	(0.040)	(0.017)	(0.022)
Low-SES with Voucher	0.022	0.014	-0.023	-0.020	-0.009	-0.008
	(0.034)	(0.037)	(0.037)	(0.042)	(0.019)	(0.028)
Random without Voucher	0.042	-0.011	0.007	-0.054	0.035	-0.003
	(0.035)	(0.036)	(0.040)	(0.040)	(0.023)	(0.028)
Random with Voucher	0.028	0.012	0.020	0.022	-0.001	0.029
	(0.033)	(0.037)	(0.039)	(0.043)	(0.020)	(0.031)
Observations	1522	1528	1522	1528	1522	1528
Control Mean	0.180	0.237	0.252	0.335	0.048	0.116
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.005***	0.062*	0.002***	0.889	0.002***	0.010**
Pooled: Voucher vs. No Voucher	0.952	0.210	0.835	0.074*	0.490	0.017**
High-SES: Voucher vs. No Voucher	0.947	0.451	0.995	0.307	0.749	0.390
Low-SES: Voucher vs. No Voucher	0.531	0.286	0.992	0.483	0.438	0.013**
Random: Voucher vs. No Voucher	0.724	0.585	0.776	0.112	0.162	0.355

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of columns (1)-(2) is an indicator variable that takes the value of 1 if the respondent reports that she earned income through self-employment in the past month. The outcome of columns (3)-(4) is an indicator variable that takes the value of 1 if there is any household member who earned income in the past month through a self-employment activity that they did not earn income from in any prior survey. The outcome of columns (5)-(6) is an indicator that takes the value of 1 if the household began engaging in the same self-employment activity as one of the women on her sharing list, and she did not report participating in this same activity in a prior survey. To construct this measure for the Control group, we construct a placebo Sharing List, so that the Control mean effectively represents the rate of switching into one neighbor's self-employment activities that we should expect absent the intervention.

Table D.5: Piecework Employment

	(1) Piecework Jobs Per Month: Growing Season	(2) Piecework Jobs Per Month: Harvest	(3) Piecework Contacts Per Job	(4)	(5) Number of Piecework Employers	(6)
	1 Year	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	-0.345	-0.082	-0.076	-0.077	-0.064*	-0.011
	(0.363)	(0.318)	(0.101)	(0.112)	(0.039)	(0.044)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	-0.541	-0.152	-0.163	-0.078	-0.139***	-0.010
	(0.441)	(0.395)	(0.122)	(0.138)	(0.043)	(0.055)
Inviter with Low-SES Guest List	-0.174	-0.048	0.028	-0.066	0.001	0.029
	(0.437)	(0.383)	(0.120)	(0.136)	(0.049)	(0.055)
Inviter with Random Guest List	-0.297	-0.063	-0.099	-0.061	-0.040	-0.010
	(0.436)	(0.383)	(0.117)	(0.129)	(0.046)	(0.053)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	-0.287	-0.024	-0.256*	0.124	-0.184***	0.034
	(0.533)	(0.462)	(0.143)	(0.179)	(0.048)	(0.070)
High-SES with Voucher	-0.802	-0.350	-0.064	-0.280*	-0.093*	-0.067
	(0.563)	(0.520)	(0.159)	(0.161)	(0.056)	(0.070)
Low-SES without Voucher	0.149	-0.021	-0.050	0.049	-0.019	0.019
	(0.524)	(0.444)	(0.145)	(0.165)	(0.056)	(0.066)
Low-SES with Voucher	-0.340	0.095	0.093	-0.097	0.020	0.060
	(0.549)	(0.494)	(0.151)	(0.172)	(0.067)	(0.072)
Random without Voucher	-0.764	-0.248	-0.168	-0.207	-0.024	-0.019
	(0.514)	(0.466)	(0.140)	(0.151)	(0.058)	(0.066)
Random with Voucher	0.314	0.129	-0.053	0.131	-0.062	0.006
	(0.549)	(0.468)	(0.142)	(0.160)	(0.055)	(0.065)
Observations	1515	1506	1522	1528	1522	1528
Control Mean	6.013	5.369	1.058	1.873	0.454	0.789
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.401	0.789	0.105	0.927	0.002***	0.489
Pooled: Voucher vs. No Voucher	0.997	0.800	0.070*	0.465	0.306	0.889
High-SES: Voucher vs. No Voucher	0.417	0.571	0.269	0.036**	0.110	0.230
Low-SES: Voucher vs. No Voucher	0.417	0.828	0.385	0.448	0.588	0.615
Random: Voucher vs. No Voucher	0.071*	0.483	0.451	0.044**	0.545	0.737

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the number of days that the respondent reports working in piecework per month during the growing season (overlaps the lean season). The outcome of column (2) is the number of days that the respondent reports working in piecework per month during the harvest. The outcome of column (3)-(4) is the number of people the respondents typically asks before finding a piecework job. The outcome of columns (5)-(6) is the number of recurring piecework employers the respondent works for.

Table D.6: Treatment Effects on Crop Diversification

	(1) Grows Maize 1 Year	(2) Grows Other Starches 1 Year	(3) Grows a Legume 1 Year	(4) Grows a Fruit or Vegetable 1 Year	(5) Grows a Cash Crop 1 Year
Panel A: Pooled Inviter Treatment Effects					
All Inviters	-0.009	0.003	0.018	0.013*	0.014
	(0.020)	(0.016)	(0.027)	(0.007)	(0.017)
Panel B: Pooled Inviter by Guest List Treatment Effects					
Inviter with High-SES Guest List	-0.014	0.009	0.018	0.015*	0.053**
	(0.026)	(0.020)	(0.033)	(0.009)	(0.022)
Inviter with Low-SES Guest List	0.002	-0.006	-0.016	0.004	-0.002
	(0.025)	(0.019)	(0.034)	(0.008)	(0.021)
Inviter with Random Guest List	0.005	0.009	0.069**	0.018*	-0.005
	(0.025)	(0.020)	(0.033)	(0.010)	(0.021)
Panel C: Inviter Sub-Group Treatment Effects					
High-SES without Voucher	-0.018	-0.010	0.005	0.027**	0.079***
	(0.033)	(0.023)	(0.041)	(0.012)	(0.029)
High-SES with Voucher	-0.014	0.031	0.018	0.009	0.017
	(0.033)	(0.026)	(0.042)	(0.011)	(0.027)
Low-SES without Voucher	0.005	0.005	-0.012	0.019*	0.003
	(0.033)	(0.023)	(0.043)	(0.012)	(0.027)
Low-SES with Voucher	0.000	0.001	-0.024	0.005	-0.001
	(0.030)	(0.022)	(0.042)	(0.007)	(0.025)
Random without Voucher	0.026	0.007	0.076*	0.024**	-0.013
	(0.031)	(0.024)	(0.040)	(0.012)	(0.025)
Random with Voucher	-0.011	0.018	0.057	0.015	0.008
	(0.030)	(0.024)	(0.042)	(0.010)	(0.027)
Observations	1528	1528	1528	1528	1528
Control Mean	0.776	0.084	0.478	0.011	0.103
P-values:					
Pooled: High-SES vs. Low-SES Guest List	0.544	0.473	0.309	0.232	0.016**
Pooled: Voucher vs. No Voucher	0.475	0.484	0.664	0.109	0.460
High-SES: Voucher vs. No Voucher	0.913	0.158	0.789	0.236	0.069*
Low-SES: Voucher vs. No Voucher	0.902	0.868	0.802	0.236	0.902
Random: Voucher vs. No Voucher	0.295	0.690	0.699	0.567	0.478

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for each column indicates if the participant grew at least one crop in the column's given crop category. Other starches include: cassava, sorghum, rice, Irish potatoes, and sweet potatoes. Legumes include: soy beans, plain beans, groudnuts, pigeon peas, and cowpeas. Fruits and vegetables include: bananas, leafy greens, tomatoes, onions and pumpkins. Cash crops include: sugar cane, tobacco, and sunflower seeds.

Table D.7: Treatment Effects on Potential for Risk-Sharing in the Network

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Number Very Comfortable Lending		Number Very Comfortable Borrowing		ROSCA Member		VSLA Member	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects								
All Inviters	0.060	0.151	0.006	0.192	-0.009	0.005	0.002	0.004
	(0.107)	(0.126)	(0.103)	(0.124)	(0.021)	(0.015)	(0.022)	(0.025)
Panel B: Pooled Inviter by Guest List Treatment Effects								
Inviter with High-SES Guest List	0.068	0.283*	0.034	0.316**	0.006	-0.015	0.029	0.037
	(0.132)	(0.162)	(0.129)	(0.161)	(0.026)	(0.017)	(0.027)	(0.030)
Inviter with Low-SES Guest List	0.018	-0.055	-0.042	-0.005	-0.014	0.023	-0.022	-0.015
	(0.127)	(0.153)	(0.121)	(0.151)	(0.026)	(0.019)	(0.029)	(0.030)
Inviter with Random Guest List	0.114	0.157	0.048	0.172	-0.019	0.008	0.001	-0.004
	(0.124)	(0.154)	(0.121)	(0.153)	(0.025)	(0.019)	(0.028)	(0.031)
Panel C: Inviter Sub-Group Treatment Effects								
High-SES without Voucher	0.086	0.304	0.089	0.327	0.009	-0.012	-0.007	0.045
	(0.162)	(0.216)	(0.162)	(0.216)	(0.032)	(0.023)	(0.035)	(0.038)
High-SES with Voucher	0.067	0.287	-0.025	0.286	-0.001	-0.024	0.062*	0.024
	(0.161)	(0.187)	(0.152)	(0.184)	(0.031)	(0.019)	(0.033)	(0.038)
Low-SES without Voucher	0.120	-0.127	0.070	-0.055	-0.025	0.029	0.002	0.009
	(0.153)	(0.190)	(0.147)	(0.186)	(0.031)	(0.024)	(0.037)	(0.037)
Low-SES with Voucher	-0.058	0.092	-0.129	0.092	0.006	0.014	-0.046	-0.038
	(0.153)	(0.186)	(0.146)	(0.185)	(0.031)	(0.023)	(0.035)	(0.039)
Random without Voucher	0.180	0.053	0.117	0.040	-0.058**	0.011	-0.008	-0.014
	(0.154)	(0.177)	(0.150)	(0.175)	(0.029)	(0.023)	(0.036)	(0.039)
Random with Voucher	-0.016	0.329	-0.078	0.351*	0.010	0.002	0.002	0.005
	(0.145)	(0.205)	(0.143)	(0.204)	(0.032)	(0.022)	(0.032)	(0.038)
Observations	1585	1528	1585	1528	1522	1528	1522	1528
Control Mean	3.808	3.979	3.846	3.997	0.164	0.069	0.430	0.459
P-values:								
Pooled: High-SES vs. Low-SES Guest List	0.691	0.039**	0.538	0.049**	0.438	0.032**	0.073*	0.093*
Pooled: Voucher vs. No Voucher	0.069*	0.278	0.037**	0.201	0.100	0.517	0.676	0.456
High-SES: Voucher vs. No Voucher	0.917	0.946	0.523	0.864	0.785	0.628	0.083*	0.627
Low-SES: Voucher vs. No Voucher	0.289	0.317	0.217	0.495	0.369	0.583	0.256	0.277
Random: Voucher vs. No Voucher	0.231	0.219	0.229	0.164	0.049**	0.741	0.809	0.682

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. For each network tie, participants report their comfort in borrowing from or lending to that person on a Likert scale. The outcomes in columns (1) and (2) report the number of those friends for whom that participant said she would be very comfortable lending to. The outcomes in columns (3) and (4) report the number of those friends for whom the participant said she would be very comfortable borrowing from. The outcome of columns (5) and (6) indicates membership with a rotating savings and credit association, a semi-formal savings and borrowing group. The outcome of columns (7) and (8) indicates membership with a Village Savings and Loan Association, a semi-formal savings and borrowing group.

SES Guest List are more likely to grow cash crops (Appendix Table D.6). The value of cash crop yields increases among Inviters with the High-SES Guest List by 30 USD (122% of the Control mean, $p < 0.05$), with a noisy off-setting reduction in the value of subsistence crop yields (a 31 USD reduction, which is a 10% decrease over the Control mean that is not statistically distinguishable) (Table D.8).

Inviters Report Financially Reliable Network Relationships

If Inviters have networks ties that they are confident they can rely on in case of a negative shock, they may be more comfortable consuming rather than saving. I create a PCA index of "potential for risk-sharing" using variables denoting subjective comfort risk-sharing with network ties and participation in organized risk-sharing groups. In Table D.9, I report the Potential for Risk-Sharing index and two representative variables entering the index.[54] Among Inviters with the High-SES Guest List, the Potential for Risk-Sharing index increases by 0.105 SD ($p < 0.10$). The index is not significantly different from Control among Inviters with the Random Guest List, but this masks Voucher-level heterogeneity. The index increases among Inviters with the Random Guest List and the Voucher by 0.183 SD, which is statistically distinguishable from Control ($p < 0.05$), and from Inviters with the Random Guest List without the Voucher ($p < 0.05$).

There are two narrow groups of Inviters who say they would be very comfortable borrowing and lending with more people in their networks: Inviters with the High-SES Guest List and the Voucher, and Inviters with the Random Guest List and the Voucher. Though not definitive, when households have exogenous access to high-SES links, they might use the Voucher to mitigate their vulnerability to shocks by establishing relationships with financially reliable ties. Indeed, Inviters with the Voucher (pooled across all three groups) are more likely than Inviters without the Voucher to say the reason they selected the Guests they invited is because they are generous (Appendix Table C.6).

Inviters are 26% more likely to join "Welfare Groups" ($p < 0.10$, Table D.9), home-grown self-help groups that are not operated by any formal institution. The most common of these groups functionally operates like informal health and life insurance.[55] This increase is driven by Inviters with the Random Guest List and the Voucher, who are 63% more likely to be Welfare Group members ($p < 0.01$).

D.1.3 Other Plausible Mediators for Increased Food Security

There are several mediators for increased food security that are plausible and for which I have suggestive evidence, but do not have the data to make conclusive statements. Thus, the following discussion is well-motivated theoretically, but speculative.

[54] I report the other variables entering the index in Appendix Table D.7.
[55] See Section D.1.3 for details on how 'Welfare Groups' operate.

Table D.8: Treatment Effects on Crop Diversification

	(1) Number of Crops Cultivated	(2) Cultivated >= 3 Crop Classes	(3) Grows a Cash Crop	(4) Cash Crop Yields USD Value	(5) Subsistence Yields USD Value
	1 Year	1 Year	1 Year	1 Year	1 Year
Panel A: Pooled Inviter Treatment Effects					
All Inviters	0.094	0.042***	0.014	12.230	-10.777
	(0.062)	(0.016)	(0.017)	(8.187)	(23.357)
Panel B: Pooled Inviter by Guest List Treatment Effects					
Inviter with High-SES Guest List	0.153*	0.059***	0.053**	31.525***	-7.974
	(0.084)	(0.021)	(0.022)	(11.954)	(29.357)
Inviter with Low-SES Guest List	0.002	0.022	-0.002	10.121	-28.174
	(0.080)	(0.020)	(0.021)	(11.017)	(27.100)
Inviter with Random Guest List	0.167**	0.045**	-0.005	-5.890	2.778
	(0.079)	(0.021)	(0.021)	(8.661)	(27.692)
Panel C: Inviter Sub-Group Treatment Effects					
High-SES without Voucher	0.140	0.055**	0.079***	28.901**	-46.341
	(0.101)	(0.027)	(0.029)	(14.479)	(32.377)
High-SES with Voucher	0.099	0.051*	0.013	36.455**	-6.161
	(0.101)	(0.027)	(0.026)	(16.996)	(35.034)
Low-SES without Voucher	0.125	0.026	0.002	0.584	-36.458
	(0.110)	(0.026)	(0.027)	(12.215)	(32.200)
Low-SES with Voucher	-0.032	0.018	-0.002	23.413	-30.018
	(0.090)	(0.023)	(0.025)	(16.015)	(33.550)
Random without Voucher	0.169*	0.048*	-0.013	-13.203	24.828
	(0.095)	(0.027)	(0.025)	(8.929)	(33.796)
Random with Voucher	0.219**	0.039	0.008	4.193	-33.330
	(0.102)	(0.027)	(0.027)	(11.865)	(33.639)
Observations	1528	1528	1528	1528	1528
Control Mean	1.604	0.087	0.103	24.487	295.187
P-values:					
Pooled: High-SES vs. Low-SES Guest List	0.100	0.104	0.016**	0.109	0.458
Pooled: Voucher vs. No Voucher	0.423	0.574	0.460	0.134	0.907
High-SES: Voucher vs. No Voucher	0.739	0.883	0.050*	0.716	0.286
Low-SES: Voucher vs. No Voucher	0.207	0.781	0.901	0.215	0.851
Random: Voucher vs. No Voucher	0.680	0.782	0.473	0.153	0.126

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for column (1) measures the number of individual crops that each participant cultivates. I then sort crops into crop-classes, where classes include: maize, other starches, fruits and vegetables, legumes, and cash crops. The outcome for column (2) is an indicator for cultivating at least one crop from at least three of the crop classes. The outcoem for column (3) is an indicator for cultivating at least one cash crop (sugar cane, tobacco, and sunflower seeds). I multiply market rates in agricultural reports by yields for the most high-yield crops: tobacco, sugar cane, sunflower seeds for Cash Crops (column (4)); and maize, soy beans, groundnuts, beans, sweet potatoes, and pumpkin for Subsistence Crops (column (5)). The outcome for column (3) of Table 3 is the sum of the outcomes for columns (4) and (5). Each of these USD value yield outcomes is winsorized at the 1st and 99th percentiles.

Table D.9: Treatment Effects on Potential for Risk-Sharing in the Network

	(1)	(2)	(3)	(4)	(5)
	Potential for Risk-Sharing Index		Number Very Comfortable Risk-Sharing		Welfare Group Member
	1 Month	1 Year	1 Month	1 Year	1 Year
Panel A: Pooled Inviter Treatment Effects					
All Inviters	0.019	0.077	0.082	0.166	0.035*
	(0.045)	(0.056)	(0.103)	(0.125)	(0.019)
Panel B: Pooled Inviter by Guest List Treatment Effects					
Inviter with High-SES Guest List	0.022	0.133*	0.140	0.284*	0.039
	(0.055)	(0.072)	(0.127)	(0.161)	(0.024)
Inviter with Low-SES Guest List	-0.006	-0.013	0.006	-0.046	0.018
	(0.053)	(0.067)	(0.123)	(0.152)	(0.024)
Inviter with Random Guest List	0.047	0.076	0.112	0.164	0.051**
	(0.053)	(0.069)	(0.122)	(0.154)	(0.025)
Panel C: Inviter Sub-Group Treatment Effects					
High-SES without Voucher	0.054	0.138	0.251	0.291	0.039
	(0.070)	(0.095)	(0.154)	(0.213)	(0.031)
High-SES with Voucher	-0.013	0.135	0.029	0.291	0.031
	(0.064)	(0.084)	(0.153)	(0.186)	(0.031)
Low-SES without Voucher	0.046	-0.031	0.122	-0.108	0.026
	(0.065)	(0.084)	(0.150)	(0.191)	(0.030)
Low-SES with Voucher	-0.053	0.046	-0.081	0.082	0.006
	(0.064)	(0.081)	(0.147)	(0.184)	(0.029)
Random without Voucher	0.078	0.026	0.190	0.027	0.019
	(0.064)	(0.078)	(0.151)	(0.175)	(0.031)
Random with Voucher	0.007	0.152	-0.008	0.359*	0.068**
	(0.063)	(0.093)	(0.142)	(0.204)	(0.031)
Observations	1522	1528	1585	1528	1528
Control Mean	0.002	-0.001	3.638	3.868	0.127
P-values:					
Pooled: High-SES vs. Low-SES Guest List	0.599	0.043**	0.272	0.043**	0.418
Pooled: Voucher vs. No Voucher	0.039**	0.281	0.011**	0.201	0.618
High-SES: Voucher vs. No Voucher	0.377	0.983	0.199	0.999	0.823
Low-SES: Voucher vs. No Voucher	0.164	0.413	0.213	0.384	0.560
Random: Voucher vs. No Voucher	0.303	0.213	0.222	0.138	0.180

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The Potential for Risk-Sharing Index (columns (1) and (2)) is a PCA index of variables subjective comfort risk-sharing in the network and participation in organized risk-sharing groups. Variables of subjective comfort risk-sharing in the network are: the number of people in the network the participant feels very comfortable lending to; the number of people in the network the participant feels very comfortable borrowing from; and the number of people in the network the participant feels very comfortable both borrowing from and lending to. Variables of participation in organized risk-sharing groups are indicators for participation in each of the following: welfare groups (endline only), ROSCA, and VSLA. For each network tie, participants report their comfort in borrowing from or lending to that person on a Likert scale. The outcomes in columns (3) and (4) report the number of those friends for whom that participant said she would be very comfortable with both lending and borrowing. Welfare Group, the outcome in column (5), is an indicator for if the respondent reports being in a home-grown self-help group, where the most common version is a mutual-insurance scheme. Other common groups involve methods of pooling resources for group purchases.

Reduced Household and Farm Expenditures

Inviters change the crops that they cultivate, so Inviters may face different input costs throughout the agricultural season. While I do not collect detailed data on farm expenditures, I do measure fertilizer use and adoption of new agricultural technologies, which both noisily decrease among Inviters (Table D.10 columns (1)-(3)). This set of results suggest that farm inputs may change and, since Inviters do not experience a reduction in total farm yields, that Inviters may use farm inputs more efficiently.

Because Inviters are more likely to join "Welfare Groups", they may incur fewer household expenditures. The most common groups operate like informal health or life insurance. Households agree in advance to an amount that they will contribute to any member who experiences illness or death of a family member. In some groups, each member contributes at the time of the shock, and in other groups, all the members contribute to a central pot of money on a regular basis which then becomes the pool of money to draw on when a household experiences a shock. Affected members do not incur any debt, and contributing members are not paid in return, but reciprocity is expected in the event that another member faces a shock. In a variation of these groups called 'Chi Sick', members agree to serve as one another's guardians in the hospital in case of illness, insuring againt non-financial risks associated with health shocks. In other variants, households pool resources to purchase goods in bulk as a group, thereby benefiting each member with lower per-unit prices. In "Kitchen Top-Up Groups", women pool resources to purchase kitchenware, which they then share as a group. In "Grocery Top-Up Groups", women pool resources to purchases groceries in bulk, enabling them to purchase goods at a lower per-unit price.

Reduced Savings

Households may have consumed more by withdrawing from their savings. Maize stores, the measure of savings I collect, is a reliable measure since it is a standard method of holding assets that enumerators visually verify. However, since the Inviter treatment changes the method through which people earn, the method that they save could change as well. Reduced maize stores could simply be a reflection of crop diversification. There is a noisy reduction in maize stores (15 fewer kilograms, 10% of the Control mean), but not statistically significant (Table D.10 columns (4)-(5)).

Table D.10: Treatment Effects on Farm Inputs and Maize Stores

	(1) Full Fertilizer Use 1 Year	(2) Partial Fertilizer Use 1 Year	(3) Adopted New Ag. Technology 1 Year	(4) Maize Stores in KGs 1 Month	(5) Maize Stores in KGs 1 Year
Panel A: Pooled Inviter Treatment Effects					
All Inviters	-0.027	-0.022	-0.034	0.115	-10.047
	(0.024)	(0.027)	(0.025)	(3.745)	(10.958)
Panel B: Pooled Inviter by Guest List Treatment Effects					
Inviter with High-SES Guest List	-0.015	-0.007	-0.022	-0.615	0.184
	(0.029)	(0.032)	(0.029)	(4.728)	(14.570)
Inviter with Low-SES Guest List	-0.044	-0.051	-0.038	-6.505	-15.931
	(0.028)	(0.034)	(0.031)	(4.419)	(12.840)
Inviter with Random Guest List	-0.019	0.006	-0.049*	6.662	-19.019
	(0.029)	(0.032)	(0.027)	(4.486)	(13.030)
Panel C: Inviter Sub-Group Treatment Effects					
High-SES without Voucher	-0.034	-0.019	-0.011	-4.487	-17.853
	(0.036)	(0.041)	(0.036)	(5.450)	(15.766)
High-SES with Voucher	-0.007	0.002	-0.036	3.710	9.107
	(0.036)	(0.040)	(0.035)	(6.297)	(19.715)
Low-SES without Voucher	-0.068*	-0.052	-0.045	-4.270	-14.287
	(0.034)	(0.042)	(0.035)	(5.436)	(16.214)
Low-SES with Voucher	-0.030	-0.037	-0.030	-7.214	-18.855
	(0.035)	(0.043)	(0.041)	(5.352)	(15.178)
Random without Voucher	0.031	0.013	-0.053*	9.173*	-15.370
	(0.035)	(0.041)	(0.030)	(5.356)	(14.789)
Random with Voucher	-0.072**	0.005	-0.047	4.112	-25.587
	(0.036)	(0.040)	(0.032)	(5.787)	(17.369)
Observations	1528	1528	1401	1522	1528
Control Mean	0.277	0.599	0.154	44.273	144.202
P-values:					
Pooled: High-SES vs. Low-SES Guest List	0.315	0.188	0.576	0.178	0.259
Pooled: Voucher vs. No Voucher	0.616	0.924	0.952	0.977	0.654
High-SES: Voucher vs. No Voucher	0.518	0.641	0.531	0.238	0.204
Low-SES: Voucher vs. No Voucher	0.332	0.771	0.731	0.609	0.785
Random: Voucher vs. No Voucher	0.012**	0.863	0.849	0.441	0.576

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Column (1) indicates if the respondent reports using fertilizer on all of her crops that require fertilizer, while column (2) indicates if she reports using fertilizer on at least some of her crops that require fertilizer. Column (3) indicates if the participant reports adopting a new agricultural technology this year, other than fertilizer use or crop diversification. Columns (4)-(5) are the kilograms of maize that the respondent reports having in stores at the time of the survey.

Table D.11: Treatment Effects on Borrowing

	(1)	(2)	(3)	(4)	(5)
	Borrowed Last Month		Borrowed Last Six Months	Amount Borrowed Last Month	
	1 Month	1 Year	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects					
All Inviters	-0.022	-0.015	-0.030	1.025	-0.984
	(0.031)	(0.028)	(0.023)	(0.906)	(1.805)
Panel B: Pooled Inviter by Guest List Treatment Effects					
Inviter with High-SES Guest List	0.006	-0.015	-0.044	1.182	-1.097
	(0.038)	(0.034)	(0.028)	(1.240)	(1.864)
Inviter with Low-SES Guest List	-0.024	-0.030	-0.040	0.657	-1.469
	(0.038)	(0.034)	(0.028)	(1.039)	(2.310)
Inviter with Random Guest List	-0.043	0.006	-0.006	1.271	-0.415
	(0.038)	(0.034)	(0.029)	(1.283)	(1.960)
Panel C: Inviter Sub-Group Treatment Effects					
High-SES without Voucher	-0.044	-0.030	-0.063*	0.788	-0.947
	(0.046)	(0.043)	(0.035)	(1.473)	(2.531)
High-SES with Voucher	0.045	0.008	-0.030	1.229	-1.468
	(0.046)	(0.041)	(0.035)	(1.599)	(1.930)
Low-SES without Voucher	-0.038	-0.085**	-0.086**	-0.275	-2.818
	(0.047)	(0.042)	(0.036)	(1.036)	(2.427)
Low-SES with Voucher	-0.021	0.029	0.016	1.196	-0.779
	(0.047)	(0.042)	(0.034)	(1.543)	(2.945)
Random without Voucher	-0.093**	-0.033	-0.036	2.358	-1.979
	(0.046)	(0.043)	(0.037)	(2.111)	(1.987)
Random with Voucher	-0.018	0.047	0.022	-0.359	0.806
	(0.049)	(0.042)	(0.035)	(1.091)	(2.690)
Observations	1310	1528	1528	1231	1244
Control Mean	0.483	0.639	0.810	4.402	10.149
P-values:					
Pooled: High-SES vs. Low-SES Guest List	0.419	0.653	0.877	0.674	0.836
Pooled: Voucher vs. No Voucher	0.047**	0.004***	0.010**	0.784	0.254
High-SES: Voucher vs. No Voucher	0.085*	0.434	0.438	0.811	0.824
Low-SES: Voucher vs. No Voucher	0.747	0.019**	0.014**	0.339	0.432
Random: Voucher vs. No Voucher	0.168	0.104	0.174	0.209	0.239

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Column2 (1)-(2) indicates if the respondent reports taking out any loan in the past month. Column (3) indicates if the respondent reports taking out any loan in the past six months. Columns (4)-(5) is the total amount she reports borrowing in the past month, reported in USD.

Table D.12: Treatment Effects on Household Bargaining

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Child Schooling		Child Health		Child Discipline		Fertility			Household Savings
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects										
All Inviters	-0.029	-0.042	0.017	-0.024	-0.006	-0.034	0.000	-0.016	-0.007	-0.060**
	(0.028)	(0.027)	(0.027)	(0.025)	(0.027)	(0.028)	(0.028)	(0.028)	(0.027)	(0.029)
Panel B: Pooled Inviter by Guest List Treatment Effects										
Inviter with High-SES Guest List	-0.035	-0.030	0.019	-0.032	-0.037	-0.041	-0.007	-0.017	-0.028	-0.049
	(0.034)	(0.033)	(0.033)	(0.031)	(0.034)	(0.034)	(0.034)	(0.034)	(0.033)	(0.035)
Inviter with Low-SES Guest List	-0.045	-0.041	0.008	-0.027	0.009	0.001	-0.009	-0.011	-0.017	-0.039
	(0.033)	(0.033)	(0.033)	(0.032)	(0.033)	(0.035)	(0.034)	(0.034)	(0.033)	(0.034)
Inviter with Random Guest List	-0.011	-0.055	0.034	-0.011	0.017	-0.060*	0.020	-0.022	0.029	-0.095***
	(0.033)	(0.034)	(0.033)	(0.031)	(0.033)	(0.035)	(0.034)	(0.035)	(0.034)	(0.035)
Panel C: Inviter Sub-Group Treatment Effects										
High-SES without Voucher	-0.015	-0.088**	0.005	-0.055	-0.008	-0.053	0.015	-0.027	-0.048	-0.055
	(0.041)	(0.042)	(0.041)	(0.040)	(0.042)	(0.043)	(0.043)	(0.042)	(0.040)	(0.044)
High-SES with Voucher	-0.054	0.025	0.035	-0.017	-0.065	-0.012	-0.041	-0.004	-0.001	-0.033
	(0.042)	(0.040)	(0.039)	(0.037)	(0.042)	(0.042)	(0.041)	(0.042)	(0.041)	(0.043)
Low-SES without Voucher	-0.071*	-0.048	-0.017	-0.053	-0.009	0.005	-0.016	0.011	-0.015	-0.036
	(0.040)	(0.042)	(0.041)	(0.039)	(0.041)	(0.042)	(0.042)	(0.041)	(0.041)	(0.041)
Low-SES with Voucher	-0.020	-0.026	0.028	0.000	0.034	0.015	-0.015	-0.018	-0.008	-0.030
	(0.040)	(0.042)	(0.042)	(0.040)	(0.042)	(0.043)	(0.042)	(0.044)	(0.041)	(0.043)
Random without Voucher	0.005	-0.095**	0.055	-0.011	0.014	-0.092**	-0.013	-0.078*	0.022	-0.121***
	(0.040)	(0.042)	(0.040)	(0.037)	(0.042)	(0.043)	(0.042)	(0.044)	(0.042)	(0.043)
Random with Voucher	-0.037	-0.004	0.010	0.004	0.034	-0.005	0.052	0.057	0.053	-0.047
	(0.041)	(0.042)	(0.040)	(0.039)	(0.041)	(0.043)	(0.042)	(0.043)	(0.042)	(0.044)
Observations	1522	1528	1522	1528	1522	1528	1522	1528	1522	1528
Control Mean	0.462	0.644	0.623	0.734	0.517	0.641	0.398	0.570	0.387	0.583
P-values:										
Pooled: High-SES vs. Low-SES Guest List	0.777	0.749	0.742	0.863	0.180	0.214	0.949	0.861	0.732	0.766
Pooled: Voucher vs. No Voucher	0.643	0.007***	0.631	0.217	0.967	0.085*	0.967	0.145	0.388	0.273
High-SES: Voucher vs. No Voucher	0.408	0.015**	0.507	0.405	0.256	0.392	0.251	0.631	0.308	0.666
Low-SES: Voucher vs. No Voucher	0.253	0.648	0.331	0.245	0.364	0.832	0.989	0.542	0.892	0.903
Random: Voucher vs. No Voucher	0.360	0.057*	0.320	0.730	0.671	0.079*	0.180	0.008***	0.526	0.139

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Each column indicates if the respondent is herself a primary decision-maker in the household for each respective household issue.

Table D.13: Treatment Effects on Objective Changes to Relationships

	(1)	(2)	(3)	(4)	(5)	(6)
	Network Degree		New Network Ties		Dropped Network Ties	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.012	0.030	-0.078	0.202*	-0.078	0.235
	(0.094)	(0.117)	(0.091)	(0.114)	(0.082)	(0.154)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	-0.080	0.038	-0.156	0.202	-0.036	0.167
	(0.114)	(0.152)	(0.111)	(0.148)	(0.100)	(0.185)
Inviter with Low-SES Guest List	-0.018	-0.043	-0.052	0.135	-0.054	0.278
	(0.117)	(0.142)	(0.109)	(0.142)	(0.100)	(0.192)
Inviter with Random Guest List	0.159	0.074	0.017	0.218	-0.146	0.277
	(0.110)	(0.145)	(0.109)	(0.145)	(0.099)	(0.193)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	-0.118	0.207	-0.101	0.388**	0.036	0.135
	(0.141)	(0.206)	(0.143)	(0.196)	(0.123)	(0.233)
High-SES with Voucher	-0.017	-0.109	-0.185	-0.022	-0.092	0.170
	(0.139)	(0.180)	(0.128)	(0.179)	(0.122)	(0.225)
Low-SES without Voucher	0.056	-0.009	-0.030	0.015	-0.096	0.141
	(0.148)	(0.181)	(0.135)	(0.177)	(0.124)	(0.238)
Low-SES with Voucher	-0.081	-0.006	-0.047	0.226	0.055	0.350
	(0.141)	(0.174)	(0.131)	(0.183)	(0.123)	(0.241)
Random without Voucher	0.214	0.105	0.016	0.182	-0.145	0.046
	(0.136)	(0.172)	(0.128)	(0.173)	(0.122)	(0.224)
Random with Voucher	0.103	0.093	0.028	0.239	-0.116	0.440*
	(0.133)	(0.193)	(0.138)	(0.194)	(0.123)	(0.258)
Observations	1585	1528	1522	1528	1522	1528
Control Mean	6.096	6.446	1.592	1.953	1.690	3.749
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.587	0.604	0.329	0.671	0.854	0.565
Pooled: Voucher vs. No Voucher	0.570	0.243	0.758	0.835	0.846	0.066*
High-SES: Voucher vs. No Voucher	0.523	0.181	0.585	0.075*	0.356	0.893
Low-SES: Voucher vs. No Voucher	0.406	0.986	0.909	0.329	0.271	0.453
Random: Voucher vs. No Voucher	0.458	0.958	0.935	0.798	0.834	0.168

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Network Degree (colums (1)-(2)) is the total number of members in the network. New Network Ties (columns (3)-(4)) is the number of network ties in any given survey who were not listed in the network in any previous survey. Dropped Network Ties (column (5)-(6)) is the number of network ties who were listed in a previous survey but who were not listed in the given survey.

Table D.14: Treatment Effects on Subjective Changes to Relationships

	(1)	(2)	(3)	(4)	(5)	(6)
	Network Churn: Weak Ties		Network Churn: Mid Ties		Network Churn: Strong Ties	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.027	0.212*	-0.117	0.201	-0.026	0.047*
	(0.096)	(0.121)	(0.081)	(0.156)	(0.018)	(0.024)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	0.099	0.201	-0.193**	0.156	-0.029	0.038
	(0.119)	(0.142)	(0.098)	(0.195)	(0.022)	(0.030)
Inviter with Low-SES Guest List	0.065	0.280*	-0.083	0.106	-0.031	0.030
	(0.117)	(0.150)	(0.099)	(0.193)	(0.022)	(0.030)
Inviter with Random Guest List	-0.055	0.099	-0.059	0.363*	-0.019	0.060**
	(0.114)	(0.148)	(0.100)	(0.196)	(0.022)	(0.030)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	0.082	0.327*	-0.097	0.137	-0.020	0.046
	(0.146)	(0.174)	(0.126)	(0.253)	(0.027)	(0.039)
High-SES with Voucher	0.167	0.035	-0.286**	0.177	-0.036	0.032
	(0.149)	(0.171)	(0.112)	(0.244)	(0.025)	(0.036)
Low-SES without Voucher	0.117	0.165	-0.124	0.026	-0.058**	-0.016
	(0.144)	(0.176)	(0.122)	(0.249)	(0.025)	(0.034)
Low-SES with Voucher	0.075	0.318*	-0.016	0.190	-0.009	0.060
	(0.143)	(0.191)	(0.123)	(0.241)	(0.027)	(0.039)
Random without Voucher	-0.057	0.001	-0.093	0.191	-0.002	0.056
	(0.142)	(0.179)	(0.119)	(0.239)	(0.028)	(0.037)
Random with Voucher	-0.025	0.168	-0.030	0.513**	-0.037	0.049
	(0.139)	(0.188)	(0.129)	(0.257)	(0.026)	(0.039)
Observations	1585	1528	1522	1528	1585	1528
Control Mean	1.666	2.240	1.501	3.303	0.114	0.158
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.768	0.582	0.258	0.809	0.904	0.785
Pooled: Voucher vs. No Voucher	0.915	0.704	0.952	0.203	0.988	0.404
High-SES: Voucher vs. No Voucher	0.612	0.134	0.160	0.892	0.579	0.746
Low-SES: Voucher vs. No Voucher	0.790	0.468	0.439	0.579	0.085*	0.073*
Random: Voucher vs. No Voucher	0.841	0.432	0.657	0.277	0.228	0.892

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Network churn among weak ties (columns (1)-(2)) is the number of ties dropped from previous survey rounds who are defined as weak ties in that survey, plus the number of ties who are new in that survey round and defined as weak ties in the relevant survey round. In other words, it is a measure of changing links within ties of a given strength (whether change is by entry or exit from the network). The analogous definition holds for mid-strength ties (columns (3)-(4)) and strong ties (columns (5)-(6)). Strength-of-ties is evaluated using five variables: the ability to trust the network link with thoughts and feelings (Likert scale), ease of laughing with the network link (Likert scale), if the link is someone with whom the participant shares secrets (binary), degree to which sharing stories in the relationship happens, and to which it is reciprocal (Likert scale), and the Inclusion-of-Self in Others (ISO) scale (1 to 7). I conduct a PCA analysis of these five variables, using the full sample in the Baseline Survey. I then apply the first principal component weights on these five variables to the same variables in all the surveys. This way, I use the levels and correlation structure of these variables from the Baseline Survey to analyze strength of ties in ensuing survey rounds. This also allows me to construct a levels comparison in strength-of-ties over time. Weak ties are defined as ties whose strength-of-ties index is below the median of tie-strength in the Baseline Survey. Strong ties are defined as ties whose strength-of-ties index is equal to the maximum tie-strength in the Baseline Survey. Mid-strength ties are defined as all ties with an intermediate tie-strength.

Table D.15: Treatment Effects on Subjective Changes to Relationships

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Dropped Links: Weak Ties		New Links: Weak Ties	Dropped Links: Mid Ties		New Links: Mid Ties	Dropped Links: Strong Ties		New Links: Strong Ties
	1 Month	1 Year	1 Year	1 Month	1 Year	1 Year	1 Month	1 Year	1 Year
Panel A: Pooled Inviter Treatment Effects									
All Inviters	-0.001	0.115	0.098	-0.075	0.106	0.068	-0.022*	0.008	0.039**
	(0.060)	(0.070)	(0.078)	(0.051)	(0.121)	(0.072)	(0.013)	(0.017)	(0.015)
Panel B: Pooled Inviter by Guest List Treatment Effects									
Inviter with High-SES Guest List	0.040	0.121	0.079	-0.080	0.030	0.091	-0.024*	0.002	0.037*
	(0.074)	(0.083)	(0.094)	(0.060)	(0.146)	(0.097)	(0.015)	(0.020)	(0.019)
Inviter with Low-SES Guest List	0.032	0.114	0.167*	-0.058	0.164	-0.057	-0.026*	0.003	0.031
	(0.074)	(0.087)	(0.097)	(0.062)	(0.151)	(0.087)	(0.014)	(0.020)	(0.019)
Inviter with Random Guest List	-0.069	0.101	0.005	-0.082	0.151	0.169*	-0.016	0.015	0.049**
	(0.071)	(0.086)	(0.099)	(0.062)	(0.152)	(0.092)	(0.015)	(0.020)	(0.020)
Observations	1585	1528	1528	1522	1528	1528	1585	1528	1528
Control Mean	0.919	1.309	0.931	0.716	2.338	0.966	0.066	0.103	0.055
P-values:									
Pooled: High-SES vs. Low-SES Guest List	0.910	0.936	0.377	0.710	0.383	0.145	0.845	0.986	0.757
Pooled: Voucher vs. No Voucher	0.920	0.259	0.666	0.722	0.124	0.893	0.882	0.524	0.504

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Strength-of-ties is evaluated using five variables: the ability to trust the network link with thoughts and feelings (Likert scale), ease of laughing with the network link (Likert scale), if the link is someone with whom the participant shares secrets (binary), degree to which sharing stories in the relationship happens, and to which it is reciprocal (Likert scale), and the Inclusion-of-Self in Others (ISO) scale (1 to 7). I conduct a PCA analysis of these five variables, using the full sample in the Baseline Survey. I then apply the first principal component weights on these five variables to the same variables in all the surveys. This way, I use the levels and correlation structure of these variables from the Baseline Survey to analyze strength of ties in ensuing survey rounds. This also allows me to construct a levels comparison in strength-of-ties over time. Weak ties are defined as ties whose strength-of-ties index is below the median of tie-strength in the Baseline Survey. Strong ties are defined as ties whose strength-of-ties index is equal to the maximum tie-strength in the Baseline Survey. The outcomes of columns (1)-(2) [(5)-(6)] is the number of ties from previous surveys that were weak [strong] in that survey, that are dropped at the given survey round. The outcomes of columns (3)-(4) [(7)-(8)] is the number of new ties (ties that did not show up in a previous survey round) in a given survey round where the tie is weak [strong] in the given survey round.

Table D.16: Treatment Effects on Presence of Own Friends in the Network

	(1) Number of Own Friends 1 Year	(2) >= 50% Own Friends 1 Year	(3) Any Own Friend Strong Tie 1 Year	(4) Index: Own Friends Network 1 Year
Panel A: Pooled Inviter Treatment Effects				
All Inviters	0.134	0.045*	0.039	0.111**
	(0.085)	(0.023)	(0.024)	(0.046)
Panel B: Pooled Inviter by Guest List Treatment Effects				
Inviter with High-SES Guest List	0.052	0.037	0.068**	0.111**
	(0.104)	(0.028)	(0.030)	(0.055)
Inviter with Low-SES Guest List	0.133	0.055*	0.033	0.109*
	(0.104)	(0.029)	(0.030)	(0.057)
Inviter with Random Guest List	0.199*	0.033	0.017	0.102*
	(0.104)	(0.027)	(0.029)	(0.055)
Panel C: Inviter Sub-Group Treatment Effects				
High-SES without Voucher	0.178	0.048	0.068*	0.144**
	(0.135)	(0.034)	(0.039)	(0.068)
High-SES with Voucher	-0.105	0.019	0.062*	0.065
	(0.125)	(0.035)	(0.038)	(0.068)
Low-SES without Voucher	0.182	0.057	0.019	0.123*
	(0.130)	(0.037)	(0.037)	(0.072)
Low-SES with Voucher	0.080	0.054	0.048	0.095
	(0.126)	(0.035)	(0.037)	(0.068)
Random without Voucher	0.144	0.030	0.043	0.097
	(0.119)	(0.033)	(0.035)	(0.067)
Random with Voucher	0.262*	0.040	-0.009	0.106
	(0.138)	(0.034)	(0.037)	(0.069)
Observations	1528	1528	1528	1528
Control Mean	2.412	0.361	0.216	0.007
P-values:				
Pooled: High-SES vs. Low-SES Guest List	0.439	0.523	0.250	0.977
Pooled: Voucher vs. No Voucher	0.206	0.788	0.688	0.363
High-SES: Voucher vs. No Voucher	0.063*	0.475	0.898	0.307
Low-SES: Voucher vs. No Voucher	0.490	0.946	0.496	0.724
Random: Voucher vs. No Voucher	0.430	0.809	0.215	0.902

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of people in the network who women report they are not related to and met themselves (not through their husband). The outcome of column (2) is a binary indicator that takes the value of 1 if at least half of the links in the network are women's own friends, and 0 otherwise. The outcome of column (3) is a binary indicator that takes the value of 1 if there is at least one person in the network who is both a strong tie and the woman's own friend. Strength-of-ties is evaluated using five variables: the ability to trust the network link with thoughts and feelings (Likert scale), ease of laughing with the network link (Likert scale), if the link is someone with whom the participant shares secrets (binary), degree to which sharing stories in the relationship happens, and to which it is reciprocal (Likert scale), and the Inclusion-of-Self in Others (ISO) scale (1 to 7). I conduct a PCA analysis of these five variables, using the full sample in the Baseline Survey. I then apply the first principal component weights on these five variables to the same variables in all the surveys. This way, I use the levels and correlation structure of these variables from the Baseline Survey to analyze strength of ties in ensuing survey rounds. This also allows me to construct a levels comparison in strength-of-ties over time. Strong ties are defined as ties whose strength-of-ties index is equal to the maximum tie-strength in the Baseline Survey. The outcome of column (4) is an index that is constructed from the variables in columns (1)-(3). The Index of Own Friends in the Network is the first PCA component of the outcomes of columns (1)-(3), in standard deviation units from the Control Mean.

Table D.17: Treatment Effects on Presence of Husband's Relatives in the Network

	(1) Number of Husband Relatives 1 Year	(2) >= 50% Husband Relatives 1 Year	(3) Any Husband Relative Strong Tie 1 Year	(4) Index: Husband Relatives Network 1 Year
Panel A: Pooled Inviter Treatment Effects				
All Inviters	-0.034	0.007	-0.013	0.000
	(0.085)	(0.022)	(0.020)	(0.042)
Panel B: Pooled Inviter by Guest List Treatment Effects				
Inviter with High-SES Guest List	-0.043	0.007	-0.021	-0.009
	(0.101)	(0.027)	(0.024)	(0.051)
Inviter with Low-SES Guest List	-0.009	0.024	-0.027	0.007
	(0.105)	(0.027)	(0.023)	(0.052)
Inviter with Random Guest List	-0.083	-0.013	0.001	-0.010
	(0.104)	(0.027)	(0.025)	(0.052)
Panel C: Inviter Sub-Group Treatment Effects				
High-SES without Voucher	-0.036	-0.016	-0.043	-0.036
	(0.131)	(0.034)	(0.029)	(0.065)
High-SES with Voucher	-0.031	0.037	-0.007	0.023
	(0.118)	(0.033)	(0.031)	(0.062)
Low-SES without Voucher	0.015	0.025	-0.040	-0.001
	(0.131)	(0.034)	(0.028)	(0.063)
Low-SES with Voucher	0.010	0.027	-0.032	0.014
	(0.132)	(0.035)	(0.029)	(0.066)
Random without Voucher	-0.068	-0.028	-0.012	-0.022
	(0.130)	(0.033)	(0.031)	(0.065)
Random with Voucher	-0.078	0.005	-0.001	-0.004
	(0.131)	(0.033)	(0.031)	(0.064)
Observations	1528	1528	1528	1528
Control Mean	2.317	0.372	0.161	-0.001
P-values:				
Pooled: High-SES vs. Low-SES Guest List	0.739	0.540	0.806	0.749
Pooled: Voucher vs. No Voucher	0.801	0.202	0.450	0.583
High-SES: Voucher vs. No Voucher	0.975	0.171	0.299	0.418
Low-SES: Voucher vs. No Voucher	0.976	0.956	0.804	0.843
Random: Voucher vs. No Voucher	0.947	0.382	0.755	0.807

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of people in the network who women report they are related to from their husband's side of the family. The outcome of column (2) is a binary indicator that takes the value of 1 if at least half of the links in the network are women's relatives from their husband's side, and 0 otherwise. The outcome of column (3) is a binary indicator that takes the value of 1 if there is at least one person in the network who is both a strong tie and the woman's relative from her husband's side. Strength-of-ties is evaluated using five variables: the ability to trust the network link with thoughts and feelings (Likert scale), ease of laughing with the network link (Likert scale), if the link is someone with whom the participant shares secrets (binary), degree to which sharing stories in the relationship happens, and to which it is reciprocal (Likert scale), and the Inclusion-of-Self in Others (ISO) scale (1 to 7). I conduct a PCA analysis of these five variables, using the full sample in the Baseline Survey. I then apply the first principal component weights on these five variables to the same variables in all the surveys. This way, I use the levels and correlation structure of these variables from the Baseline Survey to analyze strength of ties in ensuing survey rounds. This also allows me to construct a levels comparison in strength-of-ties over time. Strong ties are defined as ties whose strength-of-ties index is equal to the maximum tie-strength in the Baseline Survey. The outcome of column (4) is an index that is constructed from the variables in columns (1)-(3). The Index of Husbands Relatives in the Network is the first PCA component of the outcomes of columns (1)-(3), in standard deviation units from the Control Mean.

Table D.18: Treatment Effects on Origin of Meal-Sharing Network Links

	(1) Meal-Sharing (Hosting): Own Friends 1 Year	(2) Meal-Sharing (Hosting): Own Relatives 1 Year	(3) Meal-Sharing (Hosting): Husband Relatives 1 Year	(4) Meal-Sharing (Visiting): Own Friends 1 Year	(5) Meal-Sharing (Visiting): Own Relatives 1 Year	(6) Meal-Sharing (Visiting): Husband Relatives 1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.113*	-0.016	-0.052	0.047	0.030	-0.074
	(0.068)	(0.040)	(0.067)	(0.064)	(0.036)	(0.063)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	0.079	0.001	-0.050	0.023	0.056	-0.053
	(0.081)	(0.053)	(0.082)	(0.077)	(0.046)	(0.077)
Inviter with Low-SES Guest List	0.089	-0.017	-0.010	0.054	0.016	-0.078
	(0.082)	(0.053)	(0.081)	(0.079)	(0.046)	(0.075)
Inviter with Random Guest List	0.181**	-0.042	-0.102	0.076	0.009	-0.109
	(0.083)	(0.048)	(0.083)	(0.076)	(0.045)	(0.079)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	0.177*	-0.000	-0.085	0.094	0.082	-0.063
	(0.103)	(0.070)	(0.102)	(0.097)	(0.062)	(0.097)
High-SES with Voucher	0.001	0.000	0.008	-0.023	0.031	-0.016
	(0.096)	(0.063)	(0.102)	(0.091)	(0.055)	(0.095)
Low-SES without Voucher	0.142	-0.048	0.056	0.142	-0.018	-0.015
	(0.103)	(0.069)	(0.099)	(0.098)	(0.059)	(0.090)
Low-SES with Voucher	0.055	0.023	-0.012	0.010	0.050	-0.106
	(0.100)	(0.066)	(0.102)	(0.097)	(0.059)	(0.097)
Random without Voucher	0.128	-0.056	-0.141	0.049	-0.008	-0.184*
	(0.101)	(0.057)	(0.105)	(0.091)	(0.054)	(0.098)
Random with Voucher	0.247**	-0.018	-0.054	0.120	0.027	-0.025
	(0.102)	(0.062)	(0.102)	(0.093)	(0.058)	(0.100)
Observations	1528	1528	1528	1528	1528	1528
Control Mean	1.377	0.409	1.369	1.293	0.338	1.269
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.895	0.753	0.638	0.674	0.425	0.735
Pooled: Voucher vs. No Voucher	0.295	0.486	0.784	0.251	0.664	0.573
High-SES: Voucher vs. No Voucher	0.114	0.996	0.436	0.274	0.476	0.669
Low-SES: Voucher vs. No Voucher	0.452	0.408	0.559	0.235	0.351	0.397
Random: Voucher vs. No Voucher	0.300	0.578	0.471	0.488	0.603	0.172

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of women's own friends in the network who she hosts for shared meals. The outcome of column (2) is the total number of women's own blood relatives in the network who she hosts for shared meals. The outcome of column (3) is the total number of women's relatives on her husband's side who she hosts for shared meals. The outcome of column (4) is the total number of women's own friends in the network who she visits for shared meals. The outcome of column (5) is the total number of women's own blood relatives in the network who she visits for shared meals. The outcome of column (6) is the total number of women's relatives on her husband's side who she visits for shared meals.

Table D.19: Treatment Effects on Origin of Risk-Sharing Network Links

	(1) Borrower: Own Friends 1 Year	(2) Borrower: Own Relatives 1 Year	(3) Borrower: Husband Relatives 1 Year	(4) Lender: Own Friends 1 Year	(5) Lender: Own Relatives 1 Year	(6) Lender: Husband Relatives 1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.008	0.001	-0.132**	-0.002	0.029	-0.065
	(0.047)	(0.027)	(0.051)	(0.045)	(0.027)	(0.047)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	-0.028	0.021	-0.135**	-0.032	0.054	-0.088
	(0.059)	(0.032)	(0.060)	(0.055)	(0.033)	(0.055)
Inviter with Low-SES Guest List	0.013	-0.018	-0.146**	0.038	0.008	-0.057
	(0.058)	(0.034)	(0.059)	(0.056)	(0.033)	(0.056)
Inviter with Random Guest List	0.075	0.004	-0.096	0.018	0.031	-0.036
	(0.060)	(0.034)	(0.061)	(0.060)	(0.032)	(0.058)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	-0.039	0.047	-0.152**	-0.047	0.068	-0.121*
	(0.074)	(0.042)	(0.069)	(0.071)	(0.042)	(0.066)
High-SES with Voucher	-0.031	-0.002	-0.115	-0.030	0.046	-0.046
	(0.073)	(0.036)	(0.074)	(0.069)	(0.040)	(0.070)
Low-SES without Voucher	0.006	0.001	-0.142**	0.069	0.024	-0.117*
	(0.076)	(0.048)	(0.068)	(0.072)	(0.047)	(0.065)
Low-SES with Voucher	0.039	-0.035	-0.128*	0.026	-0.004	0.030
	(0.069)	(0.036)	(0.075)	(0.067)	(0.037)	(0.074)
Random without Voucher	0.087	-0.019	-0.174**	0.025	0.025	-0.078
	(0.077)	(0.038)	(0.073)	(0.078)	(0.039)	(0.070)
Random with Voucher	0.068	0.031	-0.024	0.015	0.040	0.017
	(0.076)	(0.046)	(0.075)	(0.075)	(0.042)	(0.075)
Observations	1528	1528	1528	1528	1528	1528
Control Mean	0.641	0.198	0.712	0.654	0.164	0.641
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.499	0.242	0.825	0.219	0.187	0.574
Pooled: Voucher vs. No Voucher	0.902	0.676	0.144	0.827	0.698	0.026**
High-SES: Voucher vs. No Voucher	0.930	0.277	0.625	0.840	0.641	0.326
Low-SES: Voucher vs. No Voucher	0.693	0.476	0.852	0.603	0.579	0.062*
Random: Voucher vs. No Voucher	0.833	0.305	0.068*	0.913	0.740	0.256

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of women's own friends in the network who she lent to in the past year. The outcome of column (2) is the total number of women's own blood relatives in the network who she lent to in the past year. The outcome of column (3) is the total number of women's relatives on her husband's side who she lent to in the past year. The outcome of column (4) is the total number of women's own friends in the network who she borrowed from in the past year. The outcome of column (5) is the total number of women's own blood relatives in the network who she borrowed from in the past year. The outcome of column (6) is the total number of women's relatives on her husband's side who she borrowed from in the past year.

Table D.20: Treatment Effects on Origin of Secret & Advice Network Links

	(1) Secret-Keeper: Own Friends 1 Year	(2) Secret-Keeper: Own Relatives 1 Year	(3) Secret-Keeper: Husband Relatives 1 Year	(4) Advice-Giver: Own Friends 1 Year	(5) Advice-Giver: Own Relatives 1 Year	(6) Advice-Giver: Husband Relatives 1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.049	0.004	-0.029	0.073**	0.012	-0.067*
	(0.033)	(0.021)	(0.031)	(0.035)	(0.020)	(0.037)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	0.009	0.011	-0.011	0.030	0.029	-0.082*
	(0.039)	(0.027)	(0.038)	(0.042)	(0.026)	(0.044)
Inviter with Low-SES Guest List	0.125***	-0.025	-0.050	0.092**	0.006	-0.041
	(0.042)	(0.025)	(0.036)	(0.047)	(0.024)	(0.042)
Inviter with Random Guest List	0.010	0.025	-0.034	0.103**	-0.001	-0.068
	(0.040)	(0.027)	(0.037)	(0.043)	(0.024)	(0.044)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	0.004	-0.032	-0.045	0.042	0.008	-0.114**
	(0.049)	(0.032)	(0.045)	(0.053)	(0.031)	(0.053)
High-SES with Voucher	0.002	0.050	0.020	0.017	0.045	-0.046
	(0.046)	(0.036)	(0.048)	(0.050)	(0.033)	(0.054)
Low-SES without Voucher	0.120**	-0.014	-0.056	0.114*	0.028	-0.043
	(0.056)	(0.032)	(0.044)	(0.064)	(0.030)	(0.050)
Low-SES with Voucher	0.135***	-0.040	-0.051	0.078	-0.009	-0.038
	(0.049)	(0.028)	(0.043)	(0.054)	(0.028)	(0.051)
Random without Voucher	0.035	0.046	-0.039	0.105*	-0.010	-0.075
	(0.050)	(0.034)	(0.049)	(0.054)	(0.029)	(0.055)
Random with Voucher	-0.004	0.006	-0.039	0.100*	0.011	-0.057
	(0.048)	(0.033)	(0.043)	(0.055)	(0.030)	(0.051)
Observations	1528	1528	1528	1528	1528	1528
Control Mean	0.567	0.158	0.398	0.330	0.140	0.443
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.004***	0.174	0.259	0.180	0.358	0.310
Pooled: Voucher vs. No Voucher	0.867	0.852	0.466	0.602	0.637	0.308
High-SES: Voucher vs. No Voucher	0.974	0.044**	0.227	0.673	0.323	0.252
Low-SES: Voucher vs. No Voucher	0.814	0.418	0.905	0.612	0.259	0.923
Random: Voucher vs. No Voucher	0.488	0.305	0.990	0.938	0.544	0.749

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of women's own friends in the network who she shared secrets with in the past year. The outcome of column (2) is the total number of women's own blood relatives in the network who she shared secrets with in the past year. The outcome of column (3) is the total number of women's relatives on her husband's side who she shared secrets with in the past year. The outcome of column (4) is the total number of women's own friends in the network who she sought advice from in the past year. The outcome of column (5) is the total number of women's own blood relatives in the network who she sought advice from in the past year. The outcome of column (6) is the total number of women's relatives on her husband's side who she sought advice from in the past year.

Table D.21: Treatment Effects on Network Meal-Sharing Relationships

	(1) Network Size: Hosted for Meals 1 Year	(2) New Ties: Hosted for Meals 1 Year	(3) Dropped Ties: Hosted for Meals 1 Year	(4) Network Churn: Hosted for Meals 1 Year	(5) Network Size: Visited for Meals 1 Year	(6) New Ties: Visited for Meals 1 Year	(7) Dropped Ties: Visited for Meals 1 Year	(8) Network Churn: Visited for Meals 1 Year
Panel A: Pooled Inviter Treatment Effects								
All Inviters	0.006	0.174***	0.117	0.284**	0.008	0.209***	0.082	0.295***
	(0.089)	(0.057)	(0.074)	(0.115)	(0.081)	(0.054)	(0.068)	(0.104)
Panel B: Pooled Inviter by Guest List Treatment Effects								
Inviter with High-SES Guest List	-0.041	0.125*	0.091	0.210	0.015	0.151**	0.039	0.190
	(0.109)	(0.071)	(0.088)	(0.137)	(0.099)	(0.066)	(0.083)	(0.126)
Inviter with Low-SES Guest List	0.051	0.177**	0.089	0.263*	-0.010	0.217***	0.082	0.301**
	(0.109)	(0.072)	(0.091)	(0.143)	(0.096)	(0.070)	(0.084)	(0.131)
Inviter with Random Guest List	0.002	0.197***	0.157*	0.340**	0.004	0.228***	0.115	0.355***
	(0.111)	(0.075)	(0.092)	(0.148)	(0.100)	(0.071)	(0.084)	(0.132)
Observations	1528	1528	1528	1528	1528	1528	1528	1528
Control Mean	3.446	0.623	1.187	1.810	3.108	0.530	1.018	1.549
P-values:								
Pooled: High-SES vs. Low-SES Guest List	0.397	0.501	0.982	0.708	0.794	0.362	0.610	0.410
Pooled: Voucher vs. No Voucher	0.492	0.499	0.195	0.240	0.659	0.337	0.436	0.338

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of people in the network who she hosted for a shared meal in the past year. The outcome of column (2) is the number of links who she hosted for shared meals with at endline, who she did not host for shared meals in previous survey rounds. The outcome of column (3) is the number of links who she hosted for shared meals in previous survey rounds, who she did not host for shared meals at endline. The outcome of column (4) is the sum of the outcomes of columns (2) and (3). The outcome of column (5) is the total number of people in the network who she visited for a shared meal in the past year. The outcome of column (6) is the number of links who she visited for shared meals with at endline, who she did not visit for shared meals at baseline. The outcome of column (7) is the number of links who she visited for shared meals at baseline, who she did not visit for shared meals at endline. The outcome of column (8) is the sum of the outcomes of columns (6) and (7).

Table D.22: Treatment Effects on Strength of Risk-Sharing Ties

	(1) Number of Lenders: Weak Ties	(2) Number of Lenders: Medium Strength	(3) Number of Lenders: Strong Ties	(4) Number of Borrowers: Weak Ties	(5) Number of Borrowers: Medium Strength	(6) Number of Borrowers: Strong Ties
	1 Year	1 Year	1 Year	1 Year	1 Year	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	-0.008	-0.058	-0.024	-0.085	-0.084*	-0.002
	(0.051)	(0.046)	(0.023)	(0.054)	(0.049)	(0.022)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	-0.050	-0.062	-0.015	-0.116*	-0.115**	0.024
	(0.060)	(0.057)	(0.029)	(0.063)	(0.058)	(0.028)
Inviter with Low-SES Guest List	0.024	-0.062	-0.033	-0.080	-0.064	-0.031
	(0.063)	(0.056)	(0.028)	(0.063)	(0.060)	(0.027)
Inviter with Random Guest List	0.021	-0.023	-0.019	-0.034	-0.036	0.007
	(0.063)	(0.058)	(0.028)	(0.067)	(0.060)	(0.027)
Observations	1528	1528	1528	1528	1528	1528
Control Mean	0.609	0.720	0.219	0.662	0.789	0.206
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.218	0.995	0.508	0.546	0.381	0.045**
Pooled: Voucher vs. No Voucher	0.549	0.369	0.135	0.789	0.119	0.111

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of weak ties in the network who she borrowed from in the past year. The outcome of column (2) is the total number of medium-strength ties in the network who she borrowed from in the past year. The outcome of column (3) is the total number of strong ties in the network who she borrowed from in the past year. The outcome of column (4) is the total number of weak ties in the network who she lent to in the past year. The outcome of column (5) is the total number of medium-strength ties in the network who she lent to in the past year. The outcome of column (6) is the total number of strong ties in the network who she lent to in the past year.

Table D.23: Treatment Effects on Borrower Network Ties

	(1) Number of Borrowers in Network 1 Year	(2) Number of Borrowers New to Network 1 Year	(3) Number of Borrowers Dropped from Network 1 Year	(4) Network Churn in Borrowers 1 Year	(5) Number of Maintained Borrowers 1 Year	(6) Number of Maintained Ties New as Borrowers 1 Year	(7) Number of Dropped Borrowers Maintained in Network 1 Year
Panel A: Pooled Inviter Treatment Effects							
All Inviters	-0.153**	0.077**	0.100**	0.171***	-0.097***	-0.117**	-0.000
	(0.063)	(0.036)	(0.047)	(0.065)	(0.037)	(0.059)	(0.046)
Panel B: Pooled Inviter by Guest List Treatment Effects							
Inviter with High-SES Guest List	-0.199***	0.057	0.059	0.117	-0.141***	-0.105	0.085
	(0.076)	(0.045)	(0.057)	(0.081)	(0.046)	(0.069)	(0.057)
Inviter with Low-SES Guest List	-0.146*	0.072*	0.094	0.160**	-0.066	-0.134*	-0.025
	(0.075)	(0.044)	(0.059)	(0.079)	(0.045)	(0.070)	(0.057)
Inviter with Random Guest List	-0.043	0.097**	0.133**	0.220***	-0.059	-0.072	-0.071
	(0.078)	(0.046)	(0.060)	(0.085)	(0.048)	(0.072)	(0.056)
Observations	1528	1528	1528	1528	1528	1528	1528
Control Mean	1.660	0.290	0.520	0.810	0.475	0.894	0.799
P-values:							
Pooled: High-SES vs. Low-SES Guest List	0.461	0.755	0.554	0.609	0.101	0.649	0.057*
Pooled: Voucher vs. No Voucher	0.730	0.772	0.173	0.316	0.922	0.396	0.183

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of ties in the network who she lent to in the past year. The outcome of column (2) is the number of people who she lent to who she did not list in her network in any previous survey round for any activity. The outcome of column (3) is the number of people who she lent to in previous survey rounds who she did not list in her endline network across any activity. The outcome of column (4) is the sum of columns (2) and (3). The outcome of column (5) is the number of people she lent to at baseline who she still lent to at endline. The outcome of column (6) is the number of people from her baseline network but who she did not lend to at baseline, to whom she now lends at endline. The outcome of column (7) is the number of people who she lent to at baseline who she no longer lends to, but who are in her network as a link who she engages in other activities with.

Table D.24: Treatment Effects on Lender Network Ties

	(1) Number of Lenders in Network	(2) Number of Lenders New to Network	(3) Number of Lenders Dropped from Network	(4) Network Churn in Lenders	(5) Number of Maintained Lenders	(6) Number of Maintained Ties New as Lenders	(7) Number of Dropped Lenders Maintained in Network
	1 Year	1 Year	1 Year	1 Year	1 Year	1 Year	1 Year
Panel A: Pooled Inviter Treatment Effects							
All Inviters	-0.056	0.070**	0.082*	0.139**	-0.048	-0.079	-0.021
	(0.058)	(0.035)	(0.045)	(0.065)	(0.037)	(0.055)	(0.045)
Panel B: Pooled Inviter by Guest List Treatment Effects							
Inviter with High-SES Guest List	-0.098	0.083*	0.050	0.114	-0.099**	-0.083	0.058
	(0.070)	(0.043)	(0.055)	(0.079)	(0.044)	(0.064)	(0.055)
Inviter with Low-SES Guest List	-0.031	0.067	0.062	0.117	-0.006	-0.097	-0.046
	(0.071)	(0.043)	(0.057)	(0.081)	(0.046)	(0.066)	(0.054)
Inviter with Random Guest List	0.018	0.055	0.115**	0.161*	-0.020	-0.019	-0.086
	(0.074)	(0.045)	(0.058)	(0.085)	(0.047)	(0.068)	(0.055)
Observations	1528	1528	1528	1528	1528	1528	1528
Control Mean	1.551	0.288	0.536	0.823	0.449	0.815	0.810
P-values:							
Pooled: High-SES vs. Low-SES Guest List	0.351	0.737	0.832	0.969	0.038**	0.824	0.048**
Pooled: Voucher vs. No Voucher	0.511	0.060*	0.056*	0.010**	0.573	0.793	0.123

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of ties in the network who she borrowed from in the past year. The outcome of column (2) is the number of people who she borrowed from who she did not list in her network for any activity in any previous survey round. The outcome of column (3) is the number of people who she borrowed from in previous survey rounds who she did not list in her endline network across any activity. The outcome of column (4) is the sum of columns (2) and (3). The outcome of column (5) is the number of people she borrowed from at baseline who she still borrowed from at endline. The outcome of column (6) is the number of people from her baseline network but from who she did not borrow from at baseline, from whom she now borrows at endline. The outcome of column (7) is the number of people who she borrowed from at baseline who she no longer borrows from, but who are in her network as a link who she engages in other activities with.

Table D.25: Treatment Effects on Loneliness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Loneliness Index		UCLA-3 Score		Lonely (UCLA-3 >=6)		I was ever lonely in the past 2 weeks		I was often lonely in the past 2 weeks	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects										
All Inviters	0.073	-0.085	-0.012	-0.031	-0.000	0.004	0.047*	-0.063**	0.022	-0.017
	(0.084)	(0.089)	(0.064)	(0.073)	(0.017)	(0.020)	(0.026)	(0.027)	(0.017)	(0.018)
Panel B: Pooled Inviter by Guest List Treatment Effects										
Inviter with High-SES Guest List	0.043	0.038	-0.002	0.026	-0.009	0.008	0.021	-0.017	0.014	0.014
	(0.101)	(0.111)	(0.078)	(0.092)	(0.020)	(0.025)	(0.031)	(0.033)	(0.020)	(0.023)
Inviter with Low-SES Guest List	0.145	-0.154	-0.006	-0.087	0.019	0.002	0.054*	-0.082**	0.036*	-0.041*
	(0.108)	(0.107)	(0.078)	(0.088)	(0.022)	(0.024)	(0.032)	(0.032)	(0.021)	(0.021)
Inviter with Random Guest List	0.062	-0.096	-0.024	-0.021	-0.008	0.003	0.071**	-0.074**	0.017	-0.011
	(0.106)	(0.108)	(0.080)	(0.089)	(0.021)	(0.025)	(0.032)	(0.033)	(0.021)	(0.022)
Observations	1358	1505	1373	1524	1522	1528	1522	1528	1522	1528
Control Mean	-0.079	0.089	3.517	3.744	0.191	0.132	0.247	0.351	0.085	0.116
P-values:										
Pooled: High-SES vs. Low-SES Guest List	0.344	0.079*	0.958	0.214	0.184	0.787	0.295	0.039**	0.311	0.011**
Pooled: Voucher vs. No Voucher	0.155	0.057*	0.474	0.066*	0.082*	0.040**	0.832	0.344	0.909	0.675

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of columns (1) and (2) is the Z-score of the first principal component of the variables in columns (3)-(10). The outcome of columns (3) and (4) reports total UCLA-3 score. The outcome of columns (5) and (6) is an indicator for if the UCLA-3 score is at least 6, a common cut-off for lonely. The outcome of columns (7) and (8) is an indicator for if participants report that they rarely, sometimes, or always felt lonely in the past two weeks (a question from the CESDR-10 questionnaire, where the alternate choice option is Never). The outcome of columns (9) and (10) is an indicator for if participants report that they sometimes or always felt lonely in the past two weeks (a question from the CESDR-10 questionnaire, where the alternate choice options are Rarely or Never).

Table D.26: Treatment Effects on Mild-to-Severe Depression: Heterogeneously by Baseline Household Decision-Making Power

	(1)	(2)	(3)	(4)	(5)	(6)
	Mild-to-Severe Depression: No Decision-Making Power		Mild-to-Severe Depression: Some Decision-Making Power		Mild-to-Severe Depression: All Decision-Making Power	
	1 Month	1 Year	1 Month	1 Year	1 Month	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	-0.034	-0.235***	-0.010	-0.042	-0.005	-0.093
	(0.071)	(0.069)	(0.039)	(0.038)	(0.064)	(0.072)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	-0.083	-0.192**	-0.021	-0.021	-0.062	-0.066
	(0.081)	(0.087)	(0.048)	(0.045)	(0.072)	(0.081)
Inviter with Low-SES Guest List	-0.021	-0.290***	0.009	-0.060	-0.024	-0.156*
	(0.076)	(0.081)	(0.048)	(0.046)	(0.078)	(0.087)
Inviter with Random Guest List	-0.010	-0.231***	-0.020	-0.052	0.079	-0.060
	(0.086)	(0.088)	(0.046)	(0.045)	(0.083)	(0.088)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	-0.100	-0.186*	-0.074	-0.048	-0.049	0.013
	(0.099)	(0.096)	(0.057)	(0.055)	(0.091)	(0.097)
High-SES with Voucher	0.030	-0.128	0.002	0.012	-0.090	-0.180*
	(0.101)	(0.113)	(0.059)	(0.057)	(0.088)	(0.102)
Low-SES without Voucher	-0.029	-0.310***	0.014	-0.058	-0.015	-0.203*
	(0.099)	(0.103)	(0.057)	(0.056)	(0.103)	(0.104)
Low-SES with Voucher	0.034	-0.255***	-0.015	-0.063	-0.054	-0.089
	(0.087)	(0.091)	(0.060)	(0.056)	(0.098)	(0.111)
Random without Voucher	-0.130	-0.251**	-0.061	-0.039	0.097	-0.106
	(0.112)	(0.127)	(0.054)	(0.056)	(0.105)	(0.108)
Random with Voucher	0.076	-0.190**	0.007	-0.058	-0.021	0.002
	(0.103)	(0.096)	(0.058)	(0.054)	(0.105)	(0.104)
Observations	310	314	838	841	339	340
Control Mean	0.300	0.444	0.299	0.316	0.291	0.395
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.404	0.224	0.518	0.368	0.610	0.240
Pooled: Voucher vs. No Voucher	0.048**	0.781	0.285	0.805	0.475	0.543
High-SES: Voucher vs. No Voucher	0.216	0.637	0.241	0.340	0.678	0.073*
Low-SES: Voucher vs. No Voucher	0.537	0.612	0.659	0.941	0.741	0.326
Random: Voucher vs. No Voucher	0.104	0.651	0.274	0.767	0.317	0.317

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Depression is measured using the CESDR-10, a 10-item revised version of the Center of Epidemiologic Studies Depression Scale. For each question, participants self-report the frequency with with they experienced the feelings that the question describes in the past two weeks. Each question is answered on a scale of 0-3: (0) Never, (1) Rarely, (2) Sometimes, (3) Always. The CESDR-10 score is the aggregate of these scores across all ten questions. Mild-to-severe depression is a score >=10. The regressions in columns (1)-(2) limit the sample to women who, at baseline, said that only her husband or someone else is the primary decision-maker about childcare (including schooling, health, and discipline), whether and when to have a child, and household savings. The regressions in columns (3)-(4) limit the sample to women who, at baseline, said that they are the primary decision-maker (jointly or alone) for between one and four of these five household decisions. The regressions in columns (5)-(6) limit the sample to women who, at baseline, said that they are the primary decision-maker (jointly or alone) for all five household decisions.

Table D.27: Treatment Effects on Network Churn in Husband's Relatives: Heterogeneously by Baseline Household Decision-Making Power

	(1) Network Churn in Husbands Relatives: No Baseline Decision-Making Power 1 Year	(2) Network Churn in Husbands Relatives: Some Baseline Decision-Making Power 1 Year	(3) Network Churn in Husbands Relatives: All Baseline Decision-Making Power 1 Year
Panel A: Pooled Inviter Treatment Effects			
All Inviters	0.592***	0.164	0.119
	(0.201)	(0.126)	(0.154)
Panel B: Pooled Inviter by Guest List Treatment Effects			
Inviter with High-SES Guest List	0.290	0.113	-0.037
	(0.253)	(0.157)	(0.183)
Inviter with Low-SES Guest List	0.670***	0.341**	0.397*
	(0.242)	(0.168)	(0.229)
Inviter with Random Guest List	0.549**	0.065	0.003
	(0.271)	(0.143)	(0.200)
Panel C: Inviter Sub-Group Treatment Effects			
High-SES without Voucher	0.360	0.342*	-0.239
	(0.318)	(0.194)	(0.208)
High-SES with Voucher	0.282	-0.151	0.168
	(0.314)	(0.184)	(0.261)
Low-SES without Voucher	0.590*	0.266	0.443
	(0.300)	(0.202)	(0.310)
Low-SES with Voucher	0.831***	0.326	0.338
	(0.272)	(0.229)	(0.293)
Random without Voucher	0.217	0.141	-0.114
	(0.426)	(0.168)	(0.234)
Random with Voucher	0.596**	-0.035	0.048
	(0.290)	(0.166)	(0.230)
Observations	314	841	340
Control Mean	1.333	1.420	0.872
P-values:			
Pooled: High-SES vs. Low-SES Guest List	0.139	0.179	0.058*
Pooled: Voucher vs. No Voucher	0.544	0.167	0.144
High-SES: Voucher vs. No Voucher	0.819	0.024**	0.146
Low-SES: Voucher vs. No Voucher	0.439	0.818	0.759
Random: Voucher vs. No Voucher	0.399	0.326	0.516

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for all specifications is the number of husbands' relatives who were listed in a previous survey round, but not listed in the endline network, plus the number of husbands' relatives who were listed in the endline network, but not listed in a previous survey round. The regression in column (1) limits the sample to women who, at baseline, said that only her husband or someone else is the primary decision-maker about childcare (including schooling, health, and discipline), whether and when to have a child, and household savings. The regression in column (2) limits the sample to women who, at baseline, said that they are the primary decision-maker (jointly or alone) for between one and four of these five household decisions. The regression in column (3) limits the sample to women who, at baseline, said that they are the primary decision-maker (jointly or alone) for all five household decisions.

Table D.28: Treatment Effects on Marital Status

	(1) Still Married 1 Year	(2) Newly Married 1 Year	(3) Divorced, Separated, or Widowed 1 Year	(4) Newly Divorced 1 Year
Panel A: Pooled Inviter Treatment Effects				
All Inviters	0.023	0.009	-0.025	-0.019
	(0.022)	(0.008)	(0.018)	(0.017)
Panel B: Pooled Inviter by Guest List Treatment Effects				
Inviter with High-SES Guest List	0.024	0.013	-0.027	-0.016
	(0.026)	(0.011)	(0.022)	(0.020)
Inviter with Low-SES Guest List	0.051**	0.011	-0.041**	-0.039**
	(0.026)	(0.011)	(0.020)	(0.019)
Inviter with Random Guest List	0.001	0.005	-0.013	-0.006
	(0.027)	(0.009)	(0.021)	(0.020)
Observations	1528	1528	1528	1528
Control Mean	0.752	0.016	0.145	0.090
P-values:				
Pooled: High-SES vs. Low-SES Guest List	0.276	0.887	0.479	0.186
Pooled: Voucher vs. No Voucher	0.927	0.901	0.263	0.169

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Still Married (column (1)) takes the value 1 for respondents who were married at baseline and are still married to the same person at endline, and 0 otherwise. Newly Married (column (2)) takes the value 1 for respondents who were single at baseline and are now married, and respondents who were married at baseline and are now married to a new person, and 0 otherwise. Divorced, Seperated, or Widowed (column (3)) indicates if a respondent is currently divorced, seperated, or widowed, and not remarried. Newly Divorced (column (4)) takes the value of 1 if a respondent was divorced over the course of the past year since baseline – women who said they were married at baseline, who at endline say they are divorced – and 0 otherwise. I ask all participants about their subjective relationships with their husbands and, whether or not they list their husband in the network, I compute strength of ties the same way for husbands that I do for other network members. The outcome for column (5) is the strength of the tie with the husband. The outceom for columdn (6) takes the value 1 if the respondent says she would be very uncomfortable to both borrow and lend to her husband, and 0 otherwise. There are only 11 people in the sample who have never been married.

Table D.29: Treatment Effects on Marital Relationships

	(1) Husband Tie Strength 1 Year	(2) Husband Risk-Sharing Discomfort 1 Year
Panel A: Pooled Inviter Treatment Effects		
All Inviters	-0.076	0.001
	(0.058)	(0.023)
Panel B: Pooled Inviter by Guest List Treatment Effects		
Inviter with High-SES Guest List	-0.089	0.014
	(0.074)	(0.027)
Inviter with Low-SES Guest List	-0.097	-0.010
	(0.070)	(0.028)
Inviter with Random Guest List	-0.035	0.000
	(0.075)	(0.028)
Observations	1176	1176
Control Mean	5.100	0.884
P-values:		
Pooled: High-SES vs. Low-SES Guest List	0.917	0.360
Pooled: Voucher vs. No Voucher	0.372	0.489

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for column (1) is the strength of the tie with the husband. The outceom for column (2) takes the value 1 if the respondent says she would be very uncomfortable to both borrow and lend to her husband, and 0 otherwise. There are only 11 people in the sample who have never been married. The sample is limited to women married at endline.

Table D.30: Treatment Effects on Husband's Relatives within the Network: by Child-Rearing Intensity

	(1) Network Churn in Husbands Relatives: No Young Child-Rearing 1 Year	(2) Network Churn in Husbands Relatives: Only Infant Caregiving 1 Year	(3) Network Churn in Husbands Relatives: Ever Pregnant 1 Year
Panel A: Pooled Inviter Treatment Effects			
All Inviters	0.251*	0.312	0.427**
	(0.129)	(0.195)	(0.185)
Panel B: Pooled Inviter by Guest List Treatment Effects			
Inviter with High-SES Guest List	0.236	0.031	0.294
	(0.153)	(0.228)	(0.227)
Inviter with Low-SES Guest List	0.348**	0.482*	0.590**
	(0.166)	(0.254)	(0.236)
Inviter with Random Guest List	0.161	0.171	0.407*
	(0.149)	(0.236)	(0.232)
Panel C: Inviter Sub-Group Treatment Effects			
High-SES without Voucher	0.279	0.333	0.236
	(0.197)	(0.280)	(0.311)
High-SES with Voucher	0.147	-0.088	0.056
	(0.190)	(0.279)	(0.267)
Low-SES without Voucher	0.331	0.610*	0.463*
	(0.206)	(0.317)	(0.268)
Low-SES with Voucher	0.313	0.573	0.810**
	(0.203)	(0.373)	(0.341)
Random without Voucher	0.190	0.493*	0.254
	(0.187)	(0.266)	(0.281)
Random with Voucher	0.173	0.093	0.686**
	(0.185)	(0.298)	(0.287)
Observations	767	394	340
Control Mean	1.194	1.481	1.198
P-values:			
Pooled: High-SES vs. Low-SES Guest List	0.476	0.079*	0.219
Pooled: Voucher vs. No Voucher	0.564	0.082*	0.540
High-SES: Voucher vs. No Voucher	0.555	0.171	0.604
Low-SES: Voucher vs. No Voucher	0.942	0.930	0.323
Random: Voucher vs. No Voucher	0.935	0.204	0.197

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for all specifications is the number of husbands' relatives who were listed in the baseline network, but not listed in the endline network, plus the number of husbands' relatives who were listed in the endline network, but not listed in the baseline network. The regression in column (1) limits the sample to women who did not report caring for any young children at baseline, midline, or endline. The regression in column (2) limits the sample to women who reported breastfeeding or that there was a child in the household under 1 year old during baseline, midline, or endline, but was never pregnant in any survey and did not give birth between baseline and endline. The regression in column (3) limits the sample to women who reported that she was pregnant in any survey, or who, at endline, reported that she had given birth since baseline.

Table D.31: Treatment Effects Among Guests

	(1)	(2)	(3)	(4)	(5)
	Food Security Index		Farm Yields USD Value	Mild-to-Severe Depressive Symptoms	
	1 Month	1 Year	1 Year	1 Month	1 Year
Inviter	0.012	0.133**	6.785	0.009	-0.076***
	(0.056)	(0.054)	(26.334)	(0.026)	(0.025)
Guest		-0.019	-2.410		-0.004
		(0.055)	(30.394)		(0.020)
Observations	1522	3447	3447	1522	3447
Low-SES Control Mean					
Spillover Estimates in Control Group					
All Inviters and Guests		-0.000	19.639		-0.007
P-value		0.999	0.424		0.795
Within-SES Inviters and Guests		0.172**	26.120		-0.017
P-value		0.037	0.482		0.606
Inviters with High-SES Guests		0.134	31.599		-0.036
P-value		0.200	0.450		0.364
High-SES Guests		-0.084	-36.850		0.008
P-value		0.494	0.514		0.884

Standard errors in parentheses

All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The Food Security Index (columns 1 and 2) is a PCA index in standard deviations from the Control mean. The Food Security Index at 1 Month is a PCA index of: the typical number of meals consumed per day in the past month, and the number of meals with fish or eggs consumed in the past month. The Food Security Index at 1 Year is a PCA index of: the typical number of meals consumed per day in the past month, and the number of meals with meat, fish or eggs consumed in the past month, the typical number of meals consumed per day during the lean season, and the number of meals with meat, fish or eggs consumed during the lean season. To calculate the Farm Yields in USD value (column 3), households estimate their harvest yields for each crop. I multiply market rates in agricultural reports by yields for the most high-yield crops: maize, soy beans, groundnuts, beans, tobacco, sugar cane, sunflower seeds, sweet potatoes, and pumpkin. Depression is calculated using the CESDR-10, a 10-item revised version of the Center of Epidemiologic Studies Depression Scale. Mild-to-Severe Depression (columns 4 and 5) indicates a score greater than or equal to 10.

* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Table D.32: Treatment Effects on Saving Guests' Phone Numbers

	(1) Any Guest Phone Numbers Saved		(2) Low-SES Guest Phone Number Saved		(3) High-SES Guest Phone Number Saved	
Random without Voucher	-0.004	(0.020)	0.025	(0.016)	-0.021	(0.018)
Random with Voucher	0.023	(0.020)	0.018	(0.016)	0.005	(0.018)
High-SES without Voucher	0.039*	(0.020)			0.058***	(0.018)
High-SES with Voucher	0.005	(0.020)			0.022	(0.018)
Low-SES without Voucher	-0.003	(0.020)	0.030*	(0.016)		
Low-SES with Voucher	-0.001	(0.020)	0.036**	(0.017)		
Observations	1600		1134		1143	
Control Mean	0.048		0.016		0.032	
P-values:						
High-SES: Voucher = No Voucher	0.142				0.083	
Low-SES: Voucher = No Voucher	0.933		0.730			
Test: *HSES without V = HSES with V = LSES without V = LSES with V*					[p= 0.533]	
DID Estimates [P-values]						
High-SES without Voucher – Low-SES without Voucher					0.028 [p= 0.291]	
High-SES with Voucher – Low-SES with Voucher					-0.014 [p= 0.561]	
Random without Voucher:						
(1) Invites High-SES – Invites Low-SES					-0.046 [p= 0.016]	
Random with Voucher:						
(1) Invites High-SES – Invites Low-SES					-0.013 [p= 0.528]	

Standard errors in parentheses
All specifications include baseline unbalanced control variables (number of garden plots and listed husband in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. Any Guest (column 1) indicates if the Inviter saved the phone number of at least one Guest. Low-SES Guest (column 2) indicates if the Inviter saved the phone number of at least one low-SES Guest (Inviters with the High-SES Guest List did not see any low-SES names, so this outcomes is mechanically zero for this group). High-SES Guest (column 3) indicates if the Inviter saved the phone number of at least one high-SES Guest (Inviters with the Low-SES Guest List did not see any high-SES names, so this outcome is mechanically zero for this group). The Control mean represents the probability of having the phone numbers of six neighbors saved, without the intervention. Control participants saw a list of six neighbors, where that list was constructed in the same way that Guest Lists were constructed for Inviters, and were asked for how many of those women they had a phone number saved.
* $p < .1$, ** $p < 0.05$, *** $p < 0.01$

Table D.33: Treatment Effects on Network Churn across Income

	(1)	(2)	(3)	(4)	(5)	(6)
	% Network Iron Sheets		> 50% Network Iron Sheets		Network Churn: Iron Sheets	Network Churn: Thatched Roof
	1 Month	1 Year	1 Month	1 Year	1 Year	1 Year
Panel A: Pooled Inviter Treatment Effects						
All Inviters	0.000	0.000	-0.012	0.032	0.029	0.431**
	(0.009)	(0.011)	(0.019)	(0.021)	(0.093)	(0.186)
Panel B: Pooled Inviter by Guest List Treatment Effects						
Inviter with High-SES Guest List	0.012	0.016	-0.019	0.047*	0.047	0.298
	(0.011)	(0.014)	(0.023)	(0.026)	(0.112)	(0.227)
Inviter with Low-SES Guest List	-0.003	-0.004	0.003	0.053**	0.022	0.451*
	(0.011)	(0.013)	(0.023)	(0.025)	(0.115)	(0.232)
Inviter with Random Guest List	-0.008	-0.008	-0.024	-0.005	0.003	0.503**
	(0.011)	(0.013)	(0.024)	(0.026)	(0.113)	(0.235)
Panel C: Inviter Sub-Group Treatment Effects						
High-SES without Voucher	-0.000	0.018	-0.035	0.052	0.186	0.306
	(0.014)	(0.018)	(0.029)	(0.033)	(0.136)	(0.297)
High-SES with Voucher	0.020	0.015	-0.011	0.050	-0.081	0.272
	(0.013)	(0.017)	(0.028)	(0.031)	(0.139)	(0.273)
Low-SES without Voucher	-0.003	-0.001	-0.008	0.064**	-0.078	0.275
	(0.013)	(0.016)	(0.027)	(0.030)	(0.135)	(0.297)
Low-SES with Voucher	-0.006	-0.006	0.008	0.045	0.123	0.584**
	(0.013)	(0.017)	(0.028)	(0.031)	(0.148)	(0.291)
Random without Voucher	-0.011	0.000	-0.047	-0.001	-0.022	0.314
	(0.013)	(0.015)	(0.029)	(0.030)	(0.139)	(0.282)
Random with Voucher	-0.006	-0.020	-0.005	-0.012	0.012	0.676**
	(0.013)	(0.018)	(0.030)	(0.033)	(0.139)	(0.311)
Observations	1501	1527	1522	1528	1528	1528
Control Mean	0.322	0.417	0.212	0.277	1.831	3.863
P-values:						
Pooled: High-SES vs. Low-SES Guest List	0.152	0.155	0.331	0.816	0.825	0.522
Pooled: Voucher vs. No Voucher	0.373	0.433	0.114	0.651	0.848	0.185
High-SES: Voucher vs. No Voucher	0.176	0.901	0.462	0.955	0.091*	0.919
Low-SES: Voucher vs. No Voucher	0.801	0.804	0.590	0.569	0.217	0.379
Random: Voucher vs. No Voucher	0.704	0.304	0.225	0.756	0.830	0.306

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of columns (1) and (2) reports the percent of current network members with an iron sheets roof, as reported by the participant in the survey where she first mentioned that person as a part of her network. The outcome of columns (3) and (4) is an indicator for if at least half of the participant's current network members have an iron sheets roof, as reported by the participant in the survey where she first mentioned that person as a part of her network. The outcome of columns (5) is the number of people with an iron sheets roof in her baseline network who are no longer a part of her endline network, plus the number of people with an iron sheets roof in her endline network who were not a part of her baseline network. The outcome of columns (6) is the number of people with a thatched roof in her baseline network who are no longer a part of her endline network, plus the number of people with a thathced roof in her endline network who were not a part of her baseline network.

Table D.34: Food Security Index: Heterogeneity by Guest Age and Guest Networks

	(1)	(2)	(3)	(4)
	Any Guest is >= Age 40	All Guests are > Age 22	Any Guest Has >= 9 Friends	All Guests Have > 3 Friends
Inviter with Guest List Heterogeneity=0	0.121*	0.086	0.156**	0.083
	(0.070)	(0.058)	(0.076)	(0.085)
Inviter with Guest List Heterogeneity=1	0.101	0.165**	0.074	0.115**
	(0.062)	(0.076)	(0.059)	(0.058)
Observations	1528	1528	1528	1528
Control Mean	-0.000	-0.000	-0.000	-0.000
P-values:				
(Inviter with Guest List Heterogeneity=0) = (Inviter with Guest List Heterogeneity=1)	0.778	0.248	0.270	0.685

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for all regressions is the food security index. The title of each column specifies the heterogeneous Guest characteristics that I test. The comparison group is the Control group. Inviter with Guest List Heterogeneity=0 is the sample who are Inviters and whose Guests do not have the characteristics specified by the column title. Inviter with Guest List Heterogeneity=1 is the sample who are Inviters and whose Guests have the characteristics specified by the column title.

Table D.35: Farm Yields USD Value: Heterogeneity by Guest Age and Guest Networks

	(1)	(2)	(3)	(4)
	Any Guest >= Age 40	All Guests > Age 22	Any Guest >= 9 Friends	All Guests > 3 Friends
Inviter with Guest List Heterogeneity=0	-14.384	7.181	-9.108	9.829
	(33.379)	(28.672)	(29.764)	(43.108)
Inviter with Guest List Heterogeneity=1	16.877	0.706	16.057	0.161
	(28.826)	(36.138)	(31.849)	(27.257)
Observations	1528	1528	1528	1528
Control Mean	324.783	324.783	324.783	324.783
P-values:				
(Inviter with Guest List Heterogeneity=0) = (Inviter with Guest List Heterogeneity=1)	0.305	0.850	0.422	0.809

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for all regressions is the total value of farm yields in US dollars. The title of each column specifies the heterogeneous Guest characteristics that I test. The comparison group is the Control group. Inviter with Guest List Heterogeneity=0 is the sample who are Inviters and whose Guests do not have the characteristics specified by the column title. Inviter with Guest List Heterogeneity=1 is the sample who are Inviters and whose Guests have the characteristics specified by the column title.

Table D.36: Mild-to-Severe Depressive Symptoms: Heterogeneity by Guest Age and Guest Networks

	(1)	(2)	(3)	(4)
	Any Guest >= Age 40	All Guests > Age 22	Any Guest >= 9 Friends	All Guests > 3 Friends
Inviter with Guest List Heterogeneity=0	-0.114***	-0.096***	-0.087***	-0.071*
	(0.033)	(0.028)	(0.032)	(0.040)
Inviter with Guest List Heterogeneity=1	-0.057**	-0.035	-0.071**	-0.079***
	(0.029)	(0.035)	(0.031)	(0.028)
Observations	1528	1528	1528	1528
Control Mean	0.361	0.361	0.361	0.361
P-values:				
(Inviter with Guest List Heterogeneity=0) = (Inviter with Guest List Heterogeneity=1)	0.055*	0.056*	0.618	0.823

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for all regressions takes the value of 1 for participants who experienced mild-to-severe depressive symptoms, and 0 otherwise. The title of each column specifies the heterogeneous Guest characteristics that I test. The comparison group is the Control group. Inviter with Guest List Heterogeneity=0 is the sample who are Inviters and whose Guests do not have the characteristics specified by the column title. Inviter with Guest List Heterogeneity=1 is the sample who are Inviters and whose Guests have the characteristics specified by the column title.

Table D.37: Mild-to-Severe Depressive Symptoms: Heterogeneity by Guest Age and Guest Networks

	(1)
	All Guests Are > Age 22
High-SES Guest List with Guest List Heterogeneity=0	-0.064
	(0.040)
High-SES Guest List with Guest List Heterogeneity=1	-0.021
	(0.042)
Low-SES Guest List with Guest List Heterogeneity=0	-0.085**
	(0.034)
Low-SES Guest List with Guest List Heterogeneity=1	-0.200***
	(0.061)
Random Guest List with Guest List Heterogeneity=0	-0.134***
	(0.035)
Random Guest List with Guest List Heterogeneity=1	0.014
	(0.055)
Observations	1528
Control Mean	0.361
P-values:	
(Inviter with Guest List Heterogeneity=0) = (Inviter with Guest List Heterogeneity=1)	
High-SES Guest List	0.371
Low-SES Guest List	0.065*
Random Guest List	0.010**

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome for all regressions takes the value of 1 for participants who experienced mild-to-severe depressive symptoms at endline, and 0 otherwise. I estimate treatment effects separately for six groups, and compare them to Control. Inviters with the High-SES Guest List whose Guest List had at least one woman who was 22 years old or younger (Guest Heterogeneity=0); Inviters with the High-SES Guest List whose Guest List had no one who was 22 years old or younger (Guest Heterogeneity=1); Inviters with the Low-SES Guest List whose Guest List had at least one woman who was 22 years old or younger (Guest Heterogeneity=0); Inviters with the Low-SES Guest List whose Guest List had no one who was 22 years old or younger (Guest Heterogeneity=1); Inviters with the Random-SES Guest List whose Guest List had at least one woman who was 22 years old or younger (Guest Heterogeneity=0); and Inviters with the Random-SES Guest List whose Guest List had no one who was 22 years old or younger (Guest Heterogeneity=1).

D.2 Treatment Effect Heterogeneity by Log-Years in the Village: Details

Treatment effects on network churn are driven by new arrivals among Inviters with the Low-SES and Random Guest Lists (where the Inviters with the Random Guest List is noisy but directionally consistent), while there is no difference in network churn by time in the village for Inviters with the High-SES Guest List (Appendix Table D.38). This suggests that the intervention speeds up network formation between low-SES women that would happen eventually, but creates links across-SES that would never happen over time. Treatment effects on depressive symptoms are not heterogeneous, suggesting that the intervention still stimulates a new *type* of interaction that is positive for mental health and would never happen absent the intervention (Appendix Table D.38).[56]

Consistent with the hypothesis that the high-SES links would not form otherwise, treatment effects on food security and self-employment are not differentiable by time in the village (Appendix Table D.38). Surprisingly, Inviters with the High-SES Guest List who are new arrivals report 144 USD greater farm yields ($p < 0.05$), equivalent to the additional farm yields associated with five years in the village among the Control group.[57] The effects on farm yields among new arrivals is driven by legume yields, which are positively but noisily associated with time in the village among Control (Appendix Table D.38). Cash crop production, which depends significantly on SES but not on time in the village, increases among Inviters with the High-SES Guest List regardless of years in the village.

[56] Treatment effects on mediating variables, such as sharing secrets and having strong-tie relationships with women's own friends, also are not heterogeneous by time in the village.

[57] This is also roughly the amount of time in the village before the treatment effect fades. Every log-year in the village is associated with a 78 USD lower treatment effect.

Table D.38: Heterogeneous Treatment Effects by Years in the Village

	(1) Network Churn	(2) Food Security Index	(3) Farm Yields USD Value	(4) Mild-to-Severe Depressive Symptoms	(5) Self-Employed Last Month	(6) Any Strong Tie Own Friend	(7) Number of Secret-Keepers Own Friends	(8) Legume Yields in KGs	(9) Cash Crop Yield USD Value
High-SES Guest List	0.543	0.025	90.132	0.002	0.086	-0.029	0.039	43.999	1.452
	(0.740)	(0.191)	(76.956)	(0.082)	(0.068)	(0.080)	(0.110)	(31.067)	(21.678)
Low-SES Guest List	1.352*	-0.055	-8.990	-0.106	0.058	-0.064	0.169	30.529	-28.194
	(0.744)	(0.186)	(69.293)	(0.082)	(0.066)	(0.076)	(0.110)	(27.198)	(18.547)
Random Guest List	1.027	-0.033	67.950	-0.033	0.076	-0.035	-0.095	35.953	-9.883
	(0.732)	(0.195)	(64.071)	(0.084)	(0.071)	(0.078)	(0.113)	(26.305)	(17.831)
Log Years in Village	-0.214	-0.129	95.489***	0.057*	0.009	-0.043	-0.012	31.133**	8.702
	(0.273)	(0.082)	(35.571)	(0.033)	(0.027)	(0.030)	(0.043)	(14.346)	(8.836)
High-SES Guest List × Log Years in Village	-0.131	0.106	-29.343	-0.029	0.006	0.062	-0.003	-28.476	17.452
	(0.378)	(0.107)	(49.126)	(0.044)	(0.037)	(0.042)	(0.058)	(18.646)	(14.939)
Low-SES Guest List × Log Years in Village	-0.564	0.064	-11.599	-0.004	-0.029	0.064	-0.010	-26.349	22.478
	(0.388)	(0.104)	(47.195)	(0.045)	(0.037)	(0.041)	(0.061)	(17.816)	(14.165)
Random Guest List × Log Years in Village	-0.315	0.074	-48.855	-0.033	-0.020	0.032	0.062	-17.152	2.913
	(0.376)	(0.105)	(42.624)	(0.045)	(0.038)	(0.041)	(0.060)	(17.129)	(11.480)
Observations	1528	1528	1528	1528	1522	1528	1528	1528	1528
Control Mean	5.702	-0.000	324.783	0.361	0.237	0.216	0.567	89.436	24.487

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All specifications include baseline unbalanced control variables (number of rented garden plots, and if the husband was listed in the baseline network), geographic cluster fixed effects, cluster-specific wealth quintile fixed effects, and lasso-selected controls. The outcome of column (1) is the total number of baseline network links who are not in the endline network, plus the total number of endline network links who are not in the baseline network. The outcome of column (2) is the food security index. The outcome of column (3) is total farm yields converted to US dollar value. The outcome of column (4) takes the value of 1 for participants who experienced mild-to-severe depressive symptoms at endline, and 0 otherwise. The outcome of column (5) takes the value of 1 for participants who, at midline, report that they earned income from self-employment in the past month, and 0 otherwise. The outcome of column (6) takes the value of 1 for participants who have any strong tie in the network who is a friend she met herself (not through her husband), and 0 otherwise. The outcome of column (7) is the number of people in the network who the respondent met herself (not through her husband) and with whom she says she shares secrets. The outcome of column (8) is the last harvest's legume yields in kilograms across all legume crops. The outcome of column (9) is last harvest's cash crop yields converted to US dollar value. I estimate treatment effects separated for each of the three main treatment groups – Inviters with the High-SES Guest List, Inviters with the Low-SES Guest List, and Inviters with the Random Guest List – and each of their interaction with log-years that the participant has lived in the village plus 1. Participants who just arrived to the village are in their first year in the village, so log years in the village is log of 1 for these participants.

D.3 Alternative explanations: Social frictions that the Inviter treatment resolves

No Evidence for Lack of Mutual Consent and Reciprocation

Conditional on sending an invitation, meal-sharing is very high and is not different by Guest-SES. Sixty-four percent of Guests who receive an invitation report that their reaction to receiving the invitation is unambiguously positive, such as feelings of happiness or excitement, while only 4% report any unambiguously negative feelings, such as discomfort or feeling obligated.[58] Furthermore, among Guests with whom Inviters actually share a meal, almost all reciprocate and host a shared meal in return (Figure C.2).

While I cannot rule out that the rate of accepting exogenous invitations would be lower outside the experiment, the reported rejection rate to invitations sent outside the experiment is similar to the within-experiment rejection rate.[59] Furthermore, Control participants who have an invitation rejected outside the experiment are balanced on observable characteristics with Inviters whose invitations are rejected within the experiment (Appendix Table B.8).

No Evidence that Low-SES Women Under-estimate Returns to Relationships

In response to hypothetical vignettes, almost all respondents report that a high-income neighbor would be helpful for finding income, and that a low-income neighbor would be helpful for reducing loneliness (Appendix Figure C.3). While the majority of respondents also believe that a low-income woman would be helpful for earning income, and a high-income neighbor would be helpful for reducing loneliness, responses are less optimistic, demonstrating that women understand the specific benefits that high-SES and low-SES relationships offer.

No Evidence that Social Norms Mitigate Negative Externalities

While I cannot comment on men or non-migrant women, who are outside my sample, neither Guests nor Control participants who are proximate to treated participants experience negative spillover effects across the main outcomes (Appendix Table D.31). It is therefore unlikely that social norms leading to low rates of social interaction serve an efficient purpose, such as maintaining an optimal allocation of friendships collectively.

[58] The remaining Guests report nervousness, surprise, no reaction, or that they don't remember.

[59] Thirty percent of respondents who say they sent invitations outside the experiment had at least one invitation rejected. In the experiment, 21% of people who send invitations don't share meals.

Appendix E Robustness

E.1 Populated Pre-Analysis Plan

Table E.1: Main Outcomes: Pre-Specified versus Paper

Main Outcomes	Pre- Specified Variable	Variable in Paper	Reasons for Deviations
Psychological well-being	Standardized first PCA component of (1) CESDR-10 score (reverse-coded), (2) UCLA-3 score (reverse-coded), (3) "is there someone you can talk to about a quarrel with your husband?"	CESDR-10 score >= 10	Variables (2) and (3) have little variation
Social capital	Standardized first PCA component of (1) Number of relatives in network, (2) Number of non-relatives in network, (3) Strength of relatives ties, (4) Strength of non-relative ties	Each variable is evaluated separately instead of together in an index as an outcome	It is not ex-ante clear which "types" of social capital we should expect to be correlated with one another, or with downstream outcomes
Savings and consumption	Standardized first PCA component of (1) Kilograms of maize in storage, (2) Participation in ROSCA or VSLA, (3) Number of meals eaten yesterday, (4) Number of meals eaten the day before yesterday, (5) Number of animal protein meals in the past month	Split into three variables: (1) Farm Yields, (2) Food Security Index, and (3) Potential for Risk-Sharing Index	Variables are conceptually distinct and represent different, individually interesting, economic strategies
Social Capital Definitions	**Pre- Specified Variable**	**Variable in Paper**	**Reasons for Deviations**
Strength of ties	Standardized first PCA component of (1) Inclusion of self-in-other score, (2) Trust in network connections, (3) Ability to laugh with network connection, (4) Comfort confiding in network connection	As described + "shares secrets with network connection"	Additional variable provides more variation, and helps identify right-tail emotionally strong ties.
Economic closeness of ties	Standardized first PCA component of (1) Comfort asking network connection to borrow money or goods, (2) Comfort lending money or goods to network connection	Both variables + "comfort both lending and borrowing with network connection" are included in the Potential for Risk-Sharing Index	Additional variable provides more variation, and helps identify right-tail economically strong ties.

Detailed Rationale for Changes in Psychological Well-Being Outcome: Variables (2) and (3) have significantly less variation than the CESDR-10 score. As such, I evaluate the CESDR-10 score on its own as the main outcome. The other variables move in the same direction, but when taken together as an index, they add more noise. Furthermore, I use a binary cutoff, rather than the continuous CESDR-10 Score, for two reasons: (1) it is a more interpretable outcome, allowing for comparison with other mental health interventions who evaluate reductions in depression using, for example, cut-offs in the PHQ-9; (2) it improves power to use a binary outcome rather than a noisy continuous outcome. Note that, while the pooled Inviter treatment effect is not statistically significant when using the CESDR-10 score, the effect among Inviters with the Low-SES Guest List is significantly lower than Control ($p < 0.05$) and as compared to Inviters with the High-SES Guest List ($p < 0.10$).

Detailed Rationale for Changes in Social Capital Outcome: It helps us understand downstream outcomes more to tease each component apart than to evaluate an overall social capital score. There are reasons to believe there could be substitution across types of social capital, or churn in social capital rather than an overall increase in the quantity or strength of ties.

Detailed Rationale for Changes in Savings and Consumption Outcome: The pre-specified index combines conceptually distinct items that are all geared towards understanding economic well-being. Specifically, I split consumption, income, and savings, rather than combining them into one index. I further replace "maize stores" with "farm yields" because maize stores was originally intended to serve as a verifiable proxy for income. However, "maize stores" combines both savings (which proxies for income from maize, the staple grain), choice of earnings method (which demotes the importance of non-maize crops and non-agricultural earnings), and choice of savings technology. Although farm yields are not verifiable, while maize stores are visually verified, it represents the simplest and most relevant proxy for income.

Detailed Rationale for Changes in Strength of Ties Outcome: The fifth variable was excluded in the pre-specified variable because it is collected in a different way than the other variables. However, it is positively correlated with the other variables, and because it is collected differently, tends to identify the fewer network members, thereby adding more information about the right-tail of emotional closeness.

Detailed Rationale for Changes in Economic Closeness Outcome: There are some people who respondents report comfort borrowing, others who they report comfort lending, and fewer for whom they report comfort both borrowing and lending. By including the combined measure, we learn more information about the right-tail of economic closeness.

Table E.2: Populated Pre-Analysis Plan

	(1)	(2)	(3)	(4)	(5)
	Psychological Well-being Index		Savings & Consumption Index		Social Capital Index
	1 Month	1 Year	1 Month	1 Year	1 Year
Panel A: Pooled Inviter Treatment Effects					
All Inviters	-0.028	0.090	-0.023	0.013	0.033
	(0.069)	(0.069)	(0.066)	(0.065)	(0.056)
Panel B: Pooled Inviter by Guest List Treatment Effects					
Inviter with High-SES Guest List	-0.015	0.008	0.015	0.086	0.072
	(0.084)	(0.085)	(0.080)	(0.081)	(0.069)
Inviter with Low-SES Guest List	-0.038	0.186**	-0.174**	-0.027	-0.019
	(0.084)	(0.083)	(0.076)	(0.078)	(0.069)
Inviter with Random Guest List	-0.058	0.060	0.078	-0.047	0.024
	(0.086)	(0.082)	(0.080)	(0.078)	(0.069)
Panel C: Inviter Sub-Group Treatment Effects					
High-SES without Voucher	0.081	0.061	-0.073	0.079	0.007
	(0.103)	(0.103)	(0.089)	(0.101)	(0.084)
High-SES with Voucher	-0.121	0.009	0.098	0.064	0.118
	(0.106)	(0.106)	(0.105)	(0.093)	(0.088)
Low-SES without Voucher	-0.110	0.246**	-0.164*	-0.084	-0.023
	(0.103)	(0.099)	(0.090)	(0.089)	(0.085)
Low-SES with Voucher	0.034	0.184*	-0.159*	0.041	-0.040
	(0.105)	(0.102)	(0.094)	(0.101)	(0.084)
Random without Voucher	-0.074	0.118	0.071	-0.044	0.135
	(0.106)	(0.102)	(0.093)	(0.093)	(0.086)
Random with Voucher	-0.010	0.053	0.068	-0.039	-0.118
	(0.112)	(0.100)	(0.101)	(0.099)	(0.082)
Observations	1373	1524	1522	1528	1528
Control Mean	0.027	-0.062	0.025	-0.010	0.067
P-values:					
Pooled: High-SES vs. Low-SES Guest List	0.784	0.032**	0.008***	0.142	0.184
Pooled: Voucher vs. No Voucher	0.883	0.300	0.404	0.518	0.208
High-SES: Voucher vs. No Voucher	0.093*	0.669	0.115	0.894	0.264
Low-SES: Voucher vs. No Voucher	0.234	0.585	0.955	0.220	0.858
Random: Voucher vs. No Voucher	0.624	0.571	0.979	0.958	0.009***

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E.2 Multiple Hypothesis Testing

To account for multiple hypothesis testing, I correct for the false discovery rate. I present FDR-sharpened q-values for a set of thirty hypotheses in Table E.3. Four of these tests are from the first stage, and twenty-eight tests are from the second stage. I test the effect of the Voucher among Inviters with the Random Guest List for three outcomes: if the Inviter sent any invitation, if she sent any invitation to a High-SES Guest, and if she sent any invitation to a Low-SES Guest. For the first outcome—if Inviters sent any invitation—I also test equality of the four other groups (Inviters with the High-SES Guest List or Low-SES Guest List, with or without the Voucher). For eleven downstream outcomes, I test two hypotheses each: the pooled Inviter treatment effect relative to the Control group, and the treatment effect among Inviters with the High-SES Guest List relative to the treatment effect among Inviters with the Low-SES Guest List. The outcomes I include in the family include the three main downstream outcomes (the food security index, farm yields in USD value, and mild-to-severe depression), and a representative outcome from each of the primary drivers of these outcomes that I test: self-employment in the past month (at midline and endline), an indicator for growing cash crops (endline), the potential for risk-sharing index (endline), the number of borrowers in the past year (endline), whether or not the Inviter borrowed from anyone in the past six months (endline), network degree (endline), network churn (endline), mild-to-severe depressive symptoms among the poorest-of-the-poor sub-sample (endline), and loneliness (an indicator for if the UCLA-3 score is $>= 6$, measured at endline).

The main first-stage effect remains statistically significant: Inviters with the Random Guest List are more likely to send invitations to High-SES Guests with the Voucher than without the Voucher (q-value = 0.077). The effects on the main outcomes also remain statistically significant, both when pooled and compared with the Control group, and when comparing Inviters with the Low-SES Guest List and Inviters with the High-SES Guest List with one another. Almost all of the tests that are statistically significant in the main analysis remain statistically significant after correcting for the false discovery rate. The only exceptions are that the differential effect on self-employment between Inviters with the High-SES Guest List and Inviters with the Low-SES Guest List, measured at endline, is marginally no longer statistically significant at conventional levels (q-value = 0.111) (the effect measured at midline remains statistically significant, both when considering the pooled effect relative to the Control group and the differential effect among Inviters with the High-SES Guest List and Inviters with the Low-SES Guest List); the pooled network churn effect is marginally no longer statistically significant (q-value=0.106); and the pooled effect on mild-to-severe depressive symptoms, among the poorest of the poor, is marginally no longer statistically significant (q-value=0.108) (the differential effect between Inviters with the High-SES Guest List and Inviters with the

Low-SES Guest List remains statistically significant with $q = 0.077$).

Table E.3: Main Estimates with FDR Sharpened Q-Values

	(1) Sent Any Invitation	(2) Sent High-SES Invitation	(3) Sent Low-SES Invitation	(4) Food Security Index	(5) Farm Yields USD Value	(6) Mild-to-Severe Depression	(7) Self-Employed Last Month	(8)	(9) Grows Cash Crops	(10) Potential for Risk-Sharing Index	(11) Number of Borrowers	(12) Borrowed Last 6 Months	(13) Network Degree	(14) Network Churn	(15) Depression: Poorest of the Poor	(16) Lonely
	1 Month	1 Month	1 Month	1 Year	1 Year	1 Year	1 Month	1 Year	1 Year	1 Year	1 Year	1 Year	1 Year	1 Year	1 Year	1 Year
Panel A: Pooled Inviter Treatment Effects																
All Inviters	–	–	–	0.140**	5.496	-0.083***	0.053**	0.017	0.014	0.077	-0.153**	-0.030	0.030	0.430*	-0.089**	0.004
	–	–	–	(0.056)	(26.984)	(0.027)	(0.022)	(0.025)	(0.017)	(0.056)	(0.063)	(0.023)	(0.117)	(0.230)	(0.038)	(0.020)
Panel B: Pooled Inviter by Guest List Treatment Effects																
Inviter with High-SES Guest List	0.006	0.238***		0.226***	31.281	-0.045	0.092***	0.049	0.053**	0.133*	-0.199***	-0.044	0.038	0.350	-0.014	0.008
	(0.026)	(0.030)		(0.071)	(35.432)	(0.033)	(0.028)	(0.031)	(0.022)	(0.072)	(0.076)	(0.028)	(0.152)	(0.285)	(0.047)	(0.025)
Inviter with Low-SES Guest List	-0.039		0.254***	0.073	-13.383	-0.104***	0.013	-0.007	-0.002	-0.013	-0.146*	-0.040	-0.043	0.429	-0.127***	0.002
	(0.028)		(0.031)	(0.065)	(32.099)	(0.032)	(0.027)	(0.030)	(0.021)	(0.067)	(0.075)	(0.028)	(0.142)	(0.290)	(0.046)	(0.024)
Inviter with Random Guest List	–	–	–	0.105	-4.031	-0.092***	0.039	0.006	-0.005	0.076	-0.043	-0.006	0.074	0.485*	-0.135***	0.003
	–	–	–	(0.068)	(31.126)	(0.032)	(0.028)	(0.030)	(0.021)	(0.069)	(0.078)	(0.029)	(0.145)	(0.294)	(0.047)	(0.025)
Observations	1200	800	800	1528	1528	1528	1522	1528	1528	1528	1528	1528	1528	1528	768	1528
Control Mean	–	–	–	—	324.783	0.361	0.180	0.237	0.103	-0.001	1.660	0.810	6.446	5.702	0.359	0.132
P-values:																
Pooled: High-SES vs. Low-SES Guest List	0.106			0.025**	0.191	0.060*	0.005***	0.062*	0.016**	0.043**	0.461	0.877	0.604	0.791	0.015**	0.787
Pooled: Voucher vs. No Voucher	0.146	0.606	0.839	0.375	0.528	0.784	0.952	0.210	0.460	0.281	0.730	0.010**	0.243	0.269	0.682	0.040**
FDR Sharpened Q-values:																
Pooled Estimate				0.079*	0.541	0.067*	0.079*	0.469	0.469	0.206	0.079*	0.206	0.541	0.098*	0.079*	0.541
High-SES vs. Low-SES Guest List				0.082*	0.206	0.098*	0.079*	0.098*	0.079*	0.092*	0.469	0.541	0.492	0.541	0.079*	0.541
H+V = H+NV = L+V = L+NV	0.146															
Random: Voucher vs. No Voucher	0.541	0.082*	0.541													

Standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Appendix F Model Appendix

F.1 Model Predictions and Empirical Tests

Each of the following tests maps directly to a theoretical prediction, using the coefficients estimated in Equations 1–4 to empirically evaluate Lemma F.2 and Proposition 6.1(a)-(b). I additionally use the following two specifications, whose results are in Appendix Table C.2:

$$I_i^{H\text{-}only} = \beta_0^{H\text{-}only} + \beta_1^{H\text{-}only} T_i^{R,V} + \beta_2^{H\text{-}only} T_i^{H,NV} + \beta_3^{H\text{-}only} T_i^{H,V} X_i + \delta_i + \gamma_i + \epsilon_i \quad (6)$$

$$I_i^{L\text{-}only} = \beta_0^{L\text{-}only} + \beta_1^{L\text{-}only} T_i^{R,V} + \beta_2^{L\text{-}only} T_i^{L,NV} + \beta_3^{L\text{-}only} T_i^{L,V} + X_i + \delta_i + \gamma_i + \epsilon_i \quad (7)$$

where I_i^{H-only} takes the value of 1 if an Inviter sends an invitation to a High-SES Guest and sends no invitations to any Low-SES Guests; and I_i^{L-only} takes the value of 1 if an Inviter sends an invitation to a Low-SES Guest and sends no invitations to any High-SES Guests. These specifications are estimated only among Inviters with the Random Guest List and the other sub-arm for whom the outcome is not mechanically zero.

I use Inviters' decisions about who to send invitations to, across treatment conditions, as a revealed preference measure of how Inviters perceive the returns to within-SES and cross-SES relationships, given different feasible levels of financial investment in the relationship.

Evaluating Model Assumptions

Model Assumption 6.1: Each Type has an Absolute Advantage

Interactions with one type is better at generating instrumental utility returns, and interactions with the other type is better at generating intrinsic utility returns.

Test: *Inviters with the High-SES Guest List have greater second-stage treatment effects on some outcomes than Inviters with the Low-SES Guest List; and Inviters with the Low-SES Guest List have greater second-stage treatment effects on other outcomes than Inviters with the High-SES Guest List*

Model Lemma F.2: Subsidies Induce Diversification

Interactions with high-SES and low-SES women generate different types of benefits, which agents both value, so agents seek economic diversity in their networks (this allows them to achieve more diversity in the potential benefits of their social interactions). Consequently, the marginal returns to a subsidy are large for mixed bundles faster than they are for homogeneous bundles [Proposition F.2]

Test: $\beta_1^{H\&L} > \beta_1^{H\text{-}only}$ and $\beta_1^{H\&L} > \beta_1^{L\text{-}only}$ (Equations 4, 6 and 7)

Among Inviters with the Random Guest List, the Voucher increases the probability of selecting a mixed bundle to a greater extent than it increases the probability of selecting a High-SES-only bundle, ro a Low-SES-only bundle.

Model Predictions in the Experiment

Model Proposition 6.1: Diversification conditional on the zero-subsidy counterfactual

Agents will be most likely to invest more in the type whose returns are more sensitive to financial investment in response to a subsidy. Thus, if the benefits of cross-SES (within-SES) relationships are more sensitive to financial investment than the benefits of within-SES (cross-SES) relationships, then prices constrain cross-SES (within-SES) linking [Proposition 6.1(a)]. Crucially, this is because the rate of substitution between initiating links within-SES and across-SES changes with the price of investment, and so we should not expect to see the same when substitution is not an option [Proposition 6.1(b)].

Test: $\beta_1^H > 0$ or $\beta_1^L > 0$, and $\beta_2 = \beta_3 = \beta_4 = \beta_5$ (Equations 1 and 2)

- $\beta_1^H > 0 \quad \Rightarrow$ high prices of social interaction contribute to income-based homophily
- $\beta_1^L > 0 \quad \Rightarrow$ high prices of social interaction contribute to economic connectedness

Among Inviters with the Random Guest List, the Voucher increases the probability of inviting either High-SES or Low-SES women. Inviters with the High-SES Guest List and Low-SES Guest List invite women at the same rate, with or without the Voucher.

This test determines if prices inhibits linking with specific types of people, which generates broader implications for network composition by income.

Ruling out Alternate Models

My model rests on the assumption that women face trade-offs between the intrinsic utility benefits of low-SES interactions and the financial benefits of high-SES interactions, and that the returns and thus this trade-off are sensitive to financial investment in the interaction. This implies that linking patterns that appear as either an intrinsic preference for homophily or a state-independent constraint from forming links across-SES are actually an optimal price response.

I can test alternate models where effort costs and prices impose feasibility constraints. I describe two alternate tests and the models they imply below:

- $\beta_2 < \beta_4$ (Equation 1) *Without the Voucher, Inviters with the High-SES Guest List send fewer invitations than Inviters with the Low-SES Guest List*: Holding returns constant, the cost of effortful initiation with high-SES women constrains cross-SES relationships

- $\beta_1^H > 0$ and $\beta_3 - \beta_2 > \beta_5 - \beta_4$ (Equations 1 and 2) *The Voucher increases invitations to high-SES women among Inviters with the High-SES Guest List and Inviters with the Random Guest List, more so than it increases invitations to low-SES Guests among Inviters with the Low-SES Guest List*: High prices constrain cross-SES relationships because high-SES links are infeasible without financial investment

Table F.1: Model Tests

Returns to Economic Diversity			
Test	**Coefs.**	**Interpretation**	**Table**
✓ (1)$\beta_1^{H\&L} - \beta_1^{L\text{-}only} > 0$, and	0.202^{***}	✓ Agents value the benefits of both types	Table C.2, Col. 2
✓ (2) $\beta_1^{H\&L} - \beta_1^{H\text{-}only} > 0$	0.128^{**}	✓ Subsidies induce diversification	Table C.2, Col. 3
Understanding Homophily			
Test	**Coefs.**	**Interpretation**	**Table**
× $\beta_2 - \beta_4 < 0$	0.08	× Effort costs do not differentially constrain cross-SES linking	Table 1, Col. 1
✓ (1) $\beta_1^H > 0$, and	0.079^{*}	✓ Price constrains cross-SES linking	Table 1, Col. 3
× (2) $(\beta_3 - \beta_2) - (\beta_5 - \beta_4) > 0$	-0.058	× Price is not a feasibility constraint	Table 1, Col. 1
✓ (3) $\beta_3 = \beta_2 = \beta_5 = \beta_4$	$p = 0.358$	✓ Price variation drives substitution ⇒ Trade-off between within-SES and cross-SES linking changes with price	Table 1, Col. 1
× $\beta_1^H - \beta_1^L > 0$	0.073	◯ High prices might constrain economic connectedness overall	Table 1, Col. 2/3

Notes: This table presents a series of empirical tests that validate assumptions and predictions of the model. Column (1) presents the empirical test, using notation from the empirical specifications outlined in Section 3.4. Column (2) presents the coefficient from the model test in column (1). Column (3) interprets these coefficients, and column (4) references the tables where the coefficients in column (2) are reported. Appendix Section F.1 outlines the relationship between the model predictions and empirical tests in more detail. The ✓ symbol denotes tests where we can reject the null, the × symbol denotes tests where we cannot reject the null, and the ◯ symbol denotes tests where the result is unclear. The last test is determined to be unclear because, while I cannot reject the null, the test is under-powered.

F.2 Model Proofs

Notation

The population consists of two groups, $g \in \{1, 2\}$, with preference weights (α_g, β_g) on consumption and intrinsic utility. By the mirror condition of the main text,

$$\psi := \alpha_1 = \beta_2, \qquad \eta := \beta_1 = \alpha_2, \qquad \eta > \psi > 0,$$

so that group G_1 has weights (ψ, η) and group G_2 has weights (η, ψ): the two groups place on their own preferred good exactly the weight the other places on the good it prefers. The functions u and h are common to both groups. No weight is normalized to one, so the two groups' utilities are built from the same pair of constants and are expressed in common units; the only remaining freedom is a common rescaling, which multiplies both groups' values by the same positive number and leaves comparisons between them unchanged.

For a bundle $S = (S_H, S_L)$ write $n_S := S_H + S_L$ and

$$Z(S, I, \tau) = \bar{y} + \mathbb{Y}(S, I) - p(\tau)\, I\, n_S, \qquad V(S, I) = \bar{v} + \mathbb{V}(S, I) - S_H c_H - S_L c_L.$$

The feasibility condition $I = 1 \Rightarrow S_H + S_L > 0$ from the main text restricts the investment levels available at a bundle: write

$$\mathbb{I}(S) := \begin{cases} \{0, 1\} & \text{if } n_S > 0, \\ \{0\} & \text{if } n_S = 0, \end{cases}$$

so that investment is a genuine choice only where there is a relationship to invest in. The value of a bundle to group g at its best investment level is then a maximum over at most two points,

$$W_S^g(\tau) \;:=\; \max_{I \in \mathbb{I}(S)} \Big[\alpha_g\, u\big(Z(S, I, \tau)\big) + \beta_g\, h\big(V(S, I)\big)\Big],$$

with maximizer $I^{*,S}(g, \tau)$ and associated consumption $Z^{*,S}(g, \tau) := Z\big(S, I^{*,S}(g, \tau), \tau\big)$.

Note that $Z(S, 0, \tau)$ does not depend on τ, while $Z(S, 1, \tau)$ is strictly increasing in τ at rate n_S.

Write

$$\tau_S^+(g) \;:=\; \inf\Big\{\tau \geq 0 :\; \alpha_g\, u\big(Z(S, 1, \tau)\big) + \beta_g\, h\big(V(S, 1)\big) \;\geq\; \alpha_g\, u\big(Z(S, 0, \tau)\big) + \beta_g\, h\big(V(S, 0)\big)\Big\}$$

for the smallest subsidy at which investing at bundle S is worthwhile for group g, defined for $n_S > 0$.

A woman holding one relationship has four candidate second relationships, distinguished by which type is added and which she already holds. Write the net gain to group g from each as

$$\begin{aligned}
\Delta^g_{H|L}(\tau) &:= W^g_{(1,1)}(\tau) - W^g_{(0,1)}(\tau) && \text{a high-SES relationship added to a low-SES one,}\\
\Delta^g_{L|H}(\tau) &:= W^g_{(1,1)}(\tau) - W^g_{(1,0)}(\tau) && \text{a low-SES relationship added to a high-SES one,}\\
\Delta^g_{H|H}(\tau) &:= W^g_{(2,0)}(\tau) - W^g_{(1,0)}(\tau) && \text{a second high-SES relationship,}\\
\Delta^g_{L|L}(\tau) &:= W^g_{(0,2)}(\tau) - W^g_{(0,1)}(\tau) && \text{a second low-SES relationship.}
\end{aligned}$$

Each side of each difference is already a maximum over $I \in \{0,1\}$, so each gain compares the value of the new bundle at its best investment level against the value of the current bundle at its own best investment level, net of the effort cost of the person added. Write $R^g(\tau) := W^g_{(1,0)}(\tau) - W^g_{(0,1)}(\tau)$ for the value to group g of one high-SES relationship over one low-SES relationship, and note that because $W^g_{(1,1)}(\tau)$ appears in the first two gains and cancels,

$$\Delta^g_{H|L}(\tau) - \Delta^g_{L|H}(\tau) \;=\; R^g(\tau). \tag{8}$$

Assumptions

Assumption F.1 (Each type has an absolute advantage in producing one type of utility)**.** *Denote types H and L with θ and θ'. θ has an absolute advantage in generating a financial return, θ' has an absolute advantage in generating intrinsic value, and no complementarity can overturn these advantages.*

1. *For both $\bar{I} \in \{0,1\}$:*

$$\begin{aligned}
\mathbb{Y}(S_\theta = 1, S_{\theta'} = 0, \bar{I}) \;&>\; \mathbb{Y}(S_\theta = 0, S_{\theta'} = 1, \bar{I})\\
\mathbb{V}(S_\theta = 1, S_{\theta'} = 0, \bar{I}) \;&<\; \mathbb{V}(S_\theta = 0, S_{\theta'} = 1, \bar{I})
\end{aligned}$$

2. *The marginal benefit of the second θ and second θ' maintain their respective advantages. For both $\bar{I} \in \{0,1\}$:*

$$\mathbb{Y}(2,0,\bar{I}) > \mathbb{Y}(1,1,\bar{I}) > \mathbb{Y}(0,2,\bar{I}), \qquad \mathbb{V}(0,2,\bar{I}) > \mathbb{V}(1,1,\bar{I}) > \mathbb{V}(2,0,\bar{I})$$

Assumption F.2 (One type is more responsive to investment)**.** *For every $g, g' \in \{1,2\}$, every $\tau \in \mathscr{T}$, and every pair of bundles S, S' with $S_H = S'_H + 1$ and $S_L = S'_L - 1$ (so*

$n_S = n_{S'}$),

$$\underbrace{\alpha_g\Big[u\big(Z(S,1,\tau)\big) - u\big(Z(S,0,\tau)\big)\Big] + \beta_g\Big[h\big(V(S,1)\big) - h\big(V(S,0)\big)\Big]}_{\textit{gain to group } g \textit{ from investing at } S}$$
$$> \underbrace{\alpha_{g'}\Big[u\big(Z(S',1,\tau)\big) - u\big(Z(S',0,\tau)\big)\Big] + \beta_{g'}\Big[h\big(V(S',1)\big) - h\big(V(S',0)\big)\Big]}_{\textit{gain to group } g' \textit{ from investing at } S'}.$$

Assumption F.3 (The two groups are symmetric absent the subsidy)**.**

$$\underbrace{\Big[\alpha_1 u\big(Z((1,1),0,0)\big) + \beta_1 h\big(V((1,1),0)\big)\Big] - \Big[\alpha_1 u\big(Z((0,1),0,0)\big) + \beta_1 h\big(V((0,1),0)\big)\Big]}_{\textit{gain to } G_1 \textit{ of adding a high-SES relationship at } \tau=0}$$
$$= \underbrace{\Big[\alpha_2 u\big(Z((1,1),0,0)\big) + \beta_2 h\big(V((1,1),0)\big)\Big] - \Big[\alpha_2 u\big(Z((1,0),0,0)\big) + \beta_2 h\big(V((1,0),0)\big)\Big]}_{\textit{gain to } G_2 \textit{ of adding a low-SES relationship at } \tau=0}.$$

Assumption F.4 (Taste heterogeneity is small relative to the investment advantage)**.** *Let $\psi := \alpha_1 = \beta_2$ and $\eta := \beta_1 = \alpha_2$, with $\eta > \psi > 0$. For every $\tau \in \mathscr{T}$, where either G_1 or G_2 invests,*

$$(\eta - \psi) \cdot \max\Big\{0,\ \Big[u\big(Z((1,1),1,\tau)\big) - u\big(Z((1,1),0,\tau)\big)\Big] - \Big[h\big(V((1,1),1)\big) - h\big(V((1,1),0)\big)\Big]\Big\}$$
$$< \max\Big\{0, \underbrace{\eta\Big[u\big(Z((1,0),1,\tau)\big) - u\big(Z((1,0),0,\tau)\big)\Big] + \psi\Big[h\big(V((1,0),1)\big) - h\big(V((1,0),0)\big)\Big]}_{\textit{gain to } G_2 \textit{ from investing at her baseline}}\Big\}$$
$$- \max\Big\{0, \underbrace{\psi\Big[u\big(Z((0,1),1,\tau)\big) - u\big(Z((0,1),0,\tau)\big)\Big] + \eta\Big[h\big(V((0,1),1)\big) - h\big(V((0,1),0)\big)\Big]}_{\textit{gain to } G_1 \textit{ from investing at her baseline}}\Big\}.$$

Assumption F.5 (The diversity premium exceeds the cross-type preference gap)**.** *For every $\tau \in \mathscr{T}$,*

$$\big|R^1(\tau)\big| < W^1_{(1,1)}(\tau) - W^1_{(2,0)}(\tau), \qquad \big|R^2(\tau)\big| < W^2_{(1,1)}(\tau) - W^2_{(0,2)}(\tau),$$

where both right-hand sides are strictly positive because $(\alpha_g, \beta_g) \in \mathscr{P}$ (Lemma F.1). That is, each group's preference for one type of relationship over the other is smaller than what she gains from holding one of each rather than two of the type she is drawn toward.

Preliminary Lemmas

Notation: Fix $\bar{I} \in \{0,1\}$ and denote,

$$\begin{aligned} Z_{11}(\bar{I}) &= \bar{y} + \mathbb{Y}(1,1,\bar{I}) - 2p(\tau)\bar{I}, \qquad & V_{11}(\bar{I}) &= \bar{v} + \mathbb{V}(1,1,\bar{I}) - c_H - c_L, \\ Z_{20}(\bar{I}) &= \bar{y} + \mathbb{Y}(2,0,\bar{I}) - 2p(\tau)\bar{I}, \qquad & V_{20}(\bar{I}) &= \bar{v} + \mathbb{V}(2,0,\bar{I}) - 2c_H, \\ Z_{02}(\bar{I}) &= \bar{y} + \mathbb{Y}(0,2,\bar{I}) - 2p(\tau)\bar{I}, \qquad & V_{02}(\bar{I}) &= \bar{v} + \mathbb{V}(0,2,\bar{I}) - 2c_L. \end{aligned}$$

Lemma F.1 (A preference space where diversity is optimal)**.** *Let Assumption F.1 hold, and for $\bar{I} \in \{0,1\}$ define*

$$\underline{r}(\bar{I}) := \frac{h\big(V_{02}(\bar{I})\big) - h\big(V_{11}(\bar{I})\big)}{u\big(Z_{11}(\bar{I})\big) - u\big(Z_{02}(\bar{I})\big)}, \qquad \overline{r}(\bar{I}) := \frac{h\big(V_{11}(\bar{I})\big) - h\big(V_{20}(\bar{I})\big)}{u\big(Z_{20}(\bar{I})\big) - u\big(Z_{11}(\bar{I})\big)}.$$

Both are well defined because u is strictly increasing and, by Assumption F.1 part (2), both denominators are strictly positive. Let

$$\underline{r} := \max_{\bar{I}\in\{0,1\},\tau\in\mathscr{T}} \underline{r}(\bar{I}), \qquad \overline{r} := \min_{\bar{I}\in\{0,1\},\tau\in\mathscr{T}} \overline{r}(\bar{I}), \qquad \mathscr{P} := \Big\{(\alpha,\beta) \in \mathbb{R}^2_{++} : \ \underline{r} < \tfrac{\alpha}{\beta} < \overline{r}\Big\}.$$

Then for every $(\alpha,\beta) \in \mathscr{P}$ and both $\bar{I} \in \{0,1\}$,

$$\alpha\, u\big(Z_{11}(\bar{I})\big) + \beta\, h\big(V_{11}(\bar{I})\big) \ > \ \max\Big\{\alpha\, u\big(Z_{20}(\bar{I})\big) + \beta\, h\big(V_{20}(\bar{I})\big),\ \ \alpha\, u\big(Z_{02}(\bar{I})\big) + \beta\, h\big(V_{02}(\bar{I})\big)\Big\},$$

so the diverse bundle is strictly preferred to both concentrated bundles at either investment level. Moreover $\mathscr{P} \neq \varnothing$ if and only if $\underline{r} < \overline{r}$ and $\overline{r} > 0$.

Lemma F.2 (Subsidies induce diversification)**.** *Fix $(\alpha,\beta) \in \mathscr{P}$. Then, $S_L^*(\tau) = 0, S_H^*(\tau) = 0, \forall \tau \in \mathscr{T}$, or $\min\{\tau_{H|L}, \tau_{L|H}\} < \infty$, and for every $\tau > \min\{\tau_{H|L}, \tau_{L|H}\}$ the unique two-relationship optimum is $(1,1)$; in particular $S_H^*(\tau) < 2$ and $S_L^*(\tau) < 2$.*

Proof of Lemmas

Proof of Lemma F.1. Proof. Fix $\bar{I}$. The diverse bundle is strictly preferred to $(2,0)$ if and only if

$$\beta\big[h(V_{11}) - h(V_{20})\big] \ > \ \alpha\big[u(Z_{20}) - u(Z_{11})\big].$$

The bracket on the right is strictly positive, since $Z_{20} > Z_{11}$ by Assumption F.1 part (2) and u is strictly increasing; the sign of the bracket on the left is not restricted. Dividing both sides by $\beta\big[u(Z_{20}) - u(Z_{11})\big] > 0$ preserves the inequality in both directions and yields the equivalent condition $\alpha/\beta < \overline{r}(\bar{I})$.

Symmetrically, the diverse bundle is strictly preferred to $(0,2)$ if and only if

$$\alpha\big[u(Z_{11}) - u(Z_{02})\big] \;>\; \beta\big[h(V_{02}) - h(V_{11})\big],$$

and dividing by $\beta\big[u(Z_{11}) - u(Z_{02})\big] > 0$, again strictly positive by Assumption F.1 part (2), yields $\alpha/\beta > \underline{r}(\bar{I})$. Imposing both conditions at both investment levels replaces the bounds by $\underline{r}$ and $\overline{r}$, which gives the displayed inequality for every $(\alpha,\beta) \in \mathscr{P}$.

For the last claim, α/β ranges over all of $(0,\infty)$ as (α,β) ranges over $\mathbb{R}^2_{++}$, so $\mathscr{P}$ is non-empty exactly when the interval $(\underline{r},\overline{r})$ meets $(0,\infty)$, that is, when it is non-degenerate and has positive upper endpoint. □

Proof of Lemma F.2. **Step 1: Values are continuous and non-decreasing in the subsidy**

Since $I \in \{0,1\}$, $W^g_S(\tau)$ is the maximum of the two functions $\alpha_g\, u\big(Z(S,0,\tau)\big)+\beta_g\, h\big(V(S,0)\big)$ and $\alpha_g\, u\big(Z(S,1,\tau)\big) + \beta_g\, h\big(V(S,1)\big)$, each continuous in τ. The first function does not depend on τ (because $I = 0$). $Z(S,1,\tau)$ is strictly increasing in τ at rate n_S and u is increasing; so each branch is non-decreasing in τ and hence so is their maximum. Thus, the second function is strictly increasing in τ and $W^g_S(\tau)$ is continuous in τ.

A maximum of finitely many continuous functions is continuous. Thus, for every bundle S and each group g, $W^g_S(\tau)$ is continuous in τ and non-decreasing in τ. Consequently each of the four gains $\Delta^g_{H|L}$, $\Delta^g_{L|H}$, $\Delta^g_{H|H}$, $\Delta^g_{L|L}$ is continuous in τ.

Step 2: Value gaps between bundles of unequal size diverge

Fix a group g and let S, S' be bundles with $n_S > n_{S'}$. Then $W^g_S(\tau) - W^g_{S'}(\tau) \to +\infty$ as $\tau \to \infty$. Consequently the threshold

$$\tau^\dagger_{S,S'}(g) := \inf\big\{\tau \geq 0 :\; W^g_S(\tau) - W^g_{S'}(\tau) \geq 0\big\}$$

is finite, and if $W^g_S(0) - W^g_{S'}(0) \;<\; 0$ it is strictly positive and satisfies $W^g_S\big(\tau^\dagger_{S,S'}\big) - W^g_{S'}\big(\tau^\dagger_{S,S'}\big) = 0$.

Evaluate S at $I = 1$, which is feasible though not necessarily optimal, so $W^g_S(\tau) \geq \alpha_g\, u\big(Z(S,1,\tau)\big)+\beta_g\, h\big(V(S,1)\big)$. Since I takes only two values, $W^g_{S'}(\tau) = \max\big\{\alpha_g\, u\big(Z(S',0)\big)+ \beta_g\, h\big(V(S',0)\big),\; \alpha_g\, u\big(Z(S',1,\tau)\big) + \beta_g\, h\big(V(S',1)\big)\big\}$. The first term is constant in τ. For the second,

$$Z(S,1,\tau) - Z(S',1,\tau) = \Big[\mathbb{Y}(S,1) - \mathbb{Y}(S',1)\Big] + \big(\tau - p^I\big)\big(n_S - n_{S'}\big),$$

a fixed constant plus a term growing at rate $n_S - n_{S'} \geq 1$; the same holds against $Z(S',0)$,

which is constant in τ. So $Z(S,1,\tau)$ exceeds both candidate consumption levels for τ beyond some finite point, and the gap diverges. Since V does not depend on τ and takes finitely many values, $\beta_g\big[h\big(V(S,1)\big) - h\big(V(S',I)\big)\big]$ is bounded below by some $-\bar{h}_0$. By the mean value theorem and $u' \geq \underline{u}' > 0$, $\alpha_g \geq \psi > 0$,

$$\alpha_g\Big[u\big(Z(S,1,\tau)\big) - u\big(Z(S',I,\tau)\big)\Big] \;\geq\; \psi\,\underline{u}' \cdot \Big[Z(S,1,\tau) - Z(S',I,\tau)\Big] \;\xrightarrow[\tau\to\infty]{}\; +\infty$$

for each $I \in \{0,1\}$, and hence for the maximum over the two. Adding $-\bar{h}_0$ leaves the divergence, so $W^g_S(\tau) - W^g_{S'}(\tau) \to +\infty$.

The defining set of $\tau^\dagger_{S,S'}$ is therefore non-empty, and bounded below by 0, so the infimum is finite. W^g_S and $W^g_{S'}$ are continuous in τ by Step 1 of Proof F.2, so if the gap is strictly negative at $\tau = 0$ the infimum is strictly positive and, taking limits from below and above, the gap is exactly zero there.

Step 3: Thresholds and the two-relationship optimum

Apply Step 2 to four pairs, writing

$$\tau_{H|L} := \tau^\dagger_{(1,1),(0,1)}, \quad \tau_{L|H} := \tau^\dagger_{(1,1),(1,0)}, \quad \tau_{H|H} := \tau^\dagger_{(2,0),(1,0)}, \quad \tau_{L|L} := \tau^\dagger_{(0,2),(0,1)},$$

each the smallest subsidy at which the corresponding gain $\Delta_{H|L}$, $\Delta_{L|H}$, $\Delta_{H|H}$, $\Delta_{L|L}$ changes sign; all four are finite by Step 2.

Since $(\alpha,\beta) \in \mathscr{P}$, the diverse bundle yields strictly higher utility than either concentrated bundle at both investment levels (Lemma F.1), and this ranking is preserved on maximizing over the two levels; so $(2,0)$ and $(0,2)$ are dominated, and the diverse bundle weakly beats is the unique two-relationship optimum. □

Lemma F.3 (A same-type addition is harder than a cross-type addition)**.** *For each group g with $(\alpha_g,\beta_g) \in \mathscr{P}$,*

$$\tau_{L|H}(g) \;<\; \tau_{H|H}(g) \qquad \textit{and} \qquad \tau_{H|L}(g) \;<\; \tau_{L|L}(g).$$

Proof. Since $(\alpha_g,\beta_g) \in \mathscr{P}$, for both investment levels the diverse bundle strictly dominates $(2,0)$ and $(0,2)$, and the ordering survives maximizing over the two levels: $W^g_{(1,1)}(\tau) > W^g_{(2,0)}(\tau)$ and $W^g_{(1,1)}(\tau) > W^g_{(0,2)}(\tau)$ for all τ. At any $\tau \geq \tau_{H|H}(g)$ we have $W^g_{(2,0)}(\tau) \geq W^g_{(1,0)}(\tau)$ by Step 2 of Proof F.2, so

$$W^g_{(1,1)}(\tau) > W^g_{(2,0)}(\tau) \;\geq\; W^g_{(1,0)}(\tau) \qquad\Longrightarrow\qquad \Delta^g_{L|H}(\tau) > 0.$$

Since $\Delta^g_{L|H}\big(\tau_{L|H}(g)\big) = 0$, the two points cannot coincide, and $\tau_{L|H}(g) < \tau_{H|H}(g)$. The

identical argument with $(0,2)$, $(0,1)$, and $\Delta^g_{H|L}$ in place of $(2,0)$, $(1,0)$, and $\Delta^g_{L|H}$ gives the second inequality. Neither uses Assumption F.2: both follow from both groups' weights lying in $\mathscr{P}$ alone.

Economically: a subsidy at which an agent is willing to add a second relationship of the type she already holds is a subsidy at which she would strictly have preferred one of each, so every subsidy that induces concentration lies above one that induces diversification. □

Proof of Proposition 6.1

Preliminaries. For a group g and a bundle S where $n_S > 0$, separate the value of the bundle into its non-investing level N^g_S and the gain from investing $G^g_S(\tau)$:

$$N^g_S := \alpha_g\, u\big(Z(S,0,\tau)\big)+\beta_g\, h\big(V(S,0)\big), \qquad G^g_S(\tau) := \Big[\alpha_g\, u\big(Z(S,1,\tau)\big)+\beta_g\, h\big(V(S,1)\big)\Big]-N^g_S,$$

where N^g_S does not depend on τ because $Z(S,0,\tau)$ does not. $G^g_S(\tau)$ is exactly the gain from investing at S that appears in Assumption F.2, and

$$W^g_S(\tau) \;=\; N^g_S + \max\big\{0,\; G^g_S(\tau)\big\}. \tag{9}$$

$Z(S,1,\tau)$ is strictly increasing in τ and u is increasing, so G^g_S is strictly increasing in τ. $G^g_S(0) \le 0$ for every S and g because $I^*_g(0) = 0$ by construction. Since $\tau^+_S(g)$ is the smallest subsidy where investment is worthwhile, it follows that $\tau^+_S(g) > 0$ for every S and g. It is convenient to abbreviate the two components of the gain from investing,

$$\Delta u_S(\tau) := u\big(Z(S,1,\tau)\big) - u\big(Z(S,0,\tau)\big), \qquad \Delta h_S := h\big(V(S,1)\big) - h\big(V(S,0)\big),$$

so that $G^g_S(\tau) = \alpha_g\, \Delta u_S(\tau) + \beta_g\, \Delta h_S$. By symmetry of G_1 and G_2,

$$G^1_S(\tau) = \psi\, \Delta u_S(\tau) + \eta\, \Delta h_S, \qquad G^2_S(\tau) = \eta\, \Delta u_S(\tau) + \psi\, \Delta h_S.$$

Proof of Proposition 6.1(a). **Step 1: The subsidy makes investment worthwhile at G_2's baseline before G_1's.** Apply Assumption F.2 with $S = (1,0)$, $S' = (0,1)$, $g = 2$ and $g' = 1$. The assumption reads

$$\begin{aligned} &\alpha_2\Big[u\big(Z((1,0),1,\tau)\big) - u\big(Z((1,0),0,\tau)\big)\Big] + \beta_2\Big[h\big(V((1,0),1)\big) - h\big(V((1,0),0)\big)\Big] \\ &> \alpha_1\Big[u\big(Z((0,1),1,\tau)\big) - u\big(Z((0,1),0,\tau)\big)\Big] + \beta_1\Big[h\big(V((0,1),1)\big) - h\big(V((0,1),0)\big)\Big]. \end{aligned}$$

By the definition of G^g_S in the Preliminaries, the left-hand side is $G^2_{(1,0)}(\tau)$ and the right-

hand side is $G^1_{(0,1)}(\tau)$, so

$$G^2_{(1,0)}(\tau) \;>\; G^1_{(0,1)}(\tau) \qquad \text{for every fixed } \tau \in \mathscr{T}.$$

Each side is strictly increasing in τ, and non-positive at $\tau = 0$ because $I^*_g(0) = 0$ at every bundle. By definition, $\tau^+_{(0,1)}(1) = \inf\{\tau \geq 0 : G^1_{(0,1)}(\tau) \geq 0\}$, so $G^1_{(0,1)}\big(\tau^+_{(0,1)}(1)\big) \geq 0$; the displayed inequality evaluated at that subsidy then gives $G^2_{(1,0)}\big(\tau^+_{(0,1)}(1)\big) > 0$. Hence $\tau^+_{(0,1)}(1)$ belongs to $\{\tau \geq 0 : G^2_{(1,0)}(\tau) \geq 0\}$, whose infimum is $\tau^+_{(1,0)}(2)$, so $\tau^+_{(1,0)}(2) \leq \tau^+_{(0,1)}(1)$. The inequality is strict: at $\tau^+_{(1,0)}(2)$ we have $G^2_{(1,0)} = 0$ by continuity and the definition of the infimum, while $G^2_{(1,0)}\big(\tau^+_{(0,1)}(1)\big) > 0$, so the two subsidies cannot coincide.

$$\tau^+_{(1,0)}(2) \;<\; \tau^+_{(0,1)}(1) \tag{10}$$

and on the interval between the two thresholds G_2 finds investment worthwhile with the single high-SES relationship she holds while G_1 does not at the single low-SES relationship she holds. This is the only role played by Assumption F.2, and it is the mechanism of the proposition: as the price of investment falls, G_2's outside option improves and G_1's does not.

Step 2: The gains to adding a second cross-type initiation begin equal and negative. At $\tau = 0$, neither G_1 nor G_2 invests, so by (9),

$$\Delta^1_{H|L}(0) = N^1_{(1,1)} - N^1_{(0,1)}, \qquad \Delta^2_{L|H}(0) = N^2_{(1,1)} - N^2_{(1,0)}.$$

These are the two sides of Assumption F.3, so

$$\Delta^1_{H|L}(0) \;=\; \Delta^2_{L|H}(0), \tag{11}$$

and both are strictly negative because every group's zero-subsidy optimum is a one-relationship bundle. Absent the subsidy, G_1 is exactly as far from wanting the high-SES relationship she lacks as G_2 is from wanting the low-SES relationship she lacks, and neither wants it.

Step 3: G_1's cross-type gain rises faster than G_2's. Substituting (9) into the definition of $\Delta^g(\tau)$:

$$\Delta^1_{H|L}(\tau) = \Big[N^1_{(1,1)} - N^1_{(0,1)}\Big] + \max\{0, G^1_{(1,1)}(\tau)\} - \max\{0, G^1_{(0,1)}(\tau)\},$$
$$\Delta^2_{L|H}(\tau) = \Big[N^2_{(1,1)} - N^2_{(1,0)}\Big] + \max\{0, G^2_{(1,1)}(\tau)\} - \max\{0, G^2_{(1,0)}(\tau)\}.$$

By Step 2, the two bracketed terms are $\Delta^1_{H|L}(0)$ and $\Delta^2_{L|H}(0)$ respectively, and by (11)

they are equal, so they cancel on subtracting:

$$\Delta^1_{H|L}(\tau) - \Delta^2_{L|H}(\tau) = \underbrace{\Big[\max\{0, G^2_{(1,0)}(\tau)\} - \max\{0, G^1_{(0,1)}(\tau)\}\Big]}_{\text{improvement in } G_2\text{'s outside option beyond } G_1\text{'s}} - \underbrace{\Big[\max\{0, G^2_{(1,1)}(\tau)\} - \max\{0, G^1_{(1,1)}(\tau)\}\Big]}_{\text{residual}}. \tag{12}$$

The first bracket is non-negative: Step 1 gives $G^2_{(1,0)}(\tau) > G^1_{(0,1)}(\tau)$, and $x \mapsto \max\{0, x\}$ is non-decreasing, so $\max\{0, G^2_{(1,0)}(\tau)\} \geq \max\{0, G^1_{(0,1)}(\tau)\}$. It is strictly positive whenever $\tau > \tau^+_{(1,0)}(2)$: there $G^2_{(1,0)}(\tau) > 0$, so the first term equals $G^2_{(1,0)}(\tau)$, which exceeds the second term whether that term is $G^1_{(0,1)}(\tau)$ or zero.

For the second bracket, note that for any reals x, y,

$$\max\{0, x\} - \max\{0, y\} \;\leq\; \max\{0, x - y\}\,: \tag{13}$$

Since $G^1_{(1,1)}$ and $G^2_{(1,1)}$ differ only in which weight multiplies which component,

$$G^2_{(1,1)}(\tau) - G^1_{(1,1)}(\tau) = (\eta - \psi)\big[\Delta u_{(1,1)}(\tau) - \Delta h_{(1,1)}\big],$$

so applying (13) with $x = G^2_{(1,1)}(\tau)$ and $y = G^1_{(1,1)}(\tau)$ and using $\eta > \psi$, the residual is at most

$$(\eta - \psi)\ \max\big\{0,\ \Delta u_{(1,1)}(\tau) - \Delta h_{(1,1)}\big\}.$$

The residual arises as an artifact of taste-heterogeneity, and vanishes when $\eta = \psi$. Assumption F.4 bounds heterogeneity in preferences between G_1 and G_2, and states precisely that this bound is strictly smaller than the first bracket of (12).

Assumption F.4 holds strictly and its left-hand side is non-negative, since $\eta > \psi$; so its right-hand side is strictly positive at every $\tau \in \mathscr{T}$. That right-hand side is the first bracket of (12), so the bracket is strictly positive throughout $\mathscr{T}$.

Hence, for every $\tau \in \mathscr{T}$,

$$\Delta^1_{H|L}(\tau) \;>\; \Delta^2_{L|H}(\tau). \tag{14}$$

Step 4: G_1's cross-type margin opens before G_2's. By Step 2 of the proof of Lemma F.2, $\tau_{H|L}(1)$ is finite; and since $\Delta^1_{H|L}(0) < 0$ by Step 2 above, the same result gives $\tau_{H|L}(1) > 0$ and $\Delta^1_{H|L}\big(\tau_{H|L}(1)\big) = 0$. Evaluating (14) at $\tau_{H|L}(1)$ gives $\Delta^2_{L|H}\big(\tau_{H|L}(1)\big) < 0$.

For $\tau < \tau_{H|L}(1)$ we have $\Delta^1_{H|L}(\tau) < 0$, since $\tau_{H|L}(1)$ is the infimum of the set on which $\Delta^1_{H|L}$ is non-negative. Applying (14) at a $\tau < \tau_{H|L}(1)$ gives $\Delta^2_{L|H}(\tau) < \Delta^1_{H|L}(\tau) < 0$. By Step 2 of the proof of Lemma F.2, $\tau_{L|H}(2)$ is the infimum of $\{\tau \geq 0 : \Delta^2_{L|H}(\tau) \geq 0\}$, a

non-empty set; every member of it therefore exceeds $\tau_{H|L}(1)$, and so does its infimum:

$$\tau_{H|L}(1) \;<\; \tau_{L|H}(2). \tag{15}$$

Step 5: The interval of subsidies. By the definitions of $\Delta^g_{H|H}(\tau)$, $\Delta^g_{H|L}(\tau)$, $\Delta^g_{L|L}(\tau)$, and $\Delta^g_{L|H}(\tau)$,

$$\begin{aligned}\Delta^g_{H|H}(\tau) - \Delta^g_{H|L}(\tau) &= \left[W^g_{(2,0)}(\tau) - W^g_{(1,1)}(\tau)\right] - R^g(\tau)\\ \Delta^g_{L|L}(\tau) - \Delta^g_{L|H}(\tau) &= \left[W^g_{(0,2)}(\tau) - W^g_{(1,1)}(\tau)\right] + R^g(\tau),\end{aligned}$$

each of which follows from the definition of R^g and canceling $W^g_{(1,1)}(\tau)$. Assumption F.5 at $g = 1$ gives $\Delta^1_{H|H}(\tau) < \Delta^1_{H|L}(\tau)$ and at $g = 2$ gives $\Delta^2_{L|L}(\tau) < \Delta^2_{L|H}(\tau)$, for every $\tau \in \mathscr{T}$.

By Step 2 of the proof of Lemma F.2, $\tau_{H|H}(g)$ is the infimum of $\{\tau \geq 0 : \Delta^g_{H|H}(\tau) \geq 0\}$ and $\Delta^g_{H|L}\big(\tau_{H|L}(g)\big) = 0$. At any $\tau \leq \tau_{H|L}(g)$ we have $\Delta^g_{H|L}(\tau) \leq 0$, implying that $\Delta^g_{H|H}(\tau) < 0$ by the inequality above. Thus, there is no $\tau \leq \tau_{H|L}(g)$ where $\Delta^g_{H|H}(\tau) \geq 0$. Hence

$$\tau_{H|H}(1) > \tau_{H|L}(1), \qquad \tau_{L|L}(2) > \tau_{L|H}(2). \tag{16}$$

the second by the identical argument with $\Delta^g_{L|L}$ and $\Delta^g_{L|H}$ in place of $\Delta^g_{H|H}$ and $\Delta^g_{H|L}$.

By Lemma F.3, $\tau_{H|H}(g) > \tau_{L|H}(g)$ and $\tau_{L|L}(g) > \tau_{H|L}(g)$. Further, note that

$\tau_{L|H}(2) > \tau_{H|L}(1)$	by (15);
$\tau_{L|L}(1) > \tau_{H|L}(1)$	by Lemma F.3 at $g = 1$;
$\tau_{H|H}(1) > \tau_{H|L}(1)$	by Assumption F.5 at $g = 1$;
$\tau_{H|H}(2) > \tau_{L|H}(2) > \tau_{H|L}(1)$	by Lemma F.3 at $g = 2$;
$\tau_{L|L}(2) > \tau_{L|H}(2) > \tau_{H|L}(1)$	by Assumption F.5 at $g = 2$.

Hence

$$\tau^\star \in \Big(\tau_{H|L}(1),\; \min\big\{\tau_{H|H}(1),\; \tau_{L|L}(1),\; \tau_{L|H}(2),\; \tau_{H|H}(2),\; \tau_{L|L}(2)\big\}\Big) \tag{17}$$

is a non-empty interval. Fix $\tau^\star$ in it. By definition of $\tau^\star$ and $\Delta^g_S(\tau)$, $\Delta^1_{H|L}(\tau^\star) > 0$, while:

$$\Delta^1_{H|H}(\tau^\star) < 0, \qquad \Delta^1_{L|L}(\tau^\star) < 0, \qquad \Delta^2_{L|H}(\tau^\star) < 0, \qquad \Delta^2_{H|H}(\tau^\star) < 0, \qquad \Delta^2_{L|L}(\tau^\star) < 0. \tag{18}$$

Of the choices of adding a new initiation that each group can make to the subsidy, only one repays its effort cost at $\tau^\star$: a high-SES relationship added to a low-SES one, by a member of G_1.

Step 6: G_1 initiates a high-SES relationship, and invests. By (18), $\Delta^1_{H|L}(\tau^\star) > 0$, so $W^1_{(1,1)}(\tau^\star) > W^1_{(0,1)}(\tau^\star)$ and $(0,1)$ is not optimal; and $\Delta^1_{L|L}(\tau^\star) < 0$, so $(0,2)$ is worse than $(0,1)$ and hence not optimal. The empty bundle is never selected, by Lemma F.2. Since $(\alpha_1, \beta_1) \in \mathscr{P}$, Lemma F.1 gives $W^1_{(1,1)}(\tau^\star) > \max\left\{W^1_{(2,0)}(\tau^\star), W^1_{(0,2)}(\tau^\star)\right\}$, so neither concentrated bundle is optimal: each is strictly beaten by the available bundle $(1,1)$.

Her optimum is therefore $(1,1)$ or $(1,0)$, and in either case $S_H = 1$ where previously $S_H = 0$. Which of the two she selects follows from the identity (8): since $\Delta^1_{H|L}(\tau^\star) > 0$, we have $\Delta^1_{L|H}(\tau^\star) = \Delta^1_{H|L}(\tau^\star) - R^1(\tau^\star)$, so $(1,1)$ beats $(1,0)$ exactly when $R^1(\tau^\star) < \Delta^1_{H|L}(\tau^\star)$. She retains her low-SES relationship if her preference for one high-SES relationship over one low-SES relationship falls short of what the high-SES addition is worth to her, and replaces it otherwise. The proposition allows both cases.

She invests at that optimum. A bundle evaluated at $I = 0$ has value N^1_S, which does not depend on τ and so equals its value at $\tau = 0$. Her baseline $(0,1)$ was optimal at $\tau = 0$, and hence weakly better than the bundle she now selects evaluated at $I = 0$; and $W^1_{(0,1)}$ is non-decreasing in τ by Step 1 of the proof of Lemma F.2. So the selected bundle at $I = 0$ is weakly worse than $(0,1)$ at $\tau^\star$, while by the previous paragraph it is strictly better. The maximum in (9) is therefore attained at $I = 1$.

Step 7: G_2 does not respond. G_2's baseline is $(1,0)$. Of the two additions available to her, $\Delta^2_{L|H}(\tau^\star) < 0$ rules out $(1,1)$ and $\Delta^2_{H|H}(\tau^\star) < 0$ rules out $(2,0)$, both by (18). To rule out switching to the low-SES singleton, apply (9) to R^2:

$$R^2(\tau) = \left[N^2_{(1,0)} - N^2_{(0,1)}\right] + \max\{0, G^2_{(1,0)}(\tau)\} - \max\{0, G^2_{(0,1)}(\tau)\}.$$

At $\tau = 0$ neither bundle invests, so the bracket equals $R^2(0)$, which is strictly positive because $(1,0)$ is G_2's zero-subsidy optimum. Assumption F.2 applied with $S = (1,0)$, $S' = (0,1)$ and $g = g' = 2$ gives $G^2_{(1,0)}(\tau) > G^2_{(0,1)}(\tau)$, and $x \mapsto \max\{0, x\}$ is non-decreasing, so the last two terms sum to a non-negative number. Hence $R^2(\tau) \geq R^2(0) > 0$ for every τ, and $(1,0)$ beats $(0,1)$ at $\tau^\star$.

Finally, $(0,2)$ is strictly beaten by $(1,1)$ by Lemma F.1, and $(1,1)$ is beaten by $(1,0)$ since $\Delta^2_{L|H}(\tau^\star) < 0$; so $(1,0)$ beats $(0,2)$. The empty bundle is excluded by Lemma F.2. Her optimum is $(1,0)$, her baseline, so she initiates no additional low-SES relationship.

Step 8: Aggregation. Every member of G_1 raises S_H from zero to one, by Step 6, and either leaves S_L at one if the selects $(1,1)$ or lowers it to zero if she selects $(1,0)$. Every member of G_2 leaves both counts unchanged, by Step 7. No group lowers S_H and no group raises S_L. Since G_1 is non-empty, high-SES initiations strictly increase across all agents, and low-SES initiations weakly decrease. □

Proof of Proposition 6.1(b). One type is infeasible for each agent. Partition agents according to the available type:

$$E_H := \{k : \ L \text{ feasible}, H \text{ infeasible}\}, \qquad E_H := \{k : \ H \text{ feasible}, L \text{ infeasible}\},$$

with equal shares and partition assignment independent of the group g. Behavior is evaluated at the $\tau^\star$ of part (a), comparing $\tau = 0$ with $\tau^\star$.

Step 1: Each agent's feasible set is a single-type chain. By construction,

$$k \in E_H : \ \mathcal{S}_k = \{(0,0),(0,1),(0,2)\}, \qquad k \in E_L : \ \mathcal{S}_k = \{(0,0),(1,0),(2,0)\}.$$

In particular $(1,1) \notin \mathcal{S}_k$ for every k, whichever group she belongs to. The one margin that opens at $\tau^\star$ in part (a) is a cross-type addition, and it is unavailable to every agent. Each agent's feasible set is a chain of three bundles in the single type she can hold: none, one, or two.

Step 2: Under the hypothesis of the proposition, every agent holds one relationship at $\tau = 0$. The hypothesis is that each agent still initiates a relationship of her feasible type absent the subsidy. So no agent holds $(0,0)$ at $\tau = 0$: an agent in E_H holds $(0,1)$ and an agent in E_L holds $(1,0)$, regardless of which group she belongs to. In particular a member of G_1 subjected to the E_L shock holds one high-SES relationship, and a member of G_2 subjected to the E_H shock holds one low-SES relationship, even though in each case this is the type she does not prefer.

Two consequences. First, the two-relationship bundle in each chain is reached, if at all, by a same-type addition from a singleton, so the only margin any agent faces at $\tau^\star$ is $\Delta^g_{H|H}$ for agents in E_L and $\Delta^g_{L|L}$ for agents in E_H. Second, the empty bundle is never selected at any $\tau \geq 0$: its value does not depend on τ, a one-relationship bundle is preferred to it at $\tau = 0$ by hypothesis, and $\max_{S \in \mathcal{S}_k, n_S \leq 1} W^g_S(\tau)$ is non-decreasing in τ by Step 1 of the proof of Lemma F.2.

Step 3: Neither same-type margin opens at $\tau^\star$. By (18), $\Delta^g_{H|H}(\tau^\star) < 0$ and $\Delta^g_{L|L}(\tau^\star) < 0$ for both $g \in \{1,2\}$: the choice of $\tau^\star$ in (17) places it strictly below $\tau_{H|H}(1)$, $\tau_{L|L}(1)$, $\tau_{H|H}(2)$ and $\tau_{L|L}(2)$. Hence no agent in E_L prefers $(2,0)$ to $(1,0)$, and no agent in E_H prefers $(0,2)$ to $(0,1)$, at $\tau^\star$.

Step 4: No agent changes her bundle. Fix an agent in E_L. Her feasible set is $\{(0,0),(1,0),(2,0)\}$; she holds $(1,0)$ at $\tau = 0$ by Step 2; the empty bundle is excluded at every subsidy by Step 2; and $(2,0)$ is excluded at $\tau^\star$ by Step 3. So she holds $(1,0)$ at $\tau^\star$ as well. The argument for an agent in E_H is identical, with $(0,1)$ and $(0,2)$ in place of $(1,0)$ and $(2,0)$.

Every agent therefore holds at $\tau^\star$ exactly the bundle she holds at $\tau = 0$. Aggregating, neither high-SES nor low-SES initiations change, and this holds whatever the shares of agents experiencing the two shocks.

Step 5: The contrast with part (a). In part (a), where the diverse bundle is feasible, the same subsidy strictly raises high-SES initiations and weakly lowers low-SES initiations. Here neither count moves. The margin the subsidy opens is a cross-type addition or switching across types, and removing one type from the feasible set removes those margin while leaving only the same-type additions, which by Step 3 the subsidy does not reach. The response in part (a) is therefore not a property of the subsidy and the types alone: it requires the possibility of substitution across types. □